

Drug Delivery: Engineering and Principles

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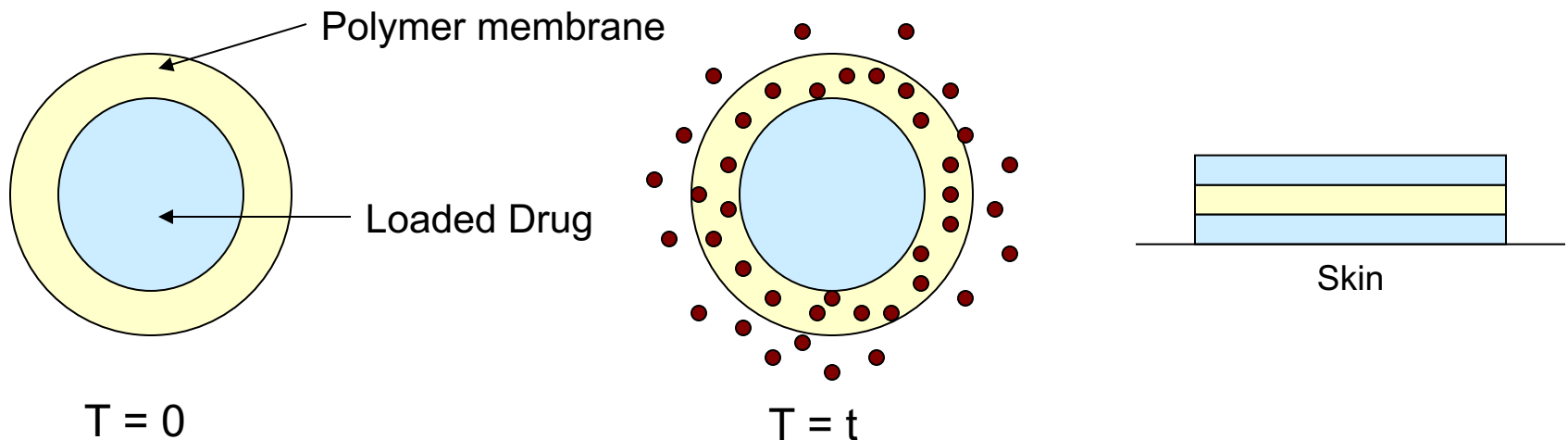
Bioengineering (BE)

Indian Institute of Science

Reservoir System

A supersaturated drug reservoir, surrounded by a non-degradable polymer membrane

- Could have planar configuration (reservoir between two membranes, e.g. Ocusert) or reservoir above a polymer membrane (e.g. skin patch)
- Could have cylindrical configuration (drug loaded in a tube like device, e.g. Norplant contraceptive device)
- Most commonly used polymers: Silicone elastomers, EVAc
- Could achieve nearly constant (zero-order) release rates



Reservoir Systems

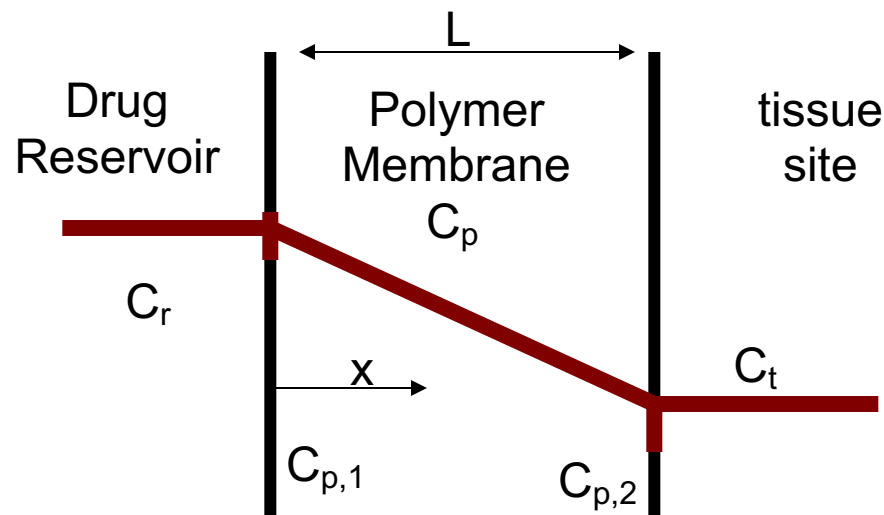
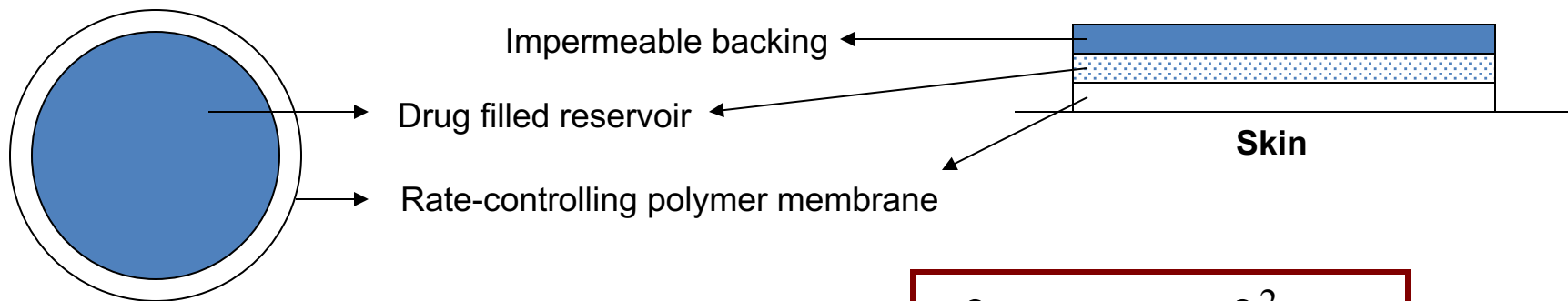
- **Advantages**

- Zero order (constant) release
- Easy to control kinetics by device design parameters

- **Disadvantages**

- Non-degradable, must be removed
- Impermeable to high molecular weight drugs (low porosities)
- Leaks can be dangerous (spilling of supersaturated drug)
- Cost of surgeries

Diffusion controlled delivery: Reservoir type system



$$\frac{\partial c_p}{\partial t} = D_{i:p} \frac{\partial^2 c_p}{\partial x^2} \quad \dots (1)$$

$$c_p = c_{p,0}; t = 0; 0 < x < L$$

$$c_p = c_{p,1}; t > 0; x = 0$$

$$c_p = c_{p,2}; t > 0; x = L$$

$C_{p,0}$ = initial concentration in membrane (uniform)

Perfect sink conditions

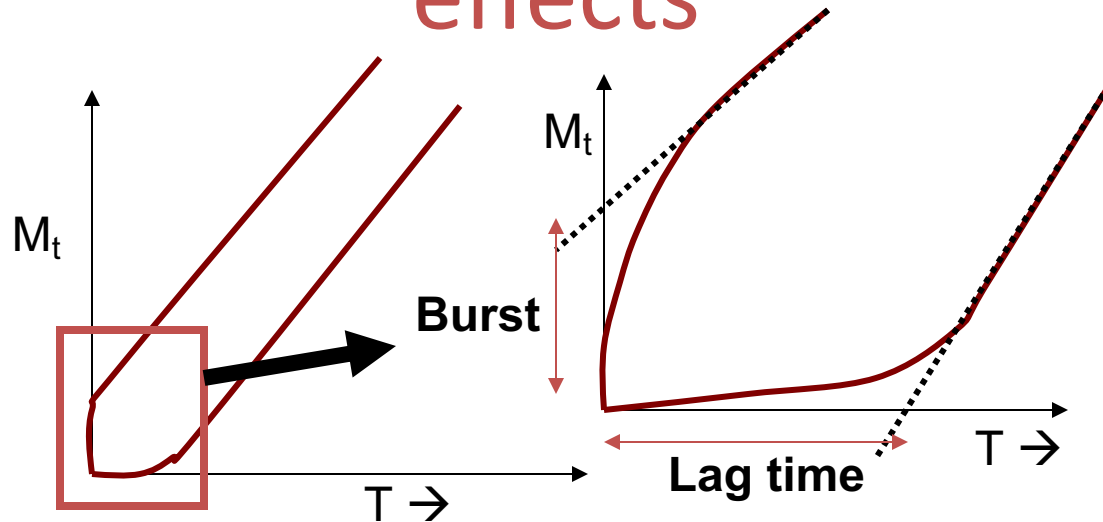
Release kinetics: Reservoir systems

- Membrane-reservoir

$$M(t) = \frac{ADKC_s}{l}t$$

- $M(t)$ is the amount of drug release at time t , A is the surface area of release, D is the drug diffusion coefficient through the polymer carrier, K is partition coefficient, C_s is the drug solubility in the polymer carrier, l is the thickness of the rate-controlling membrane, C_0 is the initial drug loading while t is the duration of release.

Reservoir systems: Lag and burst effects



- Due to the “time history” of the device
- Device used after storage $\rightarrow$ burst effect
 - Membrane is saturated with drug
 - Upon contact with tissue/aqueous medium, initial high level of drug release followed by steady state constant release
- Device used immediately after fabrication $\rightarrow$ lag effect
 - Membrane is devoid of any drug
 - Time for the drug to diffuse through the thickness of the membrane
 - Initial slow rate followed by steady state condition

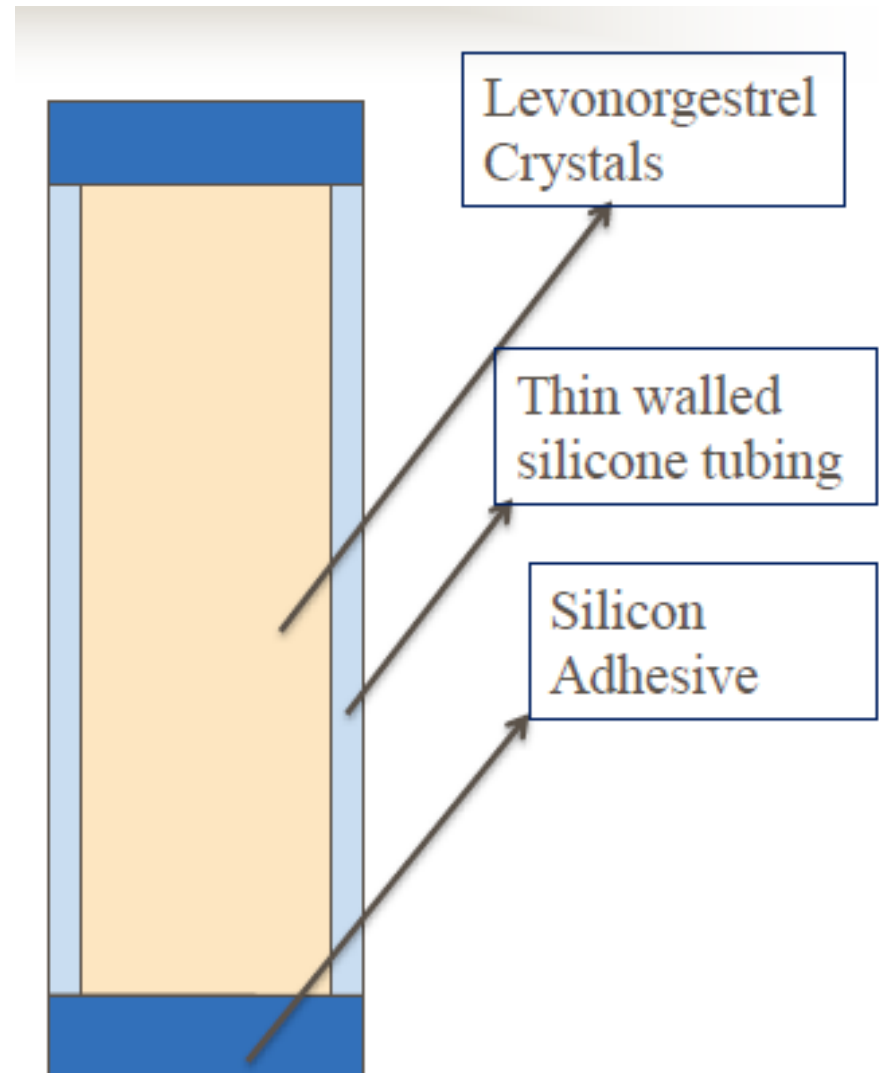
Example: Reservoir system

- Norplant Subdermal Implant:
 - Contraceptive implant
 - Six flexible closed capsules of Dimethylsiloxane / methylvinylsiloxane copolymer containing Levonorgestrel
 - Capsules are surgically implanted in the upper arm
 - Implant releases contraceptive drug continuously at the rate of $30\text{ }\mu\text{g/day}$ over a 5 year period
 - Capsules must be removed surgically
 - Upon removal patients promptly return to fertility

Norplant Contraceptive for Women

Polymer membrane
permeation-controlled
drug delivery system

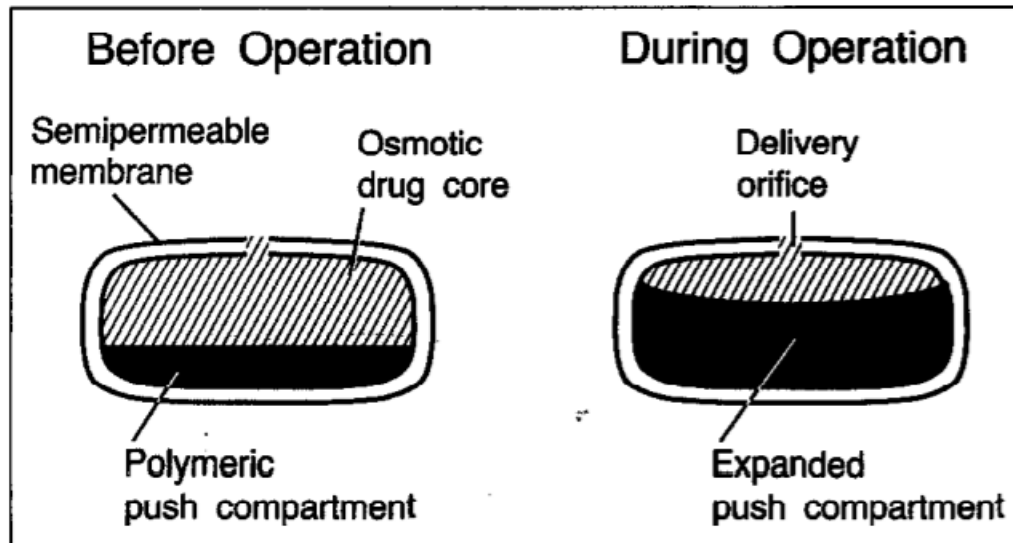
The thickness of the
wall controls the
rate of the drug release



Reservoirs with Orifice

Nifedipine delivery orally for blocking calcium channels in hypertension

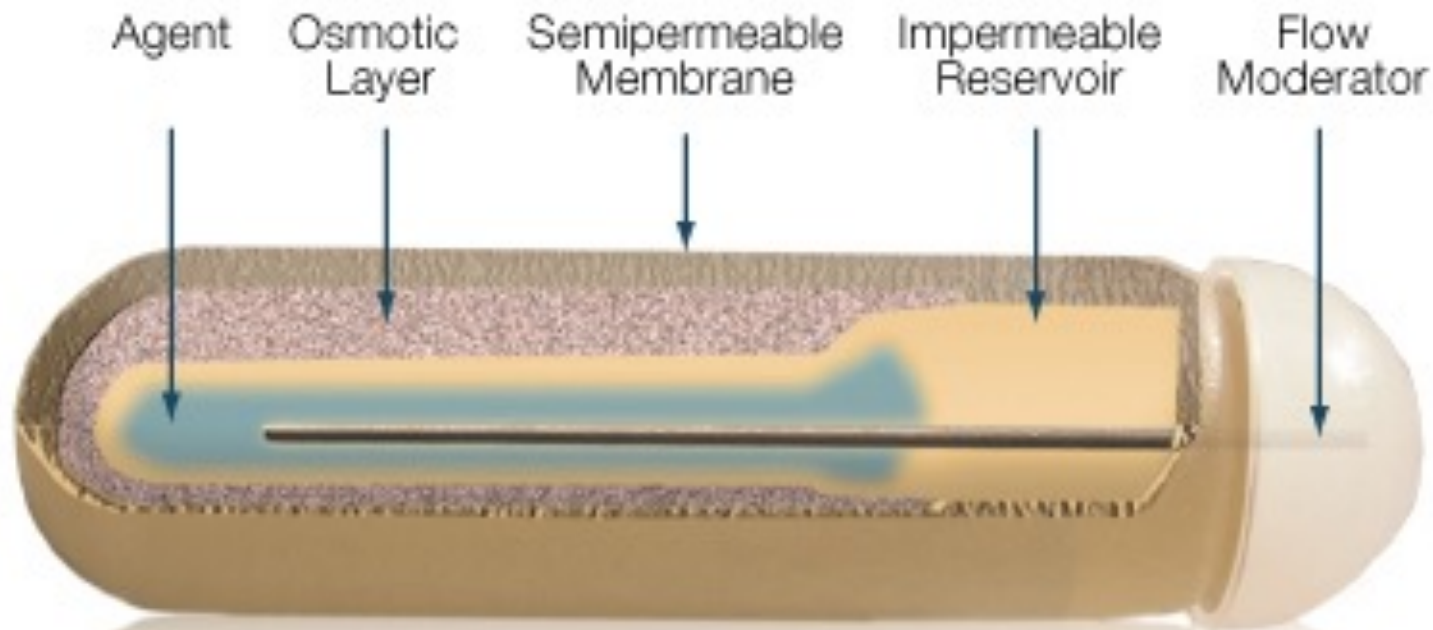
Procardia XL tablets



?

Figure 1. Cross section of the push-pull osmotic pump.

Half life of drug is 2h but this system results in sustained delivery for 24 hours. The device is then excreted in faeces

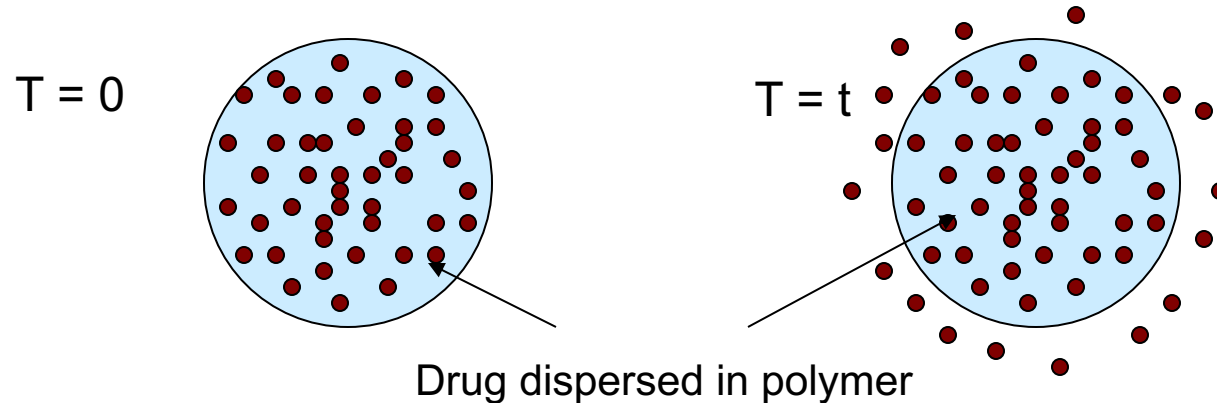


- Rate of delivery is controlled by the water permeability of the outer membrane
- Independent of the drug formulation
- Rates between 0.11 and 10 $\mu\text{L/hr}$
- Durations between 1 day and 6 weeks
- Flow modulator:
 - Minimize diffusion
 - Prevent large accidental spill



Mini osmotic pumps (Alzet Inc.)

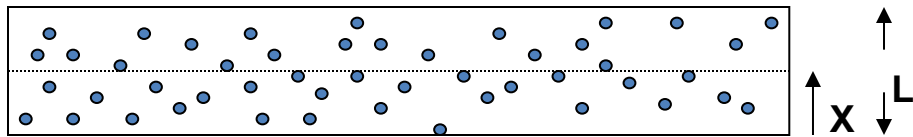
Non-erodible matrix systems



- **Advantages:**
 - Easy to fabricate
 - Less severe problem with leaks and cracks
 - Can be suitable for high molecular weight drugs
- **Disadvantages:**
 - Non-degradable, needs to be removed at end of delivery period
 - Release rate is not generally zero order

Dissolved drug, diffusion through polymer

- **Loading < solubility**
- Rate-limiting step is diffusion through polymer
- This can be made by simply soaking (absorption) the polymer matrix in a drug solution → conc. of drug solution = final conc. in matrix
- Therefore, the release can be modeled as a simple desorption problem



$$\frac{\partial c}{\partial t} = D_{i:p} \frac{\partial^2 c}{\partial x^2}$$

Initial Condition

$$c = c_0; t = 0; 0 < x < L$$

Boundary Conditions

$$c = c_{ext}; t > 0; x = 0, L$$

$$\frac{\partial c}{\partial x} = 0; t > 0; x = L/2$$

Dissolved drug, diffusion through polymer

- These equations are used to simplify the release kinetics description to early and late time approximations for a slab:
- For amount released:

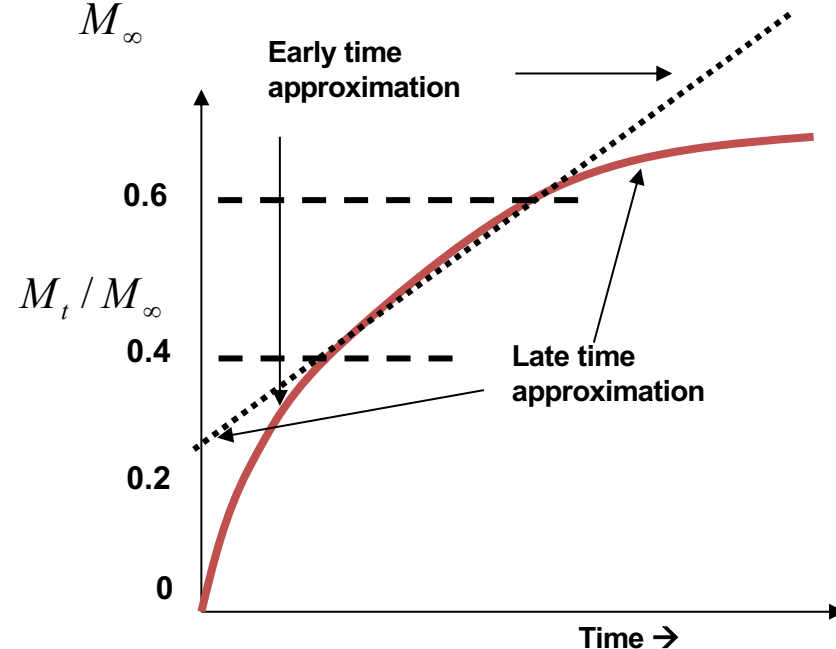
$$\text{EARLY TIME } \frac{M_t}{M_\infty} = 4\left(\frac{Dt}{\pi L^2}\right)^{1/2} \text{ for } 0 \leq \frac{M_t}{M_\infty} \leq 0.6$$

$$\text{LATE TIME } \frac{M_t}{M_\infty} = 1 - \frac{8}{\pi^2} \exp\left(-\frac{\pi^2 Dt}{L^2}\right) \text{ for } 0.4 \leq \frac{M_t}{M_\infty} \leq 1.0$$

- For release rate:

$$\text{EARLY TIME } \frac{dM_t}{dt} = 2M_\infty \left(\frac{D_{i,p}}{\pi L^2 t}\right)^{1/2}$$

$$\text{LATE TIME } \frac{dM_t}{dt} = \frac{8DM_\infty}{L^2} \exp\left(-\frac{\pi^2 Dt}{L^2}\right)$$



- A non-degradable polymer matrix slab was prepared with 100mg of drug dissolved in it. It release 5mg of drug on after 24 hours. How long will it take to release 25mg of drug?

$$\text{EARLY TIME } \frac{M_t}{M_\infty} = 4\left(\frac{Dt}{\pi L^2}\right)^{1/2} \text{ for } 0 \leq \frac{M_t}{M_\infty} \leq 0.6$$

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- A non-degradable polymer matrix slab was prepared with 100mg of drug dissolved in it. It release 5mg of drug on after 24 hours. How long will it take to release 25mg of drug?

$$\frac{5}{100} = 0.05$$

$$\frac{5}{100} = \left(4 \left(\frac{Dt}{\pi L^2} \right)^{1/2} \right) \frac{25mg}{100} = 0.25$$

$$\frac{25}{100} =$$

$$t_n = 25t \approx 25 \text{ days} \propto \frac{1}{2}$$

EARLY TIME $\frac{M_t}{M_\infty} = 4 \left(\frac{Dt}{\pi L^2} \right)^{1/2}$ for $0 \leq \frac{M_t}{M_\infty} \leq 0.6$

LATE TIME $\frac{M_t}{M_\infty} = 1 - \frac{8}{\pi^2} \exp\left(-\frac{\pi^2 Dt}{L^2}\right)$ for $0.4 \leq \frac{M_t}{M_\infty} \leq 1.0$

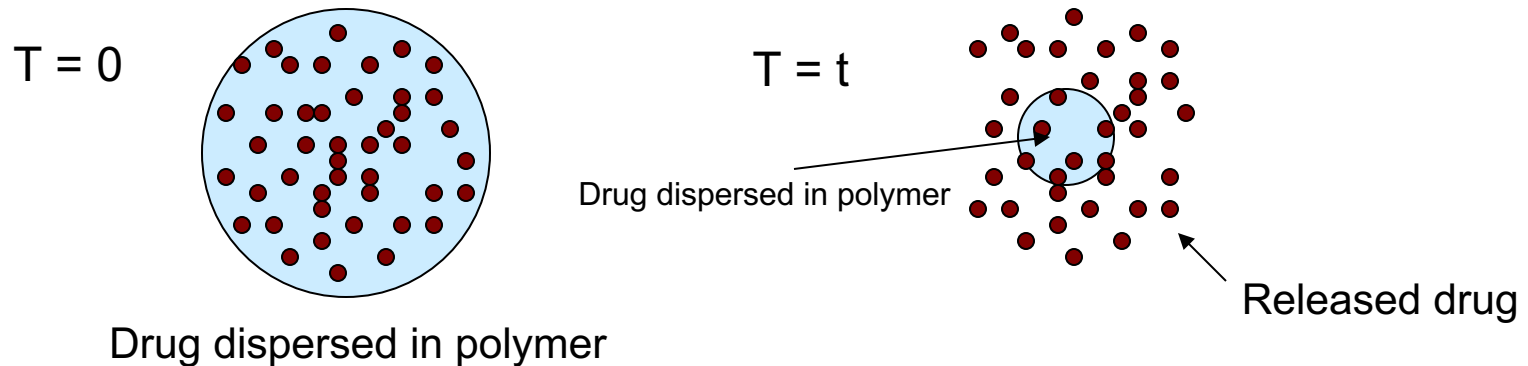
Matrix delivery: Example

- Compudose cattle growth implant
 - Microcrystalline estradiol dispersed in a silicone rubber matrix
 - Implant is placed under animal skin, estradiol is released and enters the systemic circulation, stimulates growth hormone production
 - Available to release estradiol over 200 or 400 days (by varying matrix thickness)
- EVAc vaginal implants for contraceptives
- EVAc implants for delivery of chemotherapy to the brain

Reservoir versus Matrix

- Reservoir systems
 - Zero-order release kinetics leading to constant drug release with time
 - More complicated fabrication process
- Matrix systems
 - Simple to fabricate
 - Drug release decreases with time
- Choice criteria
 - Physicochemical properties of the drug
 - Desired release rate and profile
 - Desired delivery duration
 - Ease of fabrication
 - Patient Safety

Bio-erodible systems



- **Advantages**

- Degradable, more patient compliance
- Can achieve zero order with surface eroding polymer devices

- **Disadvantages**

- Release kinetics is often difficult to control
- By-products of degradation may cause biocompatibility problems

Simplified models of erodible systems

- **Heterogeneous or Surface Erosion**

Simple slab (planar geometry) of area (of both sides) = A

Uniform drug concentration = C_0

- **For only erosion controlled release (no –diffusion),** the release rate is proportional to surface erosion rate (which is assumed constant) and can be written as:

$$\frac{dM_t}{dt} = BC_0A, \quad \text{where } B = \text{surface erosion rate} \quad \begin{array}{l} B \text{ has units of} \\ 1/\text{hr.cm}^2 \end{array}$$

Integrating and assuming initial drug release = 0 (at $t = 0$), one gets

$$M_t = BC_0At$$

- The general form of release rate holds for different geometries and yields a solution with the general form

$$\frac{M_t}{M_\infty} = 1 - \left[1 - \frac{Bt}{C_0l}\right]^n$$

where $n = 1$ for slab of thickness $2l$, $n = 2$ for a cylinder of radius l and $n = 3$ for a sphere of radius l

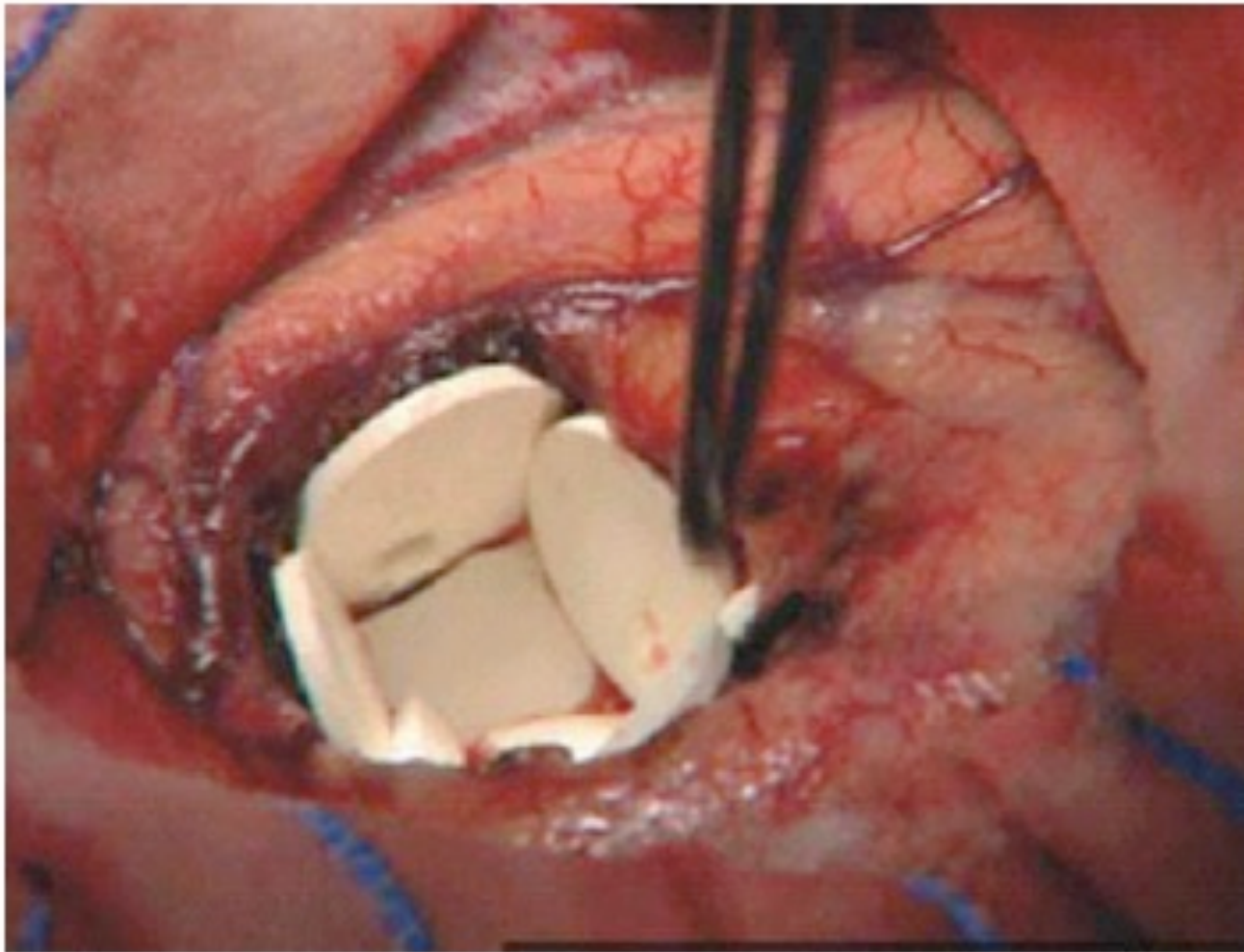


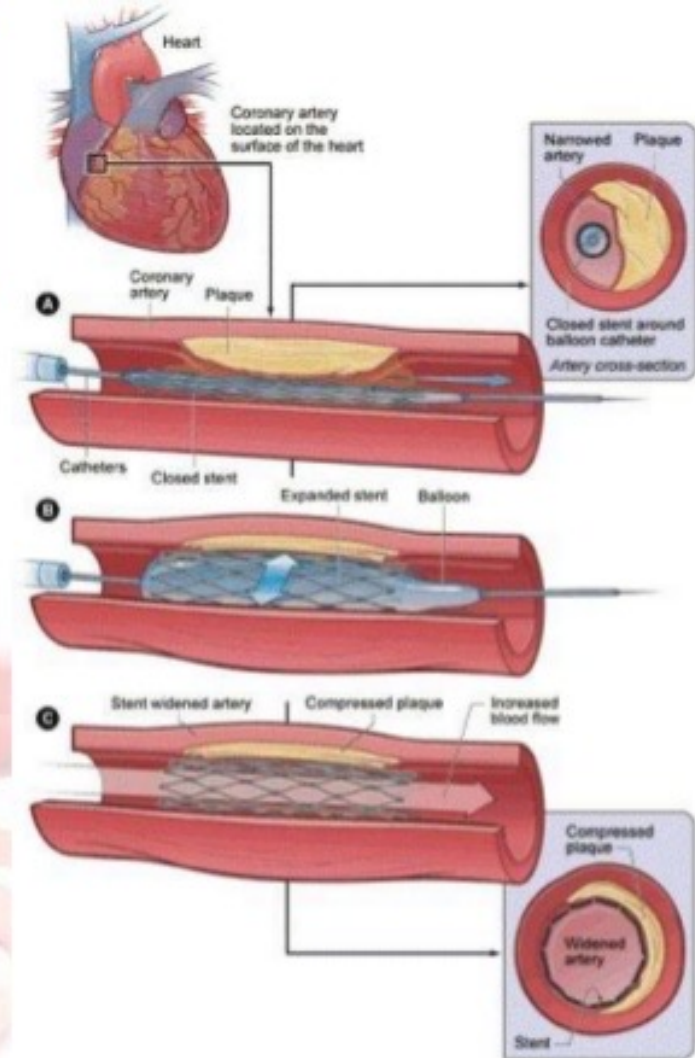
Fig. 4. Gliadel® wafer implantation at tumor resection site. Around 7–8 wafers are placed for localized delivery of BCNU via P(CPP:SA) polymer (Reproduced with permission from [22]).

Copolymer of sebacic acid and carboxy phenoxy propane, and erodes over 6–8 weeks in vivo

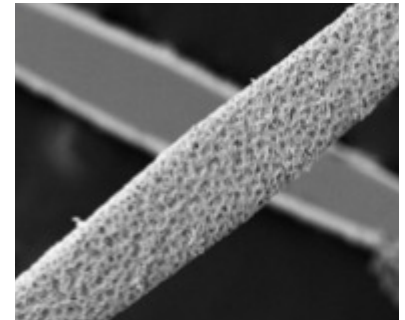
Drug Eluting stents

- A metallic stent contains drug which is gradually released over 14-30 days
- Drugs used are :
Sirolimus, Paclitaxel, everolimus

why? because they are immunosuppressants?



- Coated with matrix containing drug to prevent muscle proliferation
 - Anti-coagulants are also used



Multi-pulse drug delivery from a resorbable polymeric microchip device

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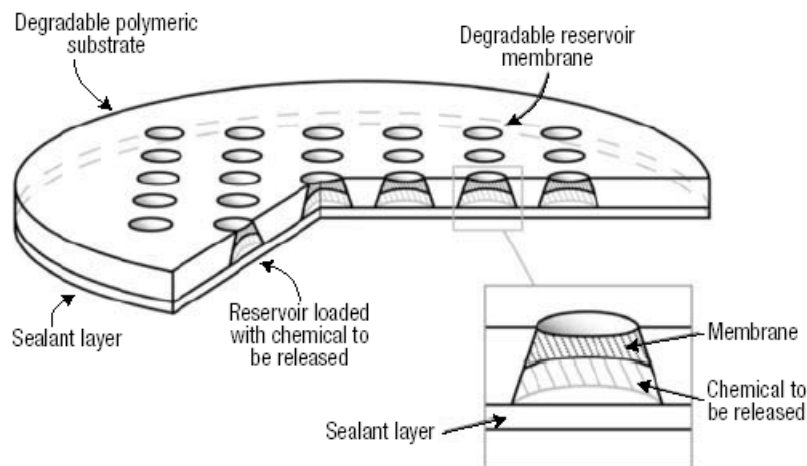
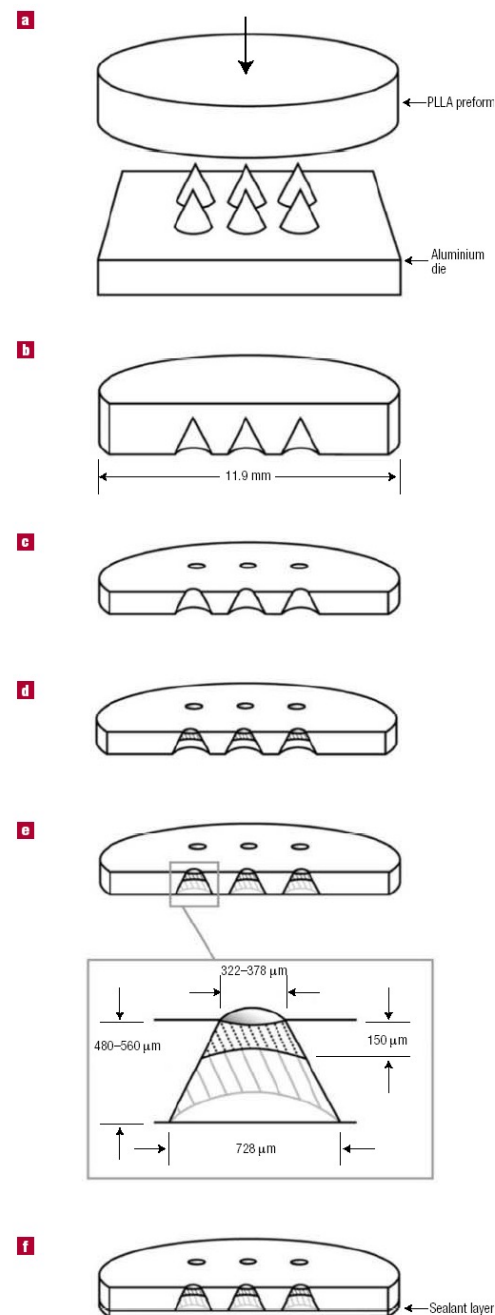


Figure 1 Diagram of polymeric microchip device. The main body of the device is composed of a reservoir-containing substrate that is fabricated from a degradable polymer. Truncated conical reservoirs in the substrate are loaded with the chemical to be released, and sealed with polymeric degradable reservoir membranes on one end and a sealant layer (polyester tape) on the opposite end. Inset, close-up of a reservoir, reservoir membrane, sealant layer and chemical to be released.



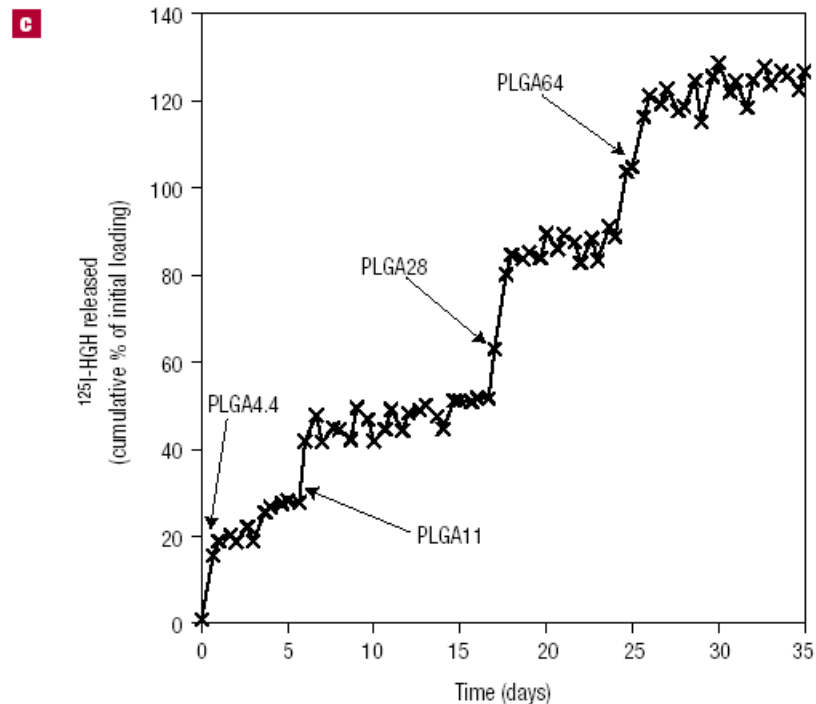
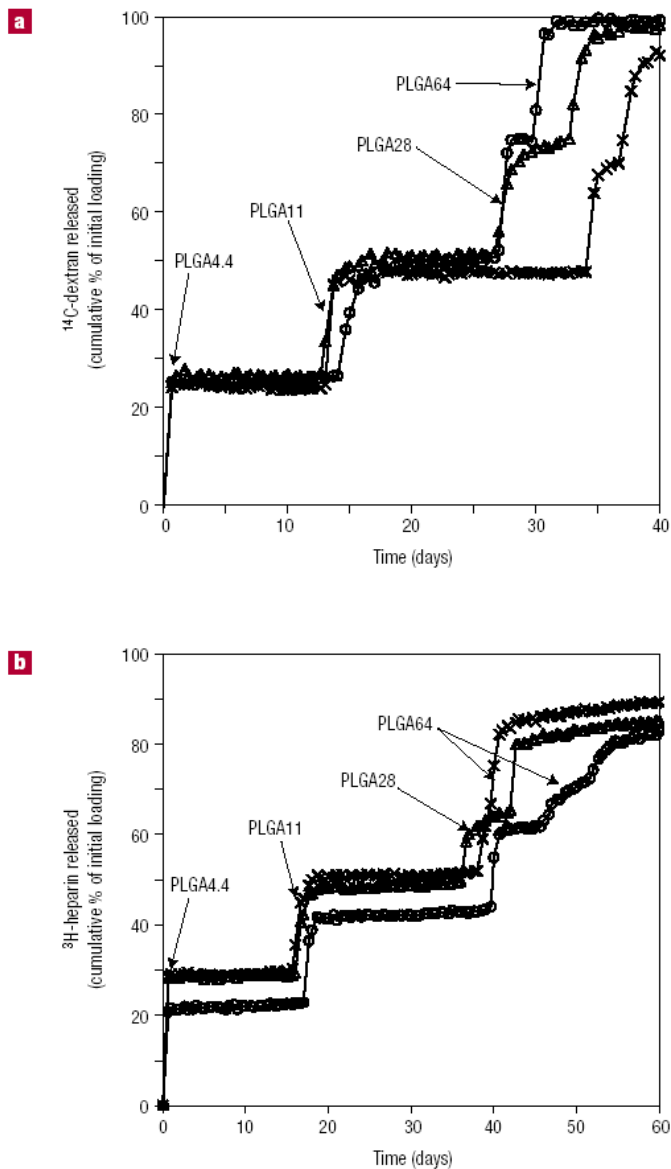


Figure 3 Cumulative percentage of initial loading released from microchip devices *in vitro*. Each symbol (triangle, cross or circle) in a panel represents data collected for a different device. Each device had a total of four reservoirs that were loaded with radiolabelled molecules and sealed with a membrane, and each reservoir had a different membrane that was fabricated from one of the PLGA4.4, PLGA11, PLGA28 or PLGA64 copolymers. The release times of the chemicals from the reservoirs increased as the molecular mass of the reservoir membrane polymers was increased, as shown by the arrows indicating the opening of each type of membrane on the devices. Experiments were conducted in saline solution at 28–33 °C *in vitro*. Devices represented by triangles are included for comparison⁴⁹. **a**, Cumulative release results for devices loaded with ¹⁴C-dextran. **b**, Cumulative release results for devices loaded with ³H-heparin. **c**, Cumulative release results for a device loaded with ¹²⁵I-HGH.

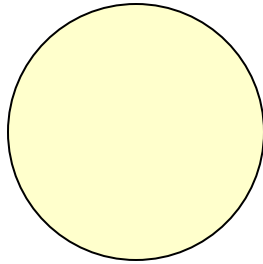
Hydrogels: What are they?

- Hydrogels are three dimensional, hydrophilic polymer networks capable of swelling by absorbing large amounts of fluid
 - Can expand to thousands of times their dry weight in fluid
- Insoluble due to physical or chemical crosslinks which provides the network structure and physical integrity
 - Could be bioerodible or non-erodible
- Numerous applications
 - Contact lenses, tissue engineering matrices, biosensor membranes, drug delivery etc.



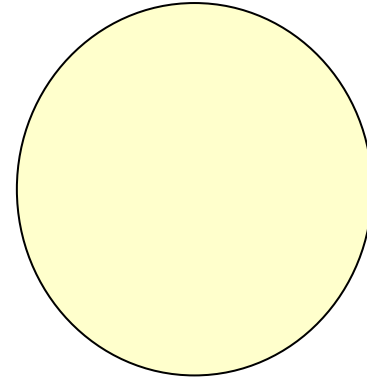
Swelling-controlled matrix (Hydrogels)

$T = 0$



Drug dissolved in polymer

$T = t$



- **Advantages**

- Low burst effects
- Reformulation of vehicle not necessary for different drugs

- **Disadvantages**

- Generally short release periods
- Not suitable for all delivery routes or targets

Hydrogels: Classifications

- **Physical hydrogels:**

- The polymer networks are held together by neutral or ionic bonds
 - Molecular entanglements
 - Secondary forces including ionic, Hydrogen-bonding or hydrophobic forces
- **Not homogeneous** → clusters of molecular entanglements, or hydrophobically- or ionically-associated domains

The entanglement will differ in different places

Physical Hydrogels

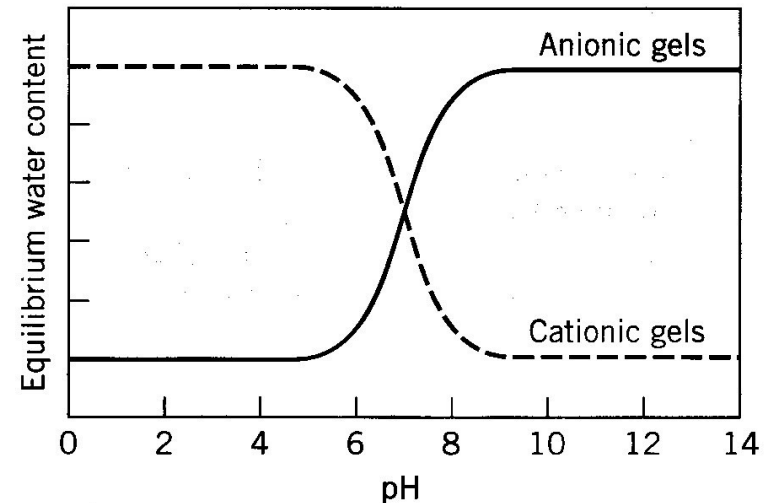
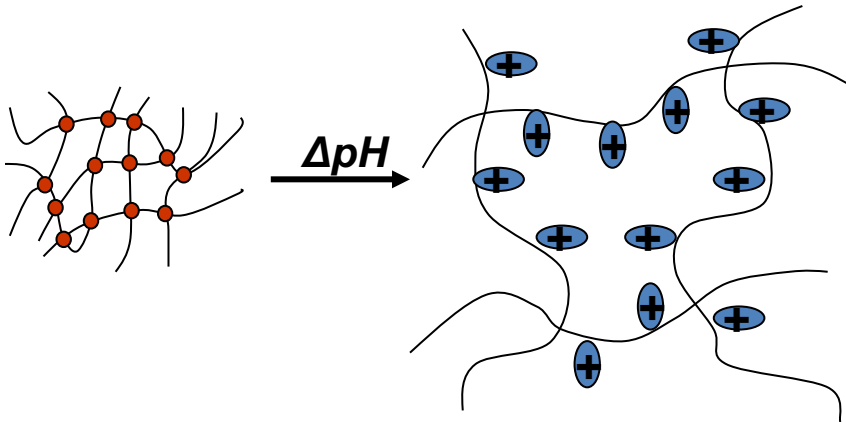
- Sometimes, physical gels can form from bio-specific recognitions,
 - Concanavalin A with a polymeric sugar or
 - Avidin with a polymeric biotin
- Reversible interactions → can be disrupted by physical factors
 - Ionic strength, pH, temperature, stress, or competing solutes
 - Can be used to release their cargo faster

Ionic Hydrogels

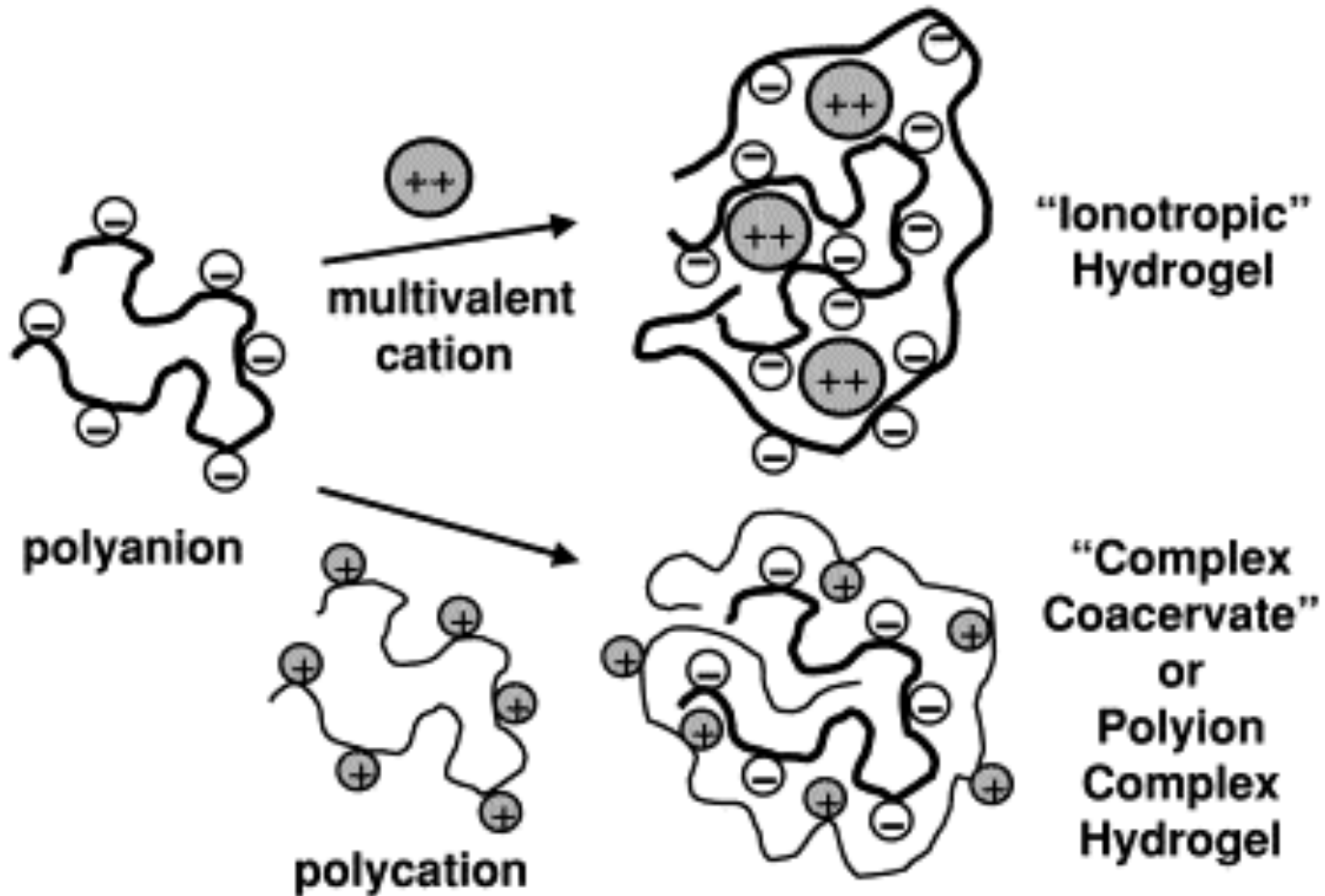
side chain in polymers

- Polymer chains within the hydrogel contains pendant groups that are cationic or anionic
- Ionic → Gels are unionized above (cationic) or below (anionic) the pKa of the groups. For pH above (anionic) or below (cationic) pKa, the pendant groups are ionized and repel each other → large swelling (completely reversible)

If they are ionized, then the groups will repel each other so that the pore sizes will increase and there will be a lot of repulsion



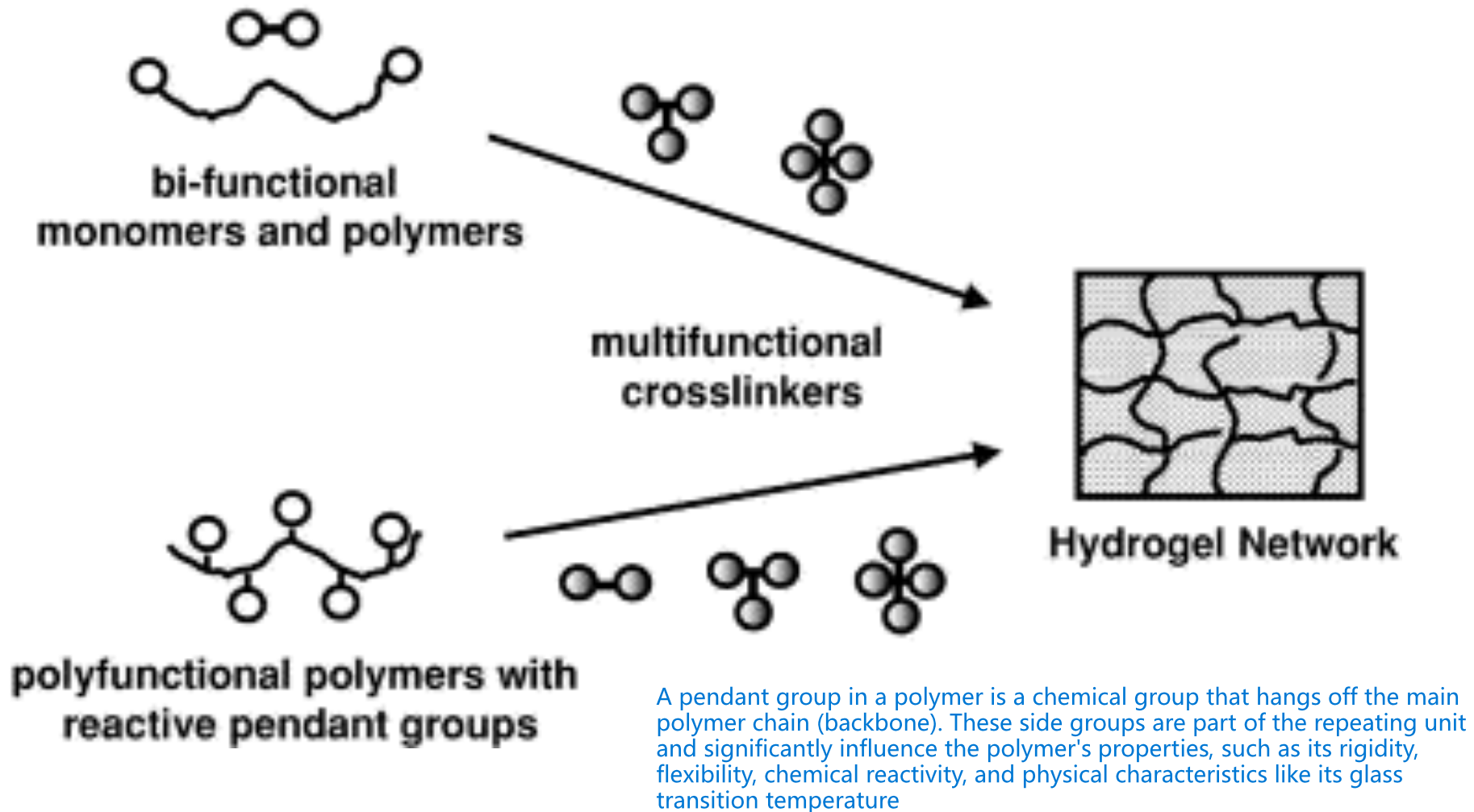
Ionic hydrogels



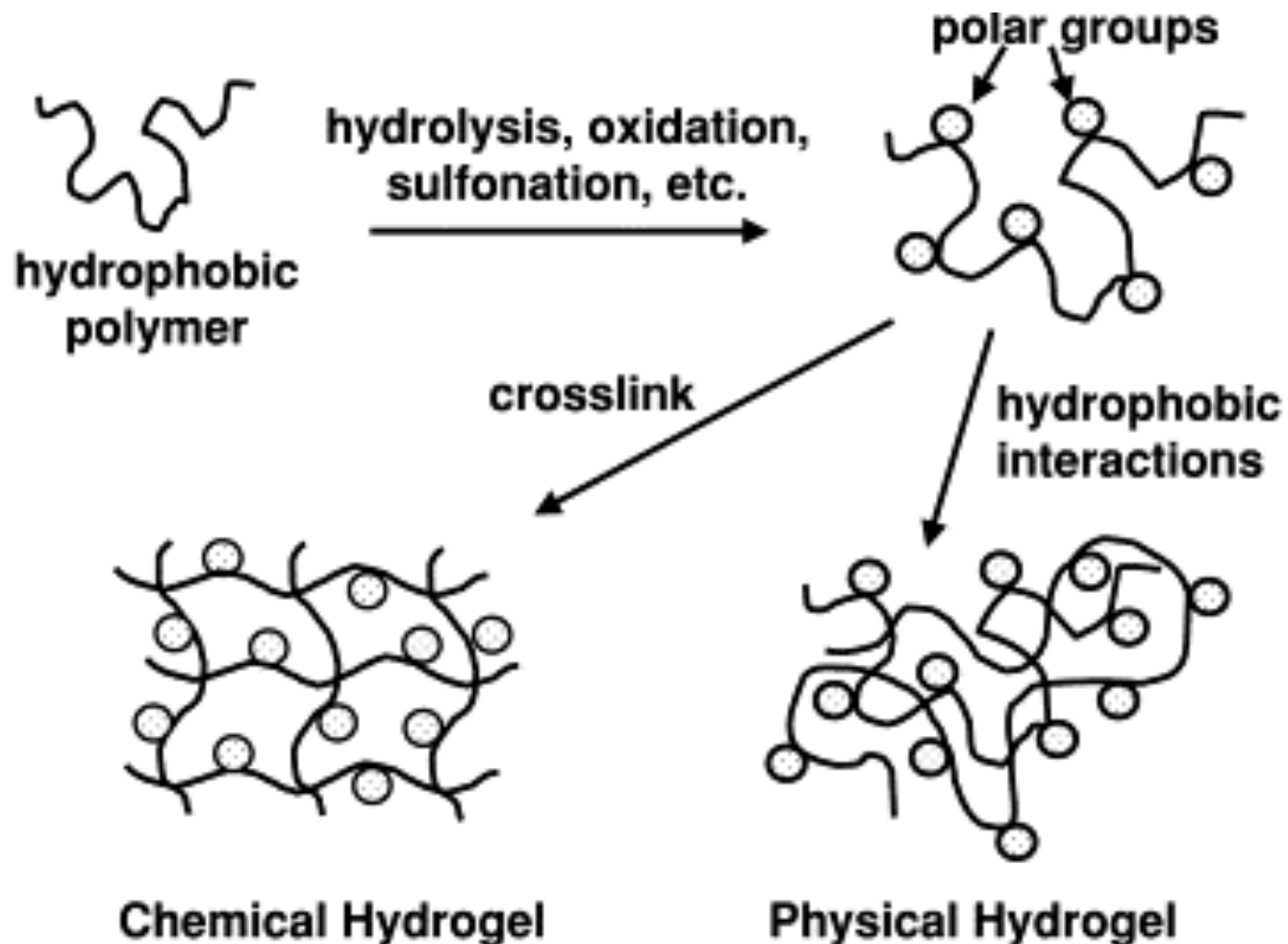
Chemical hydrogels

- Covalently-crosslinked hydrogel networks.
- May be generated by
 - directly crosslinking of water-soluble polymers, or by
 - conversion of hydrophobic polymers to hydrophilic polymers plus crosslinking to form a network.
- Like physical hydrogels, chemical hydrogels are not homogeneous.
 - Usually contain regions of low water swelling and high crosslink density, called 'clusters', that are dispersed within regions of high swelling, and low crosslink density. This may be due to hydrophobic aggregation of crosslinking agents, leading to high crosslink density clusters.

Hydrogel synthesis processes



Hydrogels from hydrophobic polymers

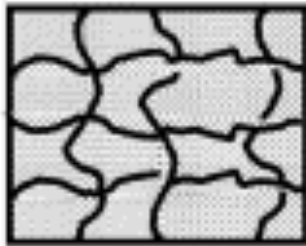


Interpenetrating network

water soluble polymer

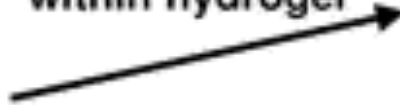


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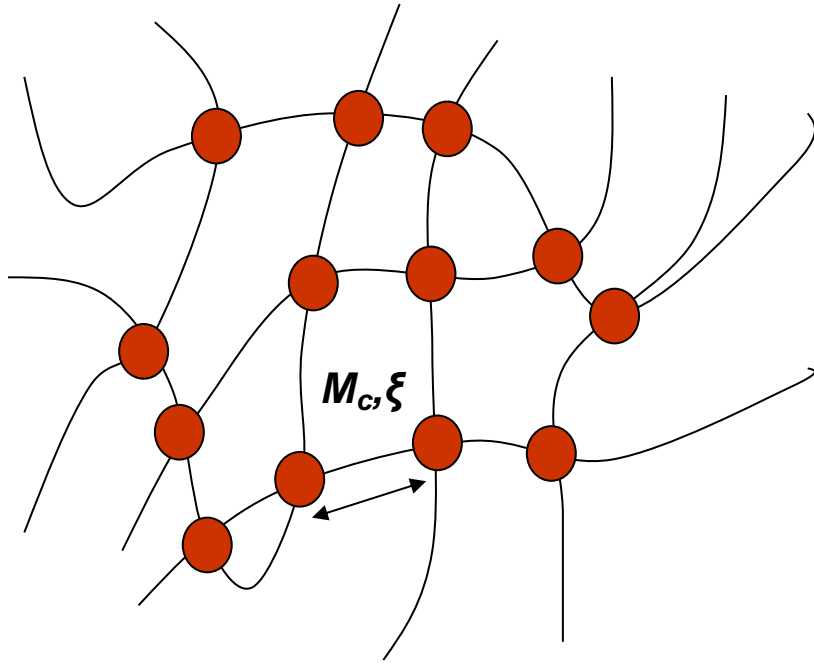
Hydrogel Network

polymerize monomer
within hydrogel



Interpenetrating
Hydrogel Network

Structure and properties of hydrogels



$$v_{2,s} = (\text{volume of polymer})/(\text{volume of swollen gel}) \\ = V_p / V_{gel} = 1/Q$$

- Most important parameters defining hydrogels
 - Polymer volume fraction in the swollen state ($v_{2,s}$)
 - A measure of the amount of fluid that a hydrogel can incorporate into its structure
 - Determined by equilibrium swelling experiments
 - Effective molecular weight of the polymer between crosslinks (M_c)
 - Measure of the degree of crosslinking in the gel
 - Network mesh or pore size (ξ)
 - Measure of the network porosity

Neutral Hydrogels: Mathematical Treatment

- **Flory-Rehner theory for crosslinked polymers gels**

- Based on **Gaussian distribution** of polymer chains
- Degree to which a polymer network swelled, was governed by (a) the elastic retractive forces of the polymer chains and (b) the thermodynamic compatibility of the polymer and solvent molecules (mixing)
- Therefore, the total free energy upon swelling is given by

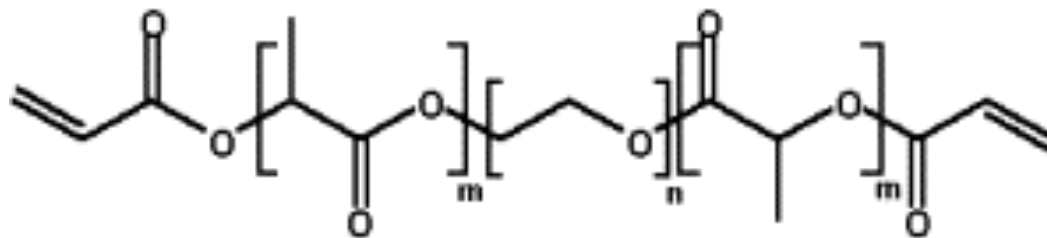
ability of polymer to be compatible with the solvent

$$\Delta G_{total} = \Delta G_{elastic} + \Delta G_{mixing}$$

ability of polymer to stretch

Example: PLA-PEG-PLA hydrogels

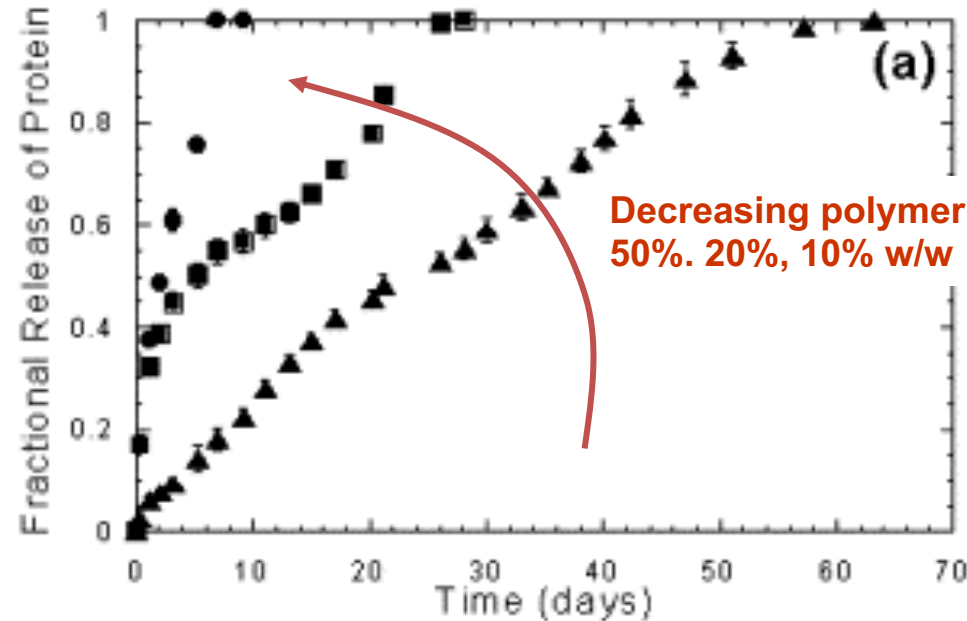
- PLA is polymerized to the hydroxyl end groups of PEG creating a a block copolymer
- The terminal acid groups of PLA are acrylated and the polymer is crosslinked using UV light to form a hydrogel



Macromer 1: $m \sim 3, n \sim 104$

Macromer 2: $m \sim 2, n \sim 104$

Release of BSA: Kinetics

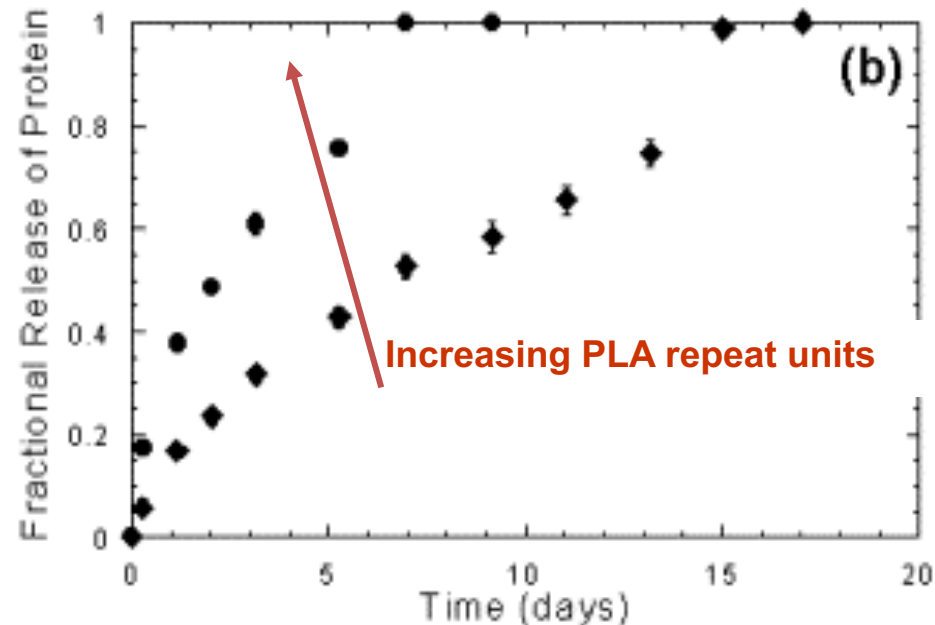


Decreasing polymer concentration
50%, 20%, 10% w/w

Effect of decreasing
polymer concentration?

more pore size, faster release

Effect of increasing the chain length of PLA
ends while keeping PEG length same?



Increasing PLA repeat units

In situ Crosslinking Hydrogels

- Cross-linked at the site – physician has a syringe and injects into body
- Forms a hydrogel inside the body
- Polymerization trigger could be:
 - Time
 - Temperature
 - pH
- Disadvantages
 - May diffuse throughout body
 - No control of shape

- Currently there are over 30 injectable hydrogel-based products that have been approved by the FDA and/or EMA (Table 1)

TABLE 1 Clinically approved injectable hydrogels, grouped by their material class and broad indication

Name (company)	Hydrogel material/payload (gelation mechanism)	Injection type	Approved indication	Approval (year)
Cancer: synthetic				
SpaceOAR [®] Hydrogel (Augmenix, Inc.)	Polyethylene glycol (chemical reaction)	Percutaneous	For protecting vulnerable tissues during prostate cancer radiotherapy	EMA (2010) FDA (2015)
Vantas [®] (Endo Pharmaceuticals)	Histrelin acetate, poly(2-hydroxyethyl methacrylate), poly(2-hydroxypropyl methacrylate) and gonadotropin releasing hormone (chemical reaction)	Subcutaneous	Palliative treatment of prostate cancer	FDA (2004) EMA (2005)
Facial correction: synthetic				
Radiesse [®] (+) (Merz Pharmaceuticals)	Hydroxylapatite, carboxymethylcellulose with Lidocaine (physical interaction)	Dermis	Correction of wrinkles and folds, stimulation of natural collagen production	FDA (2015)
Radiesse [®] (Bioform Medical, Inc.)	Hydroxylapatite, carboxymethylcellulose (physical interaction)	Dermis	For correction of facial folds and wrinkles, signs of facial fat loss and volume loss	EMA (2004) FDA (2006 for first indication)
Artefill [®] (Suneva Medical, Inc.)	Polymethylmethacrylate beads, collagen and lidocaine (physical interaction)	Dermis	Facial wrinkles and folds	FDA (2006)
Sculptra [®] (Sanofi Aventis U.S.)	Poly-L-lactic acid (physical interaction)	Dermis	For correction of signs of facial fat loss, shallow to deep contour deficiencies and facial wrinkles	EMA (2000) FDA (2004 for first indication)

Facial correction: natural

Belotero balance® (+) Lidocaine (Merz Pharmaceuticals)	Hyaluronic acid with lidocaine (chemical reaction)	Dermis	Moderate to severe facial wrinkles and folds	FDA (2019)
Revanesse® Versa+	Hyaluronic acid with lidocaine (chemical reaction)	Dermis	Moderate to severe facial wrinkles and creases	FDA (2018)
Teosyal® RHA (Teoxane SA)	Hyaluronic acid (chemical reaction)	Dermis	Facial wrinkles and folds	EMA (2015) FDA (2017)
Revanesse® Versa/Revanesse® Ultra (Prollenium Medical Technologies Inc.)	Hyaluronic acid (chemical reaction)	Dermis	Moderate to severe facial wrinkles and creases	FDA (2017)
Restylane® Lyft, Restylane® Refyne, Restylane® Defyne (Galderma Laboratories, L.P.)	Hyaluronic acid with Lidocaine (chemical reaction)	Subcutaneous, dermis, lips	For correction of volume deficit, facial folds and wrinkles, midface contour deficiencies, and perioral rhytids	EMA (2010) FDA (2012 for first indication)
Restylane® Silk (Valeant Pharmaceuticals North America LLC/Medicis)				
Restylane® Injectable Gel (Medicis Aesthetics Holdings, Inc.)				
Belotero balance® (Merz Pharmaceuticals)	Hyaluronic acid (chemical reaction)	Dermis	Moderate to severe facial wrinkles and folds	EMA (2004) FDA (2011)
Juvéderm® XC (Allergan, Inc.)	Hyaluronic acid with lidocaine (chemical reaction)	Facial tissue	Correction of facial wrinkles and folds	FDA (2010)
Evolence® Collagen Filler (Colbar Lifescience I)	Collagen (chemical reaction)	Dermis	Moderate to deep facial wrinkles and folds	EMA (2004) FDA (2008)
Eleveess® (Anika Therapeutics)	Hyaluronic acid with lidocaine (chemical reaction)	Dermis	Moderate to severe facial wrinkles and folds	FDA (2006) EMA (2007)

TABLE 1 (Continued)

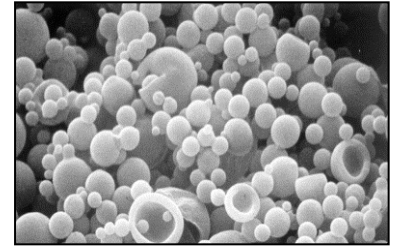
Name (company)	Hydrogel material/payload (gelation mechanism)	Injection type	Approved indication	Approval (year)
Juvéderm [®] /Voluma XC/Ultra XC/Volbella XC/ Vollure XC (Allergan, Inc)	Hyaluronic acid (chemical reaction)	Facial tissue, cheek, lips	For correction of facial wrinkles and folds, volume loss, and lip augmentation.	EMA (2000) FDA (2006 for first indication)
Hylaform [®] (Hylan B gel), Captique Injectable Gel, Prevelle Silk (Genzyme Biosurgery)	Modified hyaluronic acid derived from a bird (avian) source (chemical reaction)	Dermis	Correction of moderate to severe facial wrinkles and folds	EMA (1995) FDA (2004)
Collagen Implant, CosmoDerm [®] 1 human-based collagen, CosmoDerm [®] 2 human-based collagen CosmoPlast [®] human-based collagen (Inamed Corporation/Allergan, Inc.)	Human collagen (CosmoDerm: physical interaction, CosmoPlast: chemical reaction)	Superficial papillary dermis	For correction of soft tissue contour deficiencies, such as wrinkles and acne scars	FDA & EMA (2003)
Fibrel [®] (Serono Laboratories)	Collagen (physical interaction)	Dermis	For correction of depressed cutaneous scars	FDA (1988)
Zyplast(R) [®] and Zyderm(R) [®] (Inamed Corporation/Allergan, Inc.)	Bovine collagen (chemical reaction)	Dermis	For correction of contour deficiencies	FDA and EMA (1981)
Spinal fusion: natural				
EUFLEXA [®] (Ferring Pharmaceuticals Inc.)	Hyaluronic acid (physical interaction)	Intra-articular	Knee osteoarthritis	FDA (2004) EMA (2005)
INFUSE [®] bone graft (Medtronic Sofamor Danek USA, Inc.)	Collagen and recombinant human bone morphogenetic protein-2 (physical interaction)	Spinal injection	Spinal fusion, and spine, oral-maxillofacial and orthopedic trauma surgeries	FDA (2002 for first indication)

Osteogenic protein 1(OP-1®) implant, OP-1® Putty (Stryker Biotech)	Collagen, carboxymethylcellulose, and recombinant OP-1 (physical interaction)	Spinal injection	Posterolateral lumbar spinal fusion	FDA (2001)
Other: synthetic				
TracelT® Hydrogel Tissue Marker (Augmenix, Inc.)	Polyethylene glycol (chemical reaction)	Percutaneous	Improved soft tissue alignment for image guided therapy	FDA (2013)
Supprelin LA® (Indevus Pharmaceuticals, Inc.)	Histrelin acetate, Poly (2-hydroxyethyl methacrylate) (chemical reaction)	Subcutaneous	Central precocious puberty	EMA (2005) FDA (2007)
Bulkamid® hydrogel (Searchlight Pharma)	Polyacrylamide (chemical reaction)	Transurethral	Female stress urinary incontinence	EMA (2003) FDA (2006)
Coaptite® (BioForm Medical, Inc.)	Calcium hydroxylapatite, sodium carboxymethylcellulose, glycerin (physical interaction)	Submucosal	Female stress urinary incontinence	EMA (2001) FDA (2005)
Other: natural				
Algisyl-LVR® Hydrogel Implant (LoneStar Heart, Inc.)	Alginate (physical interaction)	Percutaneous	Advanced heart failure	EMA (2014)

Nano- and Micro-particles

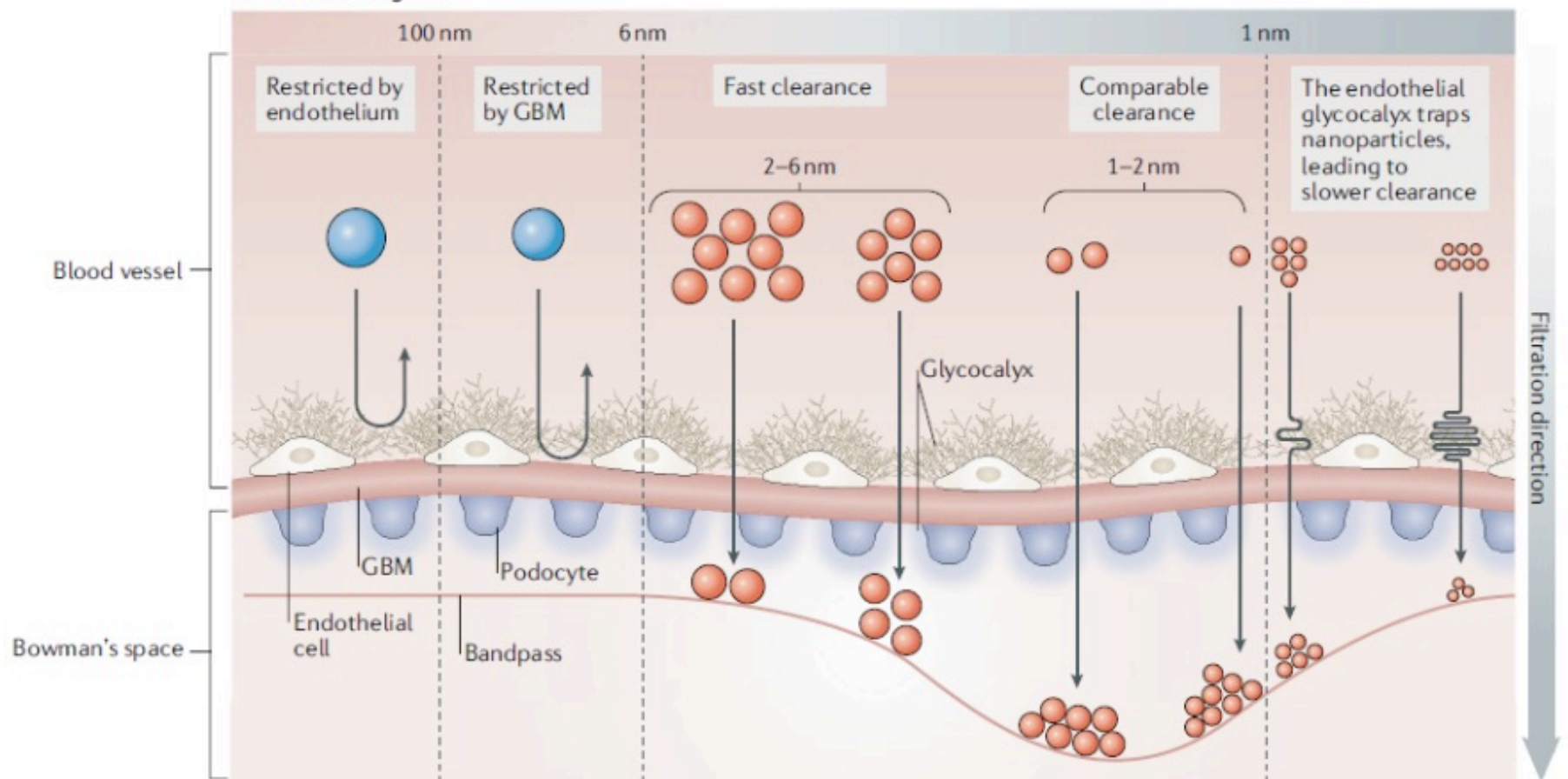
- Acts as a controlled release depot system similar to matrices, reservoirs or hydrogels
- Decreasing the non-specific delivery of the drug to non- target tissues
- Injectable (no need for surgical implantation)
- Convenient way for delivering large hydrophilic molecules inside cells
- Particles can be adapted for targeted delivery to cells and tissues
- Improving the stability of the drug in vivo
- Improving shelf life of the product.

Particles in drug delivery

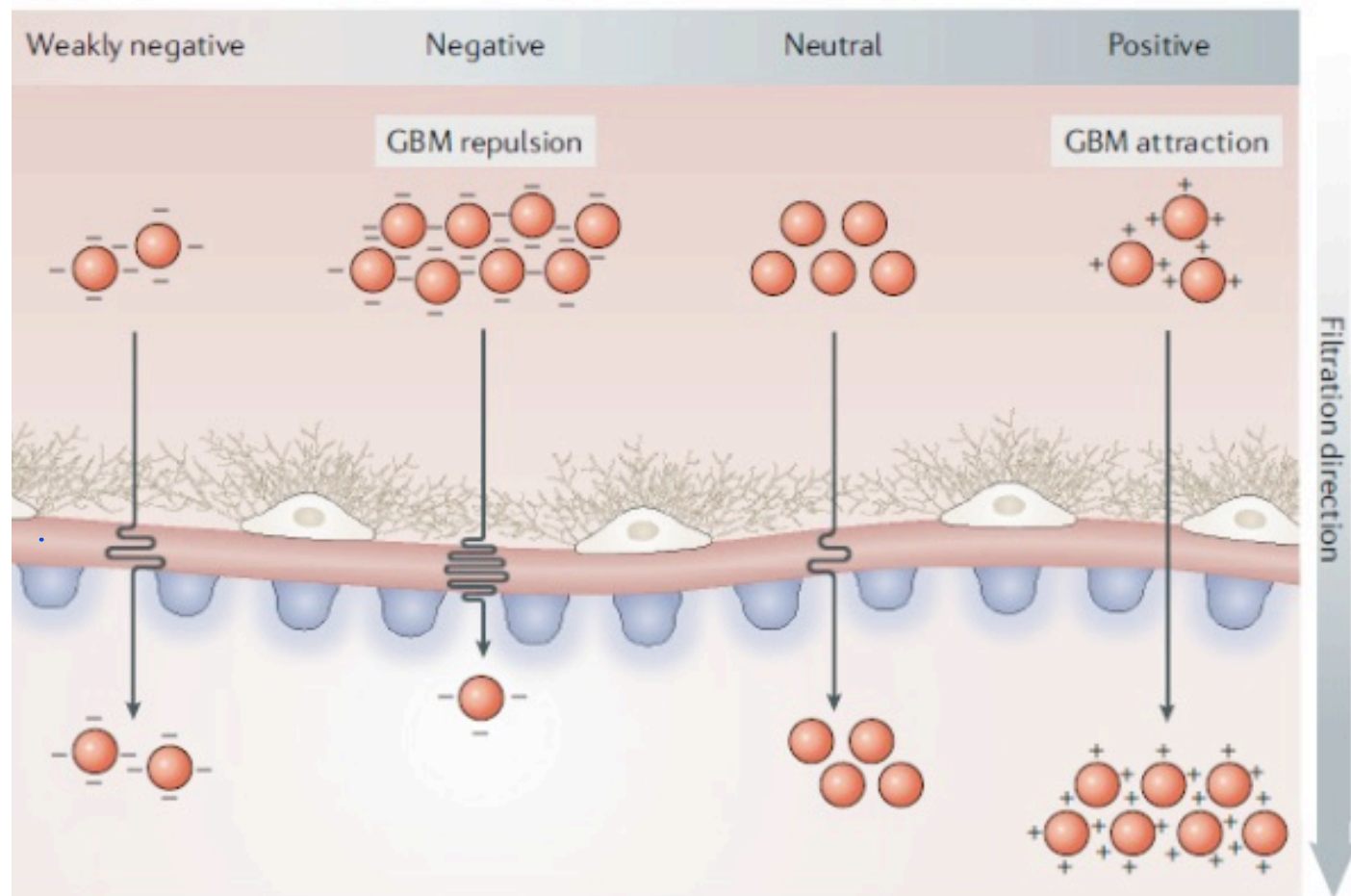


- Microparticles: $> 1 \mu\text{m}$ in size (typically $1\text{-}100 \mu\text{m}$)
- Nanoparticles: Typically $10\text{nm} - 1000\text{nm}$ in size
- Drugs (small molecules, proteins, peptides, nucleic acids) are generally encapsulated within these particles, but they can also be adsorbed or covalently linked to the particle surface
- These particles can be in the form of
 - Micro or nano capsules (well defined polymer envelop surrounding a hollow core)
 - Solid micro or nanoparticles (particles that are essentially a matrix)

a Size-scaling law



- Nanoparticles with a hydrodynamic diameter (HD) > 100 nm cannot traverse the endothelium.
- Nanoparticles with HD = 6–100 nm cannot cross the **glomerular basement membrane** (GBM).
- Nanoparticles with HD = 1–6 nm can traverse the glomerular filtration membrane, with smaller nanoparticles crossing faster than larger ones.
- Nanoparticles with HD < 1 nm **physically interact with the endothelial glycocalyx**, resulting in inverse size- dependent filtration, with smaller particles being cleared more slowly than larger ones.



- The endothelial glycocalyx (charge layer 1), GBM (charge layer 2) and podocyte glycocalyx (charge layer 3) are negatively charged and thus make the glomerular filtration of engineered nanoparticles charge-dependent
- Positively charged nanoparticles are cleared faster than neutral ones, followed by negatively charged nanoparticles, owing mainly to interactions with the GBM

Particle-mediated delivery: Definitions

- **Particle size and shape**

- Qualitatively: microscopic techniques. Most commonly by electron microscopic techniques.
- Quantitative: coulter counting or light scattering techniques. Gives a distribution of diameters from which number average, weight average, volume average sizes can be calculated.

- **Polydispersity**

- Calculated from the size distribution. Has same definition as polymers. A measure of how broad or narrow the size distribution is.

- **Carrier Composition**

- Percentage of polymer, drug, solvents, surfactants and any other additives remaining in the final particle formulation

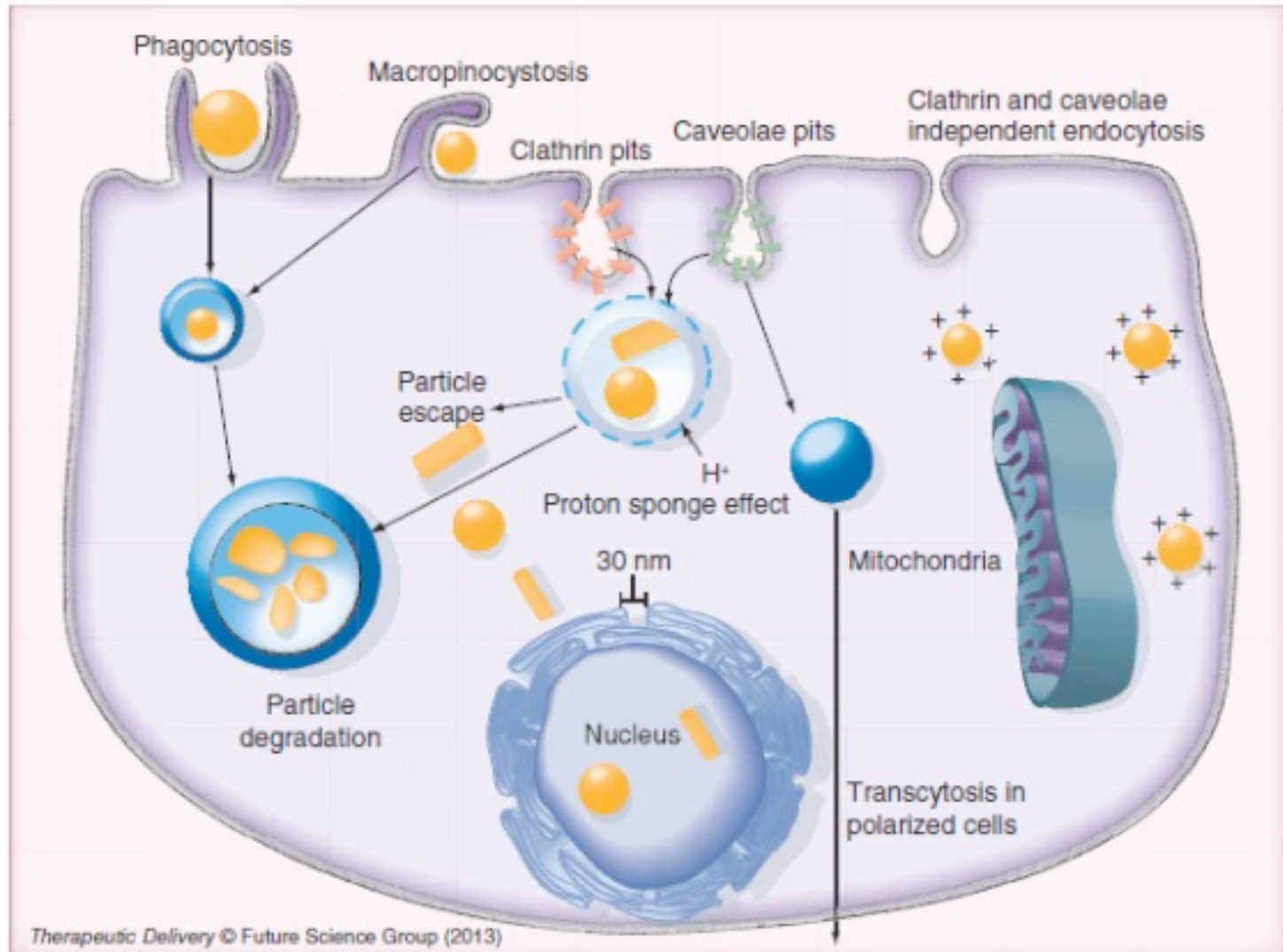
- **Encapsulation efficiency**

- Percentage of the starting encapsulant (drug) that is captured in the finished polymer particles

- **Loading level or encapsulation ratio**

- The weight percentage of drug in the particle formulation

Intra-cellular Delivery

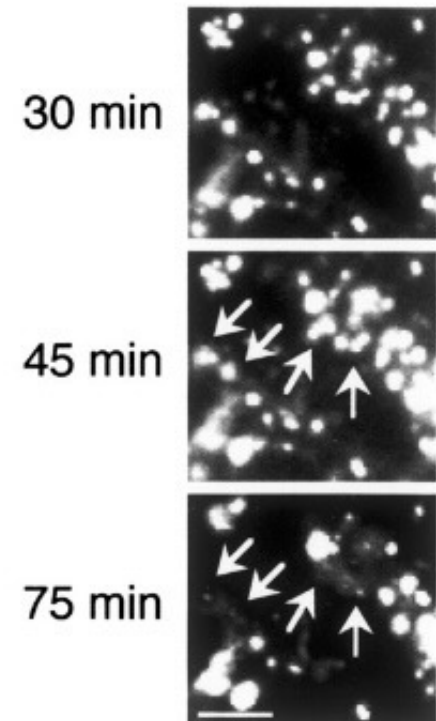
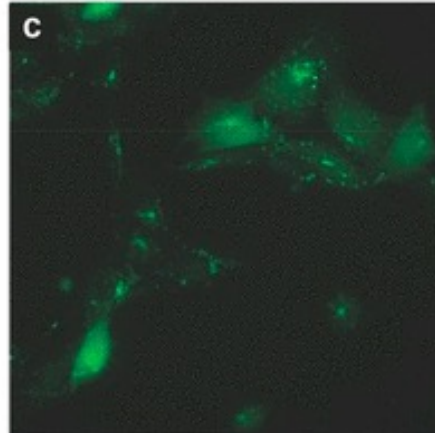
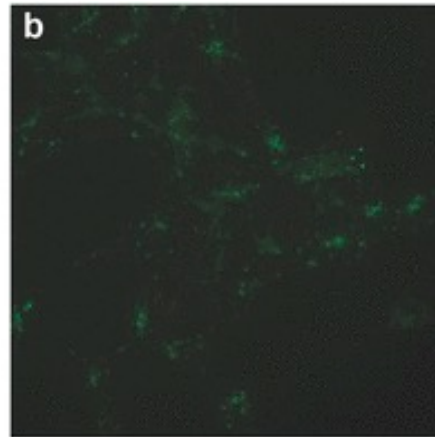
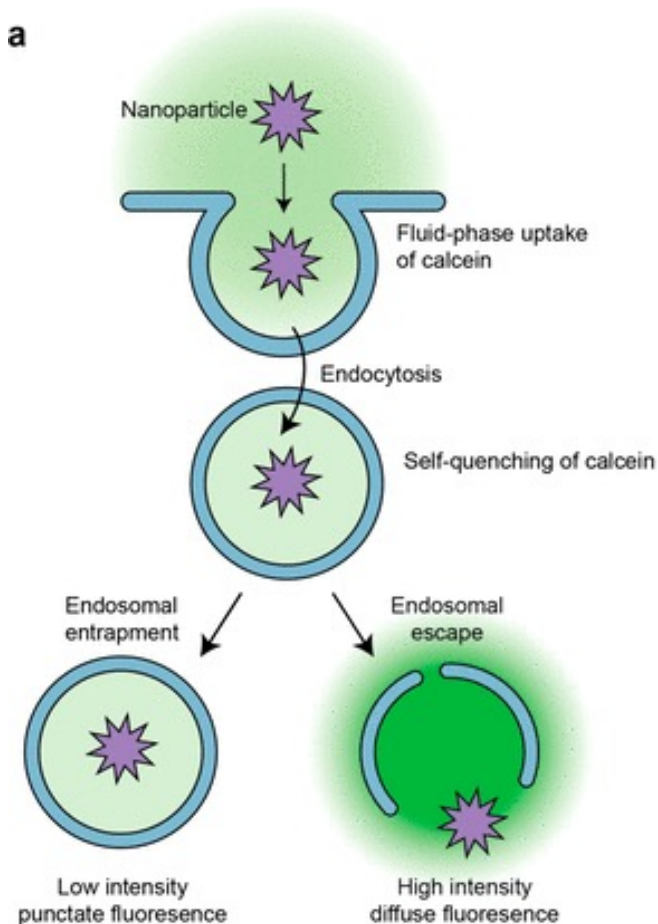


Proton Sponge effect

- Particles uptaken through endocytosis
- Get localized in endosomes and lysosomes where the environment is very harsh
- To help escape the particles, “proton sponge” effect is used
- Particles are designed to carry secondary or tertiary amines
 - Buffer the pH of endosomes
 - Results in proton influx through the endosomal proton pumps,
 - Followed by chloride ion transport
 - Leads to osmotic swelling and bursting of vesicles
- PEI (poly ethylene Imine) is a widely used polymer for this phenomenon
- Conjugation of peptides that can create pores in endosomal membrane also been proposed
- However, the maximum reported efficacy of endosomal escape is <10%

Proton Sponge effect

Membrane impermeable calcein and has low fluorescence at low pH



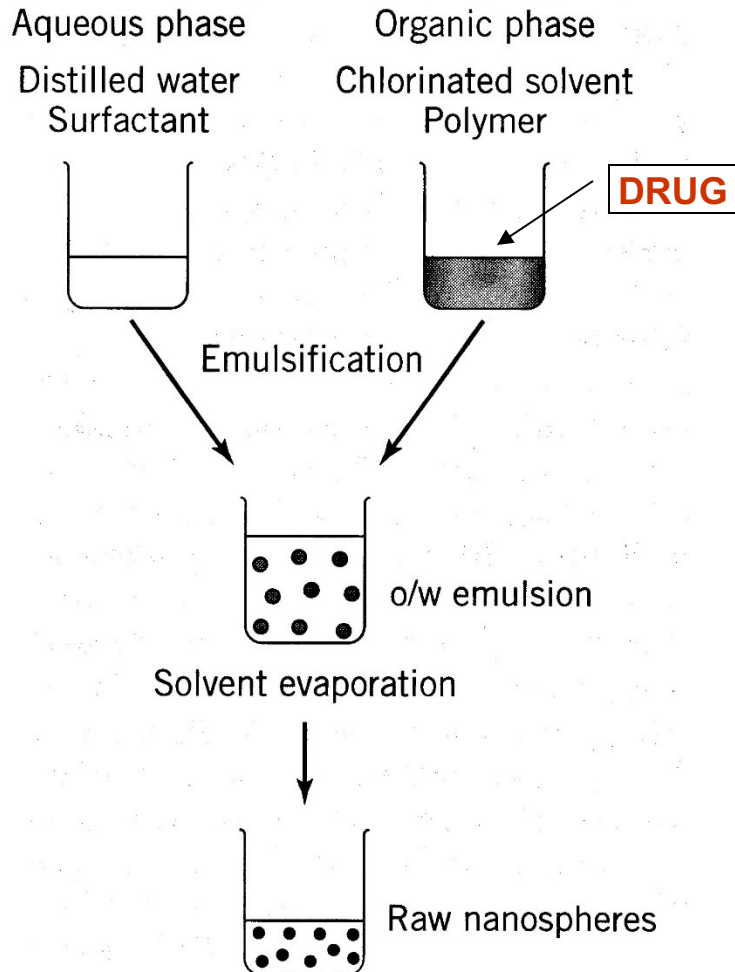
- Ensures that signal is presented is through MHCI and MHCII both

Particle Synthesis

- Chemical methods:
 - For polymer particles: Involves polymerization (interfacial or emulsion) of monomers during the synthesis of particles
 - For Metal particles: Reduction/oxidation/Crystallization from salts
- Physical methods:
 - Controlled “precipitation” of polymer emulsions (single or double emulsions)
 - Complex coacervation / coacervation
 - Solvent evaporation
 - Solvent removal (extraction)
 - Hot melt
 - Spray-drying
 - Phase separation
 - Salting out

PARTICLE SYNTHESIS: SOLVENT EVAPORATION

SINGLE EMULSION PROCESS

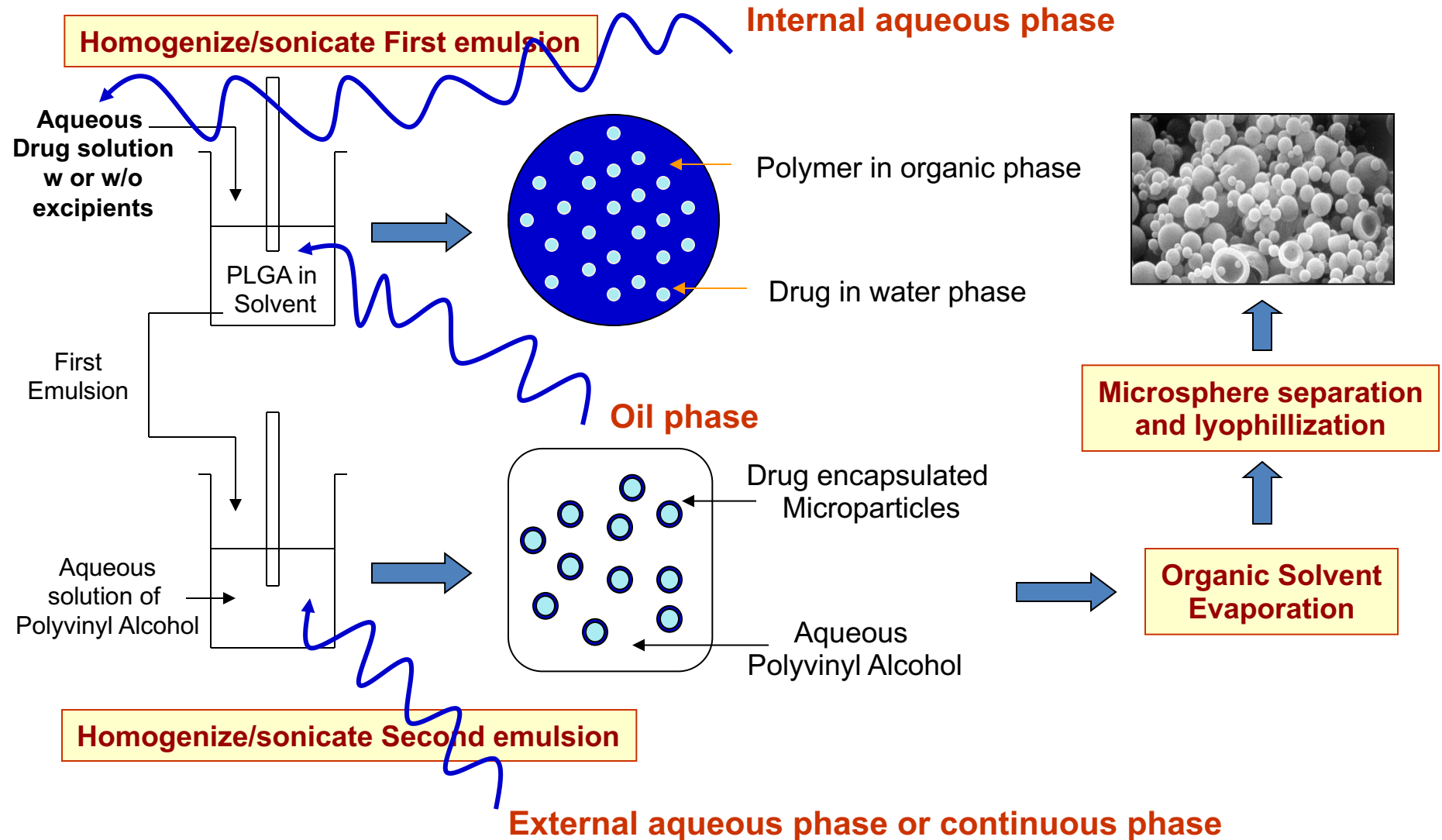


- PLGA/PLA microparticles as example
- Oil-in-water (o/w) emulsion (could also be oil-in-oil emulsion) → first oil phase must not be soluble or miscible in second oil phase
- Produces a solid matrix type particles (no capsules or hollow particles)
- Drug must be soluble or dispersible in organic solvent phase.
- Polymer dissolved in a volatile organic solvent → most commonly methylene chloride or dichloromethane (DCM). Other solvents used are chloroform, ethyl acetate etc.
- The polymer-drug solution/dispersion is emulsified in a large volume of aqueous phase → water containing surfactants, most commonly polyvinyl alcohol (PVA)
- Emulsion is stirred (could be under reduced pressure and elevated temperature) for several hours to evaporate organic solvent and “harden” the emulsion droplets → droplets solidifies into particles

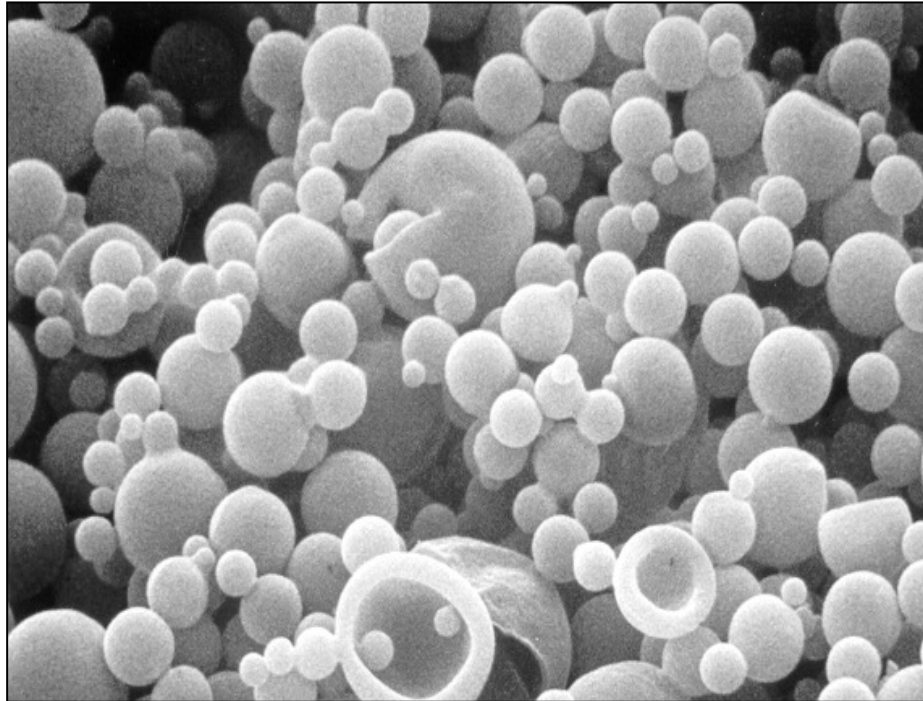
PARTICLE SYNTHESIS: SOLVENT EVAPORATION

TYPICAL W/O/W DOUBLE EMULSION PROCESS

26



PARTICLE SYNTHESIS: SOLVENT EVAPORATION



Parameters affecting particles

- Polymer type and molecular weight
- Polymer concentration in the oil phase
- Type of drug and method of incorporation (solid, liquid, suspension, hydrophilic, hydrophobic etc.)
- Organic solvent used and polymer solubility
- Type and amount of surfactant in the external phase
- Internal aqueous phase / organic solvent ratio
- Rate of stirring and type of mechanical force –
Increased stirring, decreased particle size
- Evaporation rate / temperature / pressure
- Extraction rate (volume, temperature, etc.)

Single or double emulsion processes:

Solvent extraction/removal method

- **Most solvents used to dissolve the polymer [Oil phase] has some solubility (generally low) in water.**
- **Emulsion can be added to a very large volume of aqueous solution (with or without surfactant) → volume should be large enough for all the organic solvent to be “soluble” in the water phase**
- **Solvent is rapidly extracted out of the polymer phase into the external continuous phase**
- **Hardens the microparticles rapidly**
- **Internal particle structures / porosities etc. can be altered by choosing evaporation versus extraction.**
- **Disadvantage: large volume required for solvents like DCM → difficult to scale up for industrial productions**
 - **For example: DCM is soluble in water up to ~1.5% w/w. So to “extract” 10 ml of DCM rapidly by this process, the volume of the external aqueous phase must be greater than 660 ml**

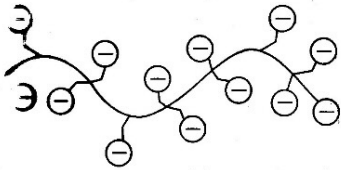
Emulsion based methods: Short comings

- High polydispersity
- Biomolecule denaturation
- Low encapsulation efficiency for water soluble drugs
- Difficult to synthesize sizes below 150nm

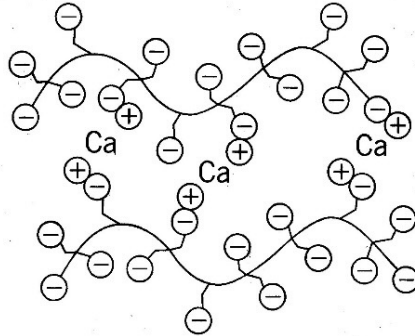
PARTICLE SYNTHESIS: GELATION

Ionic gelation process

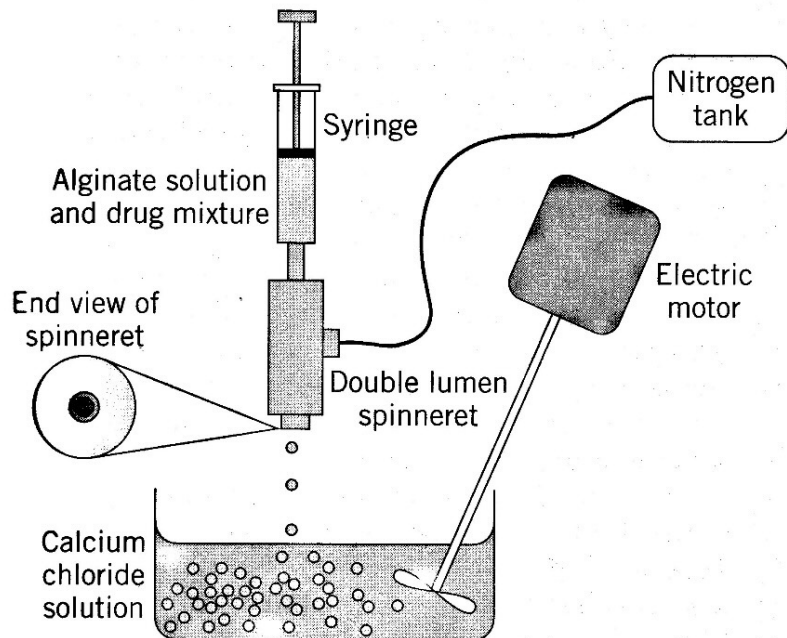
Alginate in solution:
anionic polyelectrolyte



Alginate cross-linked with calcium ions



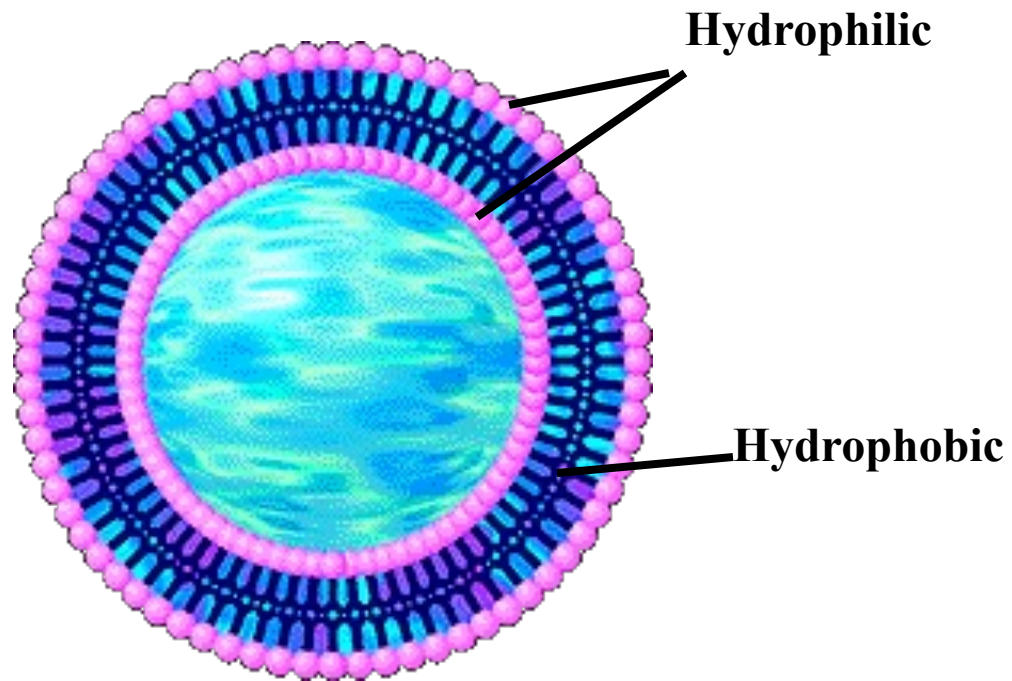
Encapsulation procedure



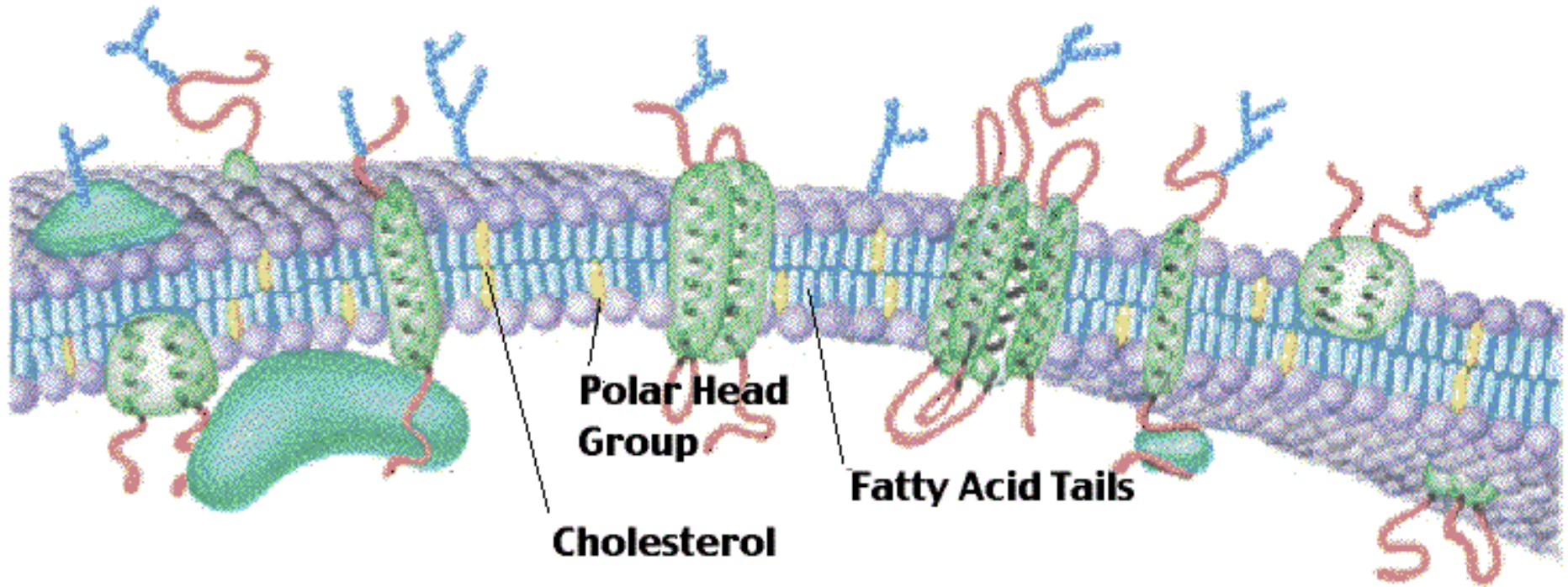
- A hydrogel microparticle: Physical ionic hydrogel composed of anionic polyelectrolyte (alginate) crosslinked with a bivalent cation (calcium)
- Drugs, cells, tissues (islets) etc. have been encapsulated within these microparticles
- Fairly mild conditions (except exposure to high calcium concentration could be detrimental to cells)

What is a liposome?

phospholipid bilayer

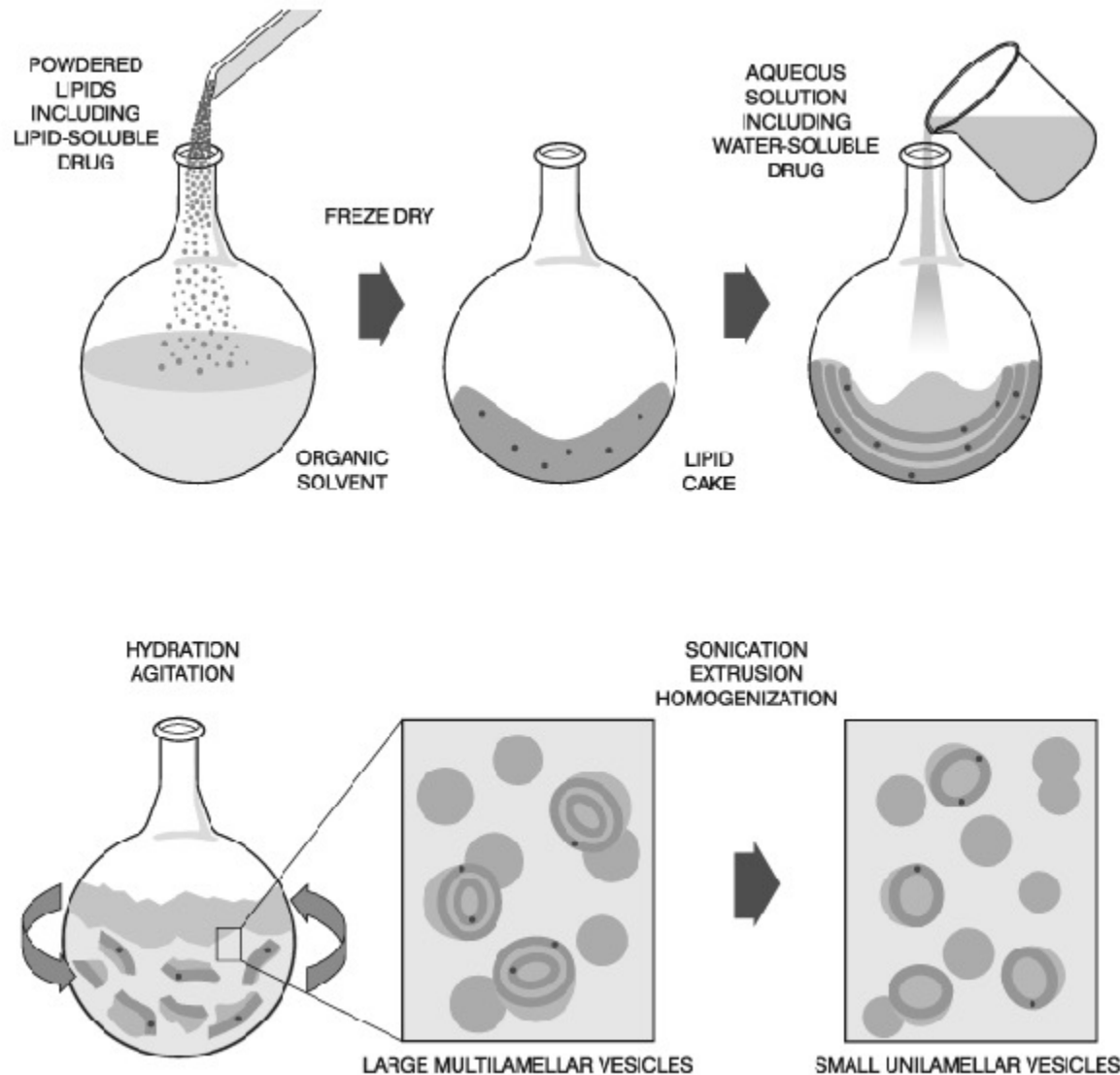


Cell Membrane Phospholipids



Phospholipids are the major structural components of biological membranes such as the cell membrane

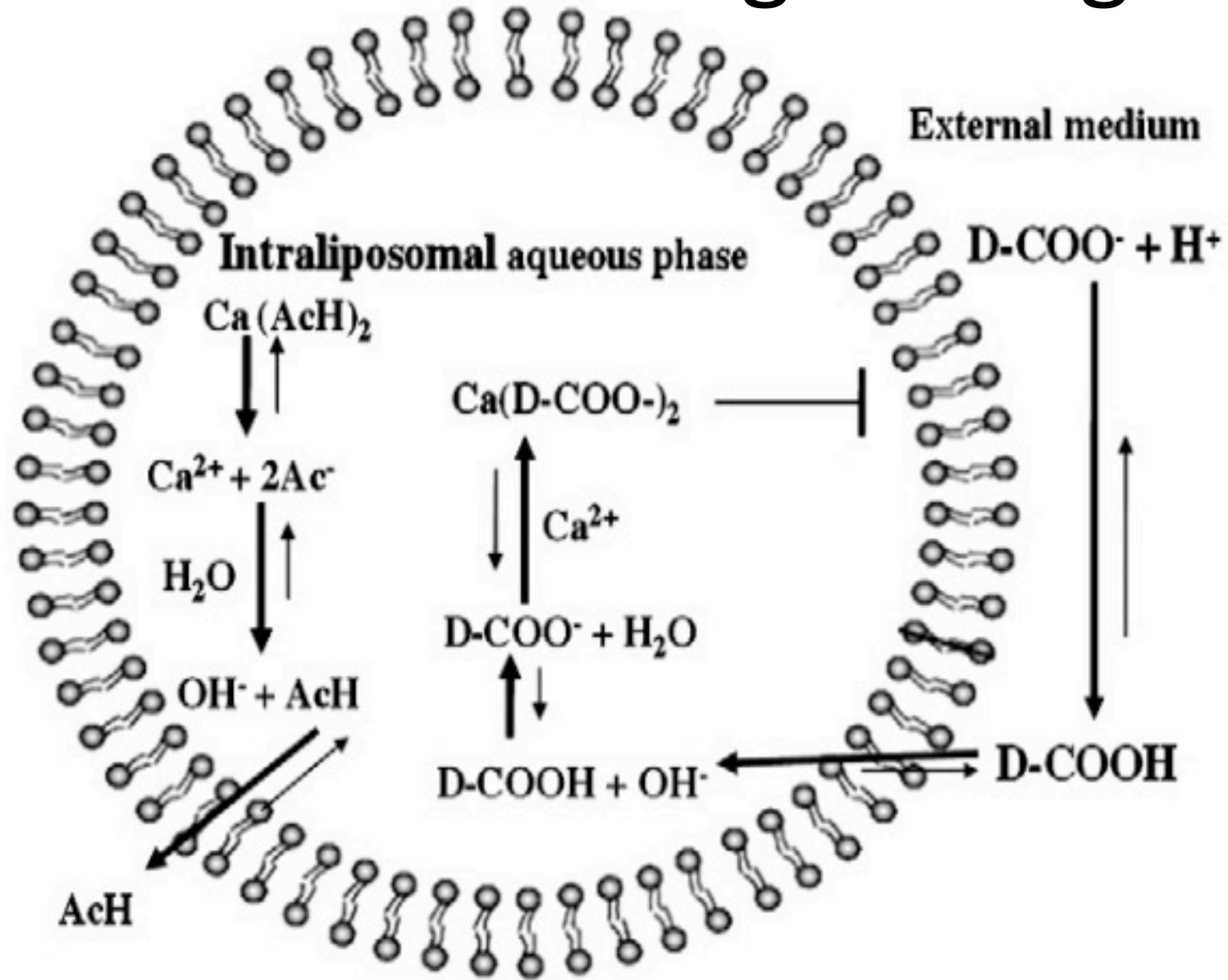
Liposome synthesis and drug encapsulation



Release from liposome

- Lipid membranes are fluidic at high temperature (transition temperature) and hence release the drug
- Lipids are chosen such that they are fluidic just above 37°C
- Liposomes can also burst when in contact with large amphipathic compounds such as proteins

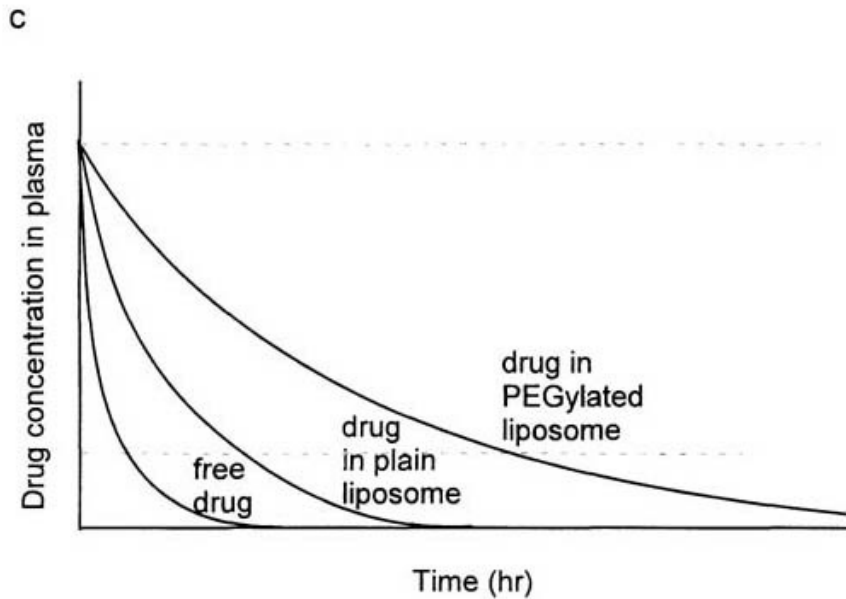
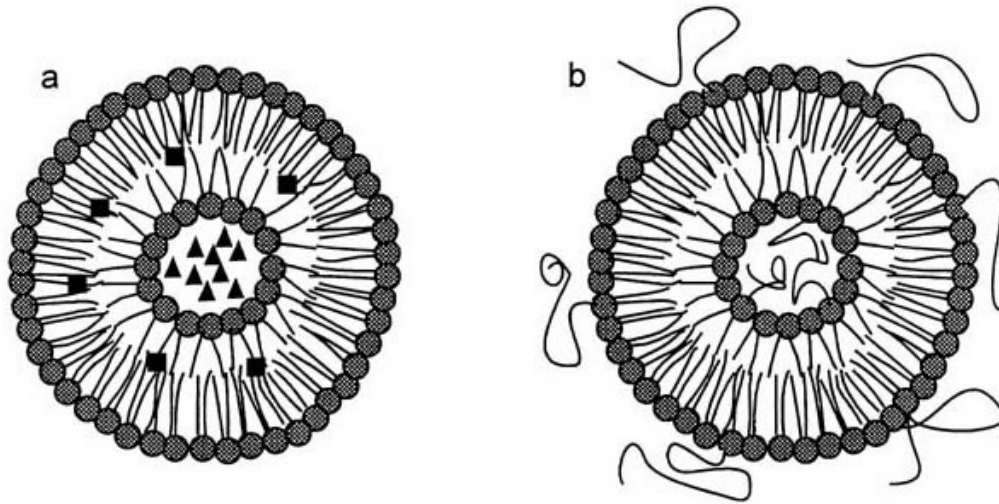
Remote loading of drug



Remote loading of drug

- Loading is done on preformed liposomes
- Neutral drug will diffuse through the membrane and will be converted to charged species incapable of diffusing out
- Only amphipathic weak acids and bases fit these requirements
- pH and Ion gradients were created by salts of either weak bases (e.g., ammonium) or weak acids (e.g., acetate). These ions can be present (depending on pH) as charged and uncharged species. Such ions can traverse the liposome membrane only in their uncharged form.
- Hydrated in either ammonium sulfate (for bases' loading), or calcium acetate (for acids' loading)

Stealth Liposomes: Half lives



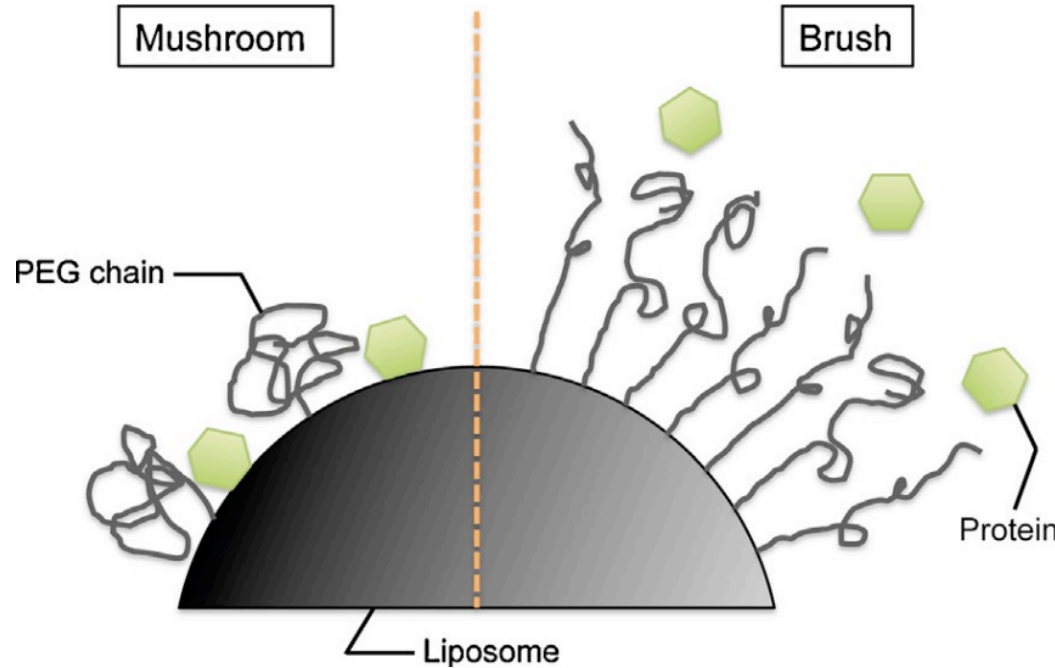
Stealth Liposomes: Half lives

Liposomes ^b		Half life (min)
3 mol%	PEG/PE	80
	PEG2/PE	140
	PACM/PE	40
	PVP/PE	90
7 mol%	PEG/PE	230
	PEG2/PE	230
	PACM/PE	130
	PEV/PE	170
Without any polymer		10

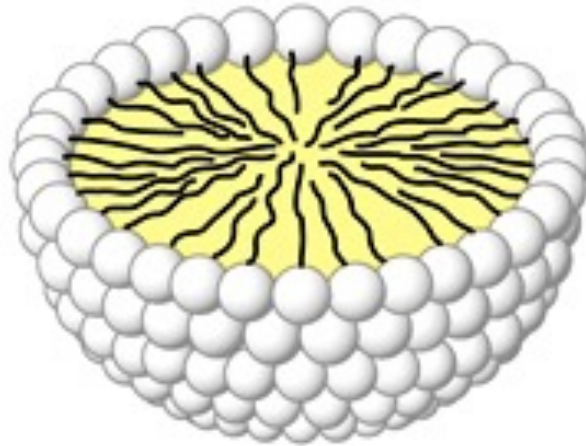
Half life of "stealth" Liposomes in mice circulation after incorporation of linear PEG, branched PEG, PACM and PVP

PEGylation: Brush vs Mushroom

- Brush type conformation is considered better
- Applicable for any type of particle/surface



Micelles



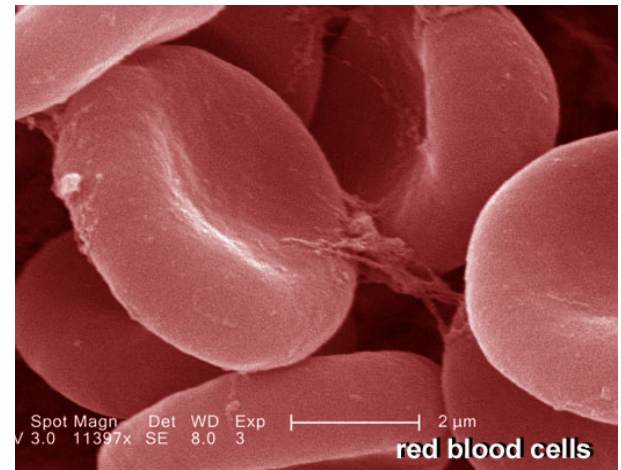
- Aggregate of amphipathic molecules in water, with the nonpolar portions in the interior and the polar portions at the exterior surface, exposed to water.
- ▶ Amphiphilic molecules form micelle above a particular concentration which is called as critical micellar concentration (CMC).
- ▶ Hydrophobic drugs can be encapsulated/solubilized, into inner core
- ▶ Micelle formation is favored when the cross-sectional area of the head group is greater than the acyl side chains

Effect of particle shape

- Natural occurrence of diversely shaped entities like virus, bacteria, red blood cells
- Non-spherical shapes: large contact area to adhere to surfaces
- Rod: allows fast motility in bacteria
- Disc shaped RBCs: marginate in blood vessels



Bacteria



Red Blood Cells

Making polymeric micro- and nanoparticles of complex shapes

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Department of Chemical Engineering, University of California, Santa Barbara, CA 93106

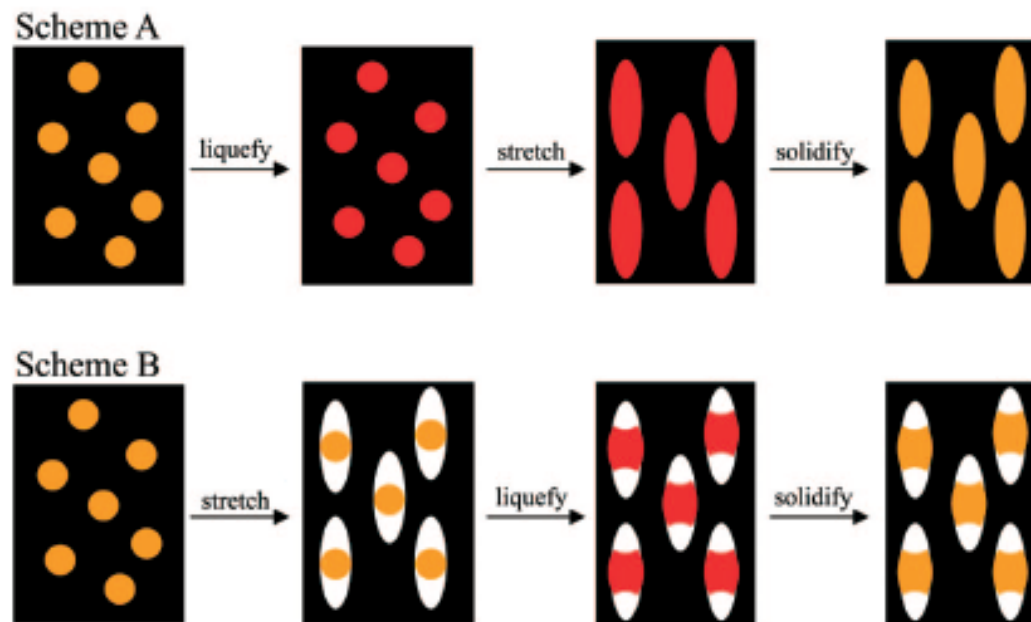


Fig. 1. Methods used for making particles with different shapes can be categorized into two general schemes. (*Upper*) Scheme A involves liquefaction of particles by using heat or toluene, stretching the film in one or two dimensions and solidifying the particles by extracting toluene or cooling. The example shown here produces elliptical disks. (*Lower*) Scheme B involves stretching the film in air to create voids around the particle, followed by liquefaction using heat or toluene and solidification. The example shown here produces barrels.

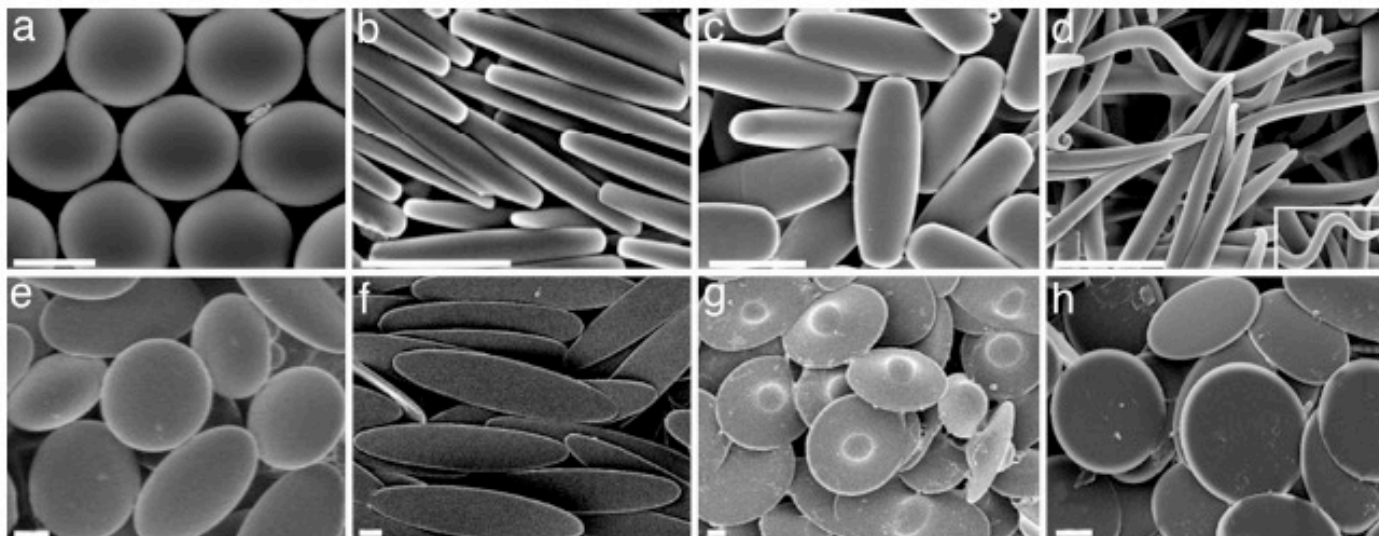
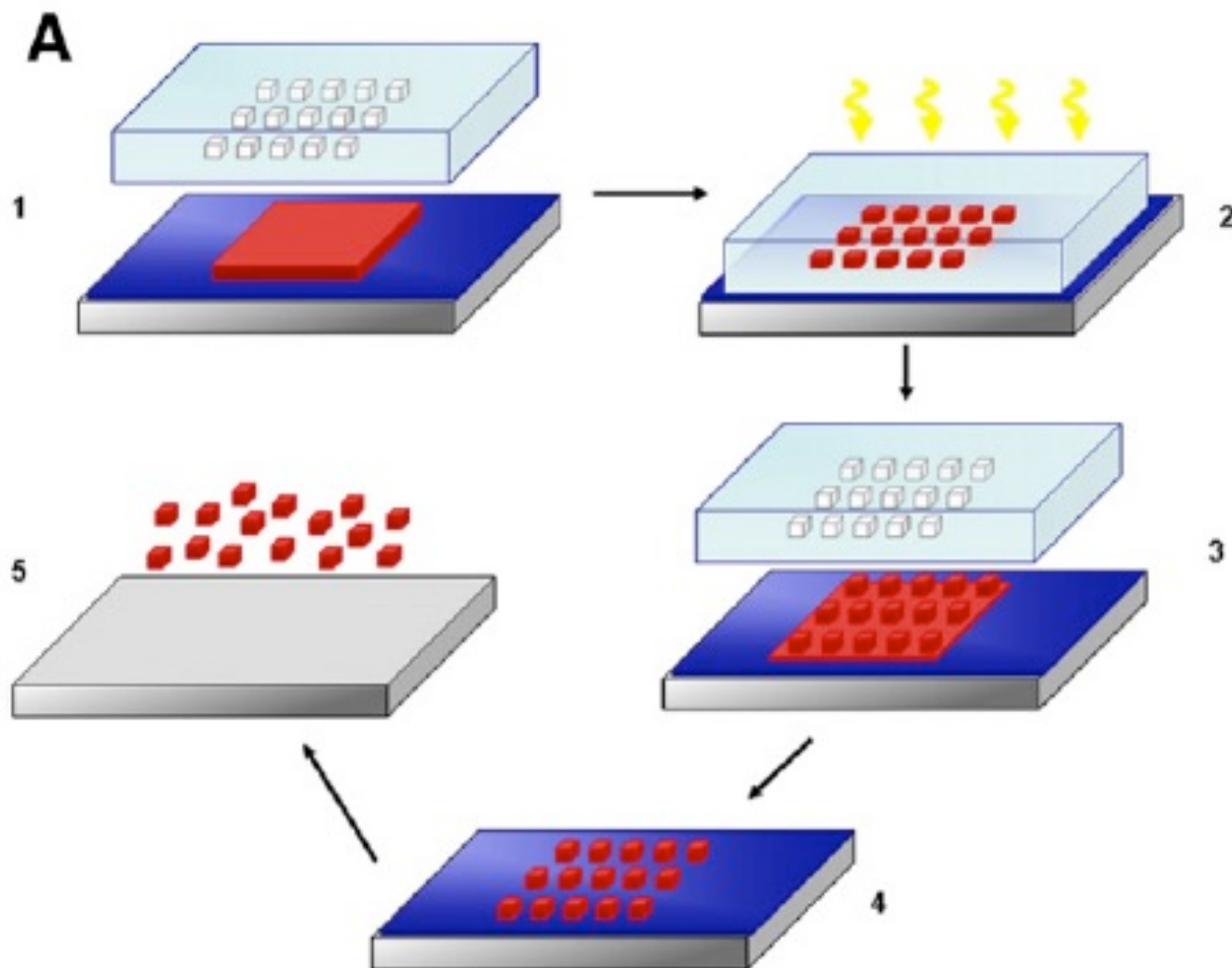


Fig. 2. Micrographs of shapes made by using scheme A. (a) Spheres. (b) Rectangular disks. (c) Rods. (d) Worms. (e) Oblate ellipses. (f) Elliptical disks. (g) UFOs. (h) Circular disks. (Scale bars: 2 μm .)

Nanoimprint lithography based fabrication of shape-specific, enzymatically-triggered smart nanoparticles

Luz Cristal Glangchai ^{a,1}, Mary Caldorera-Moore ^{a,1}, Li Shi ^{b,c,d,*}, Krishnendu Roy ^{a,c,d,*}



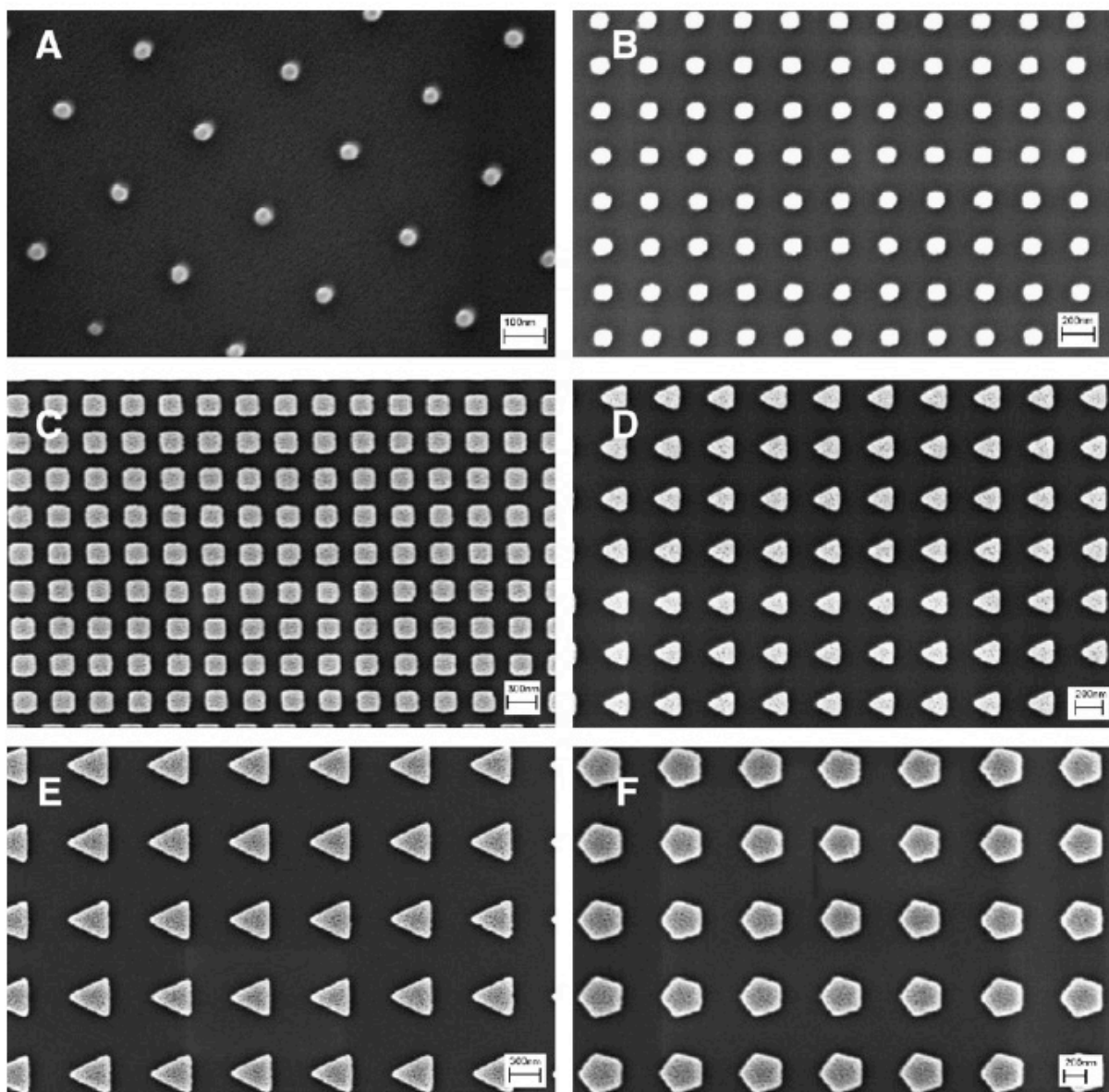
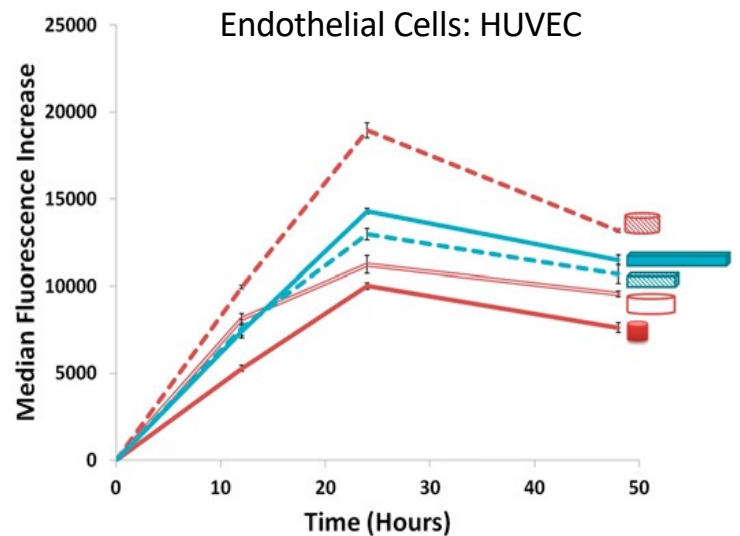
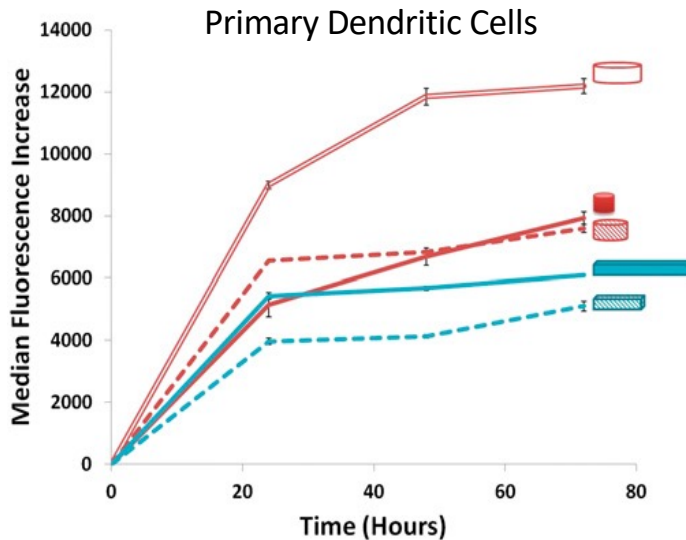
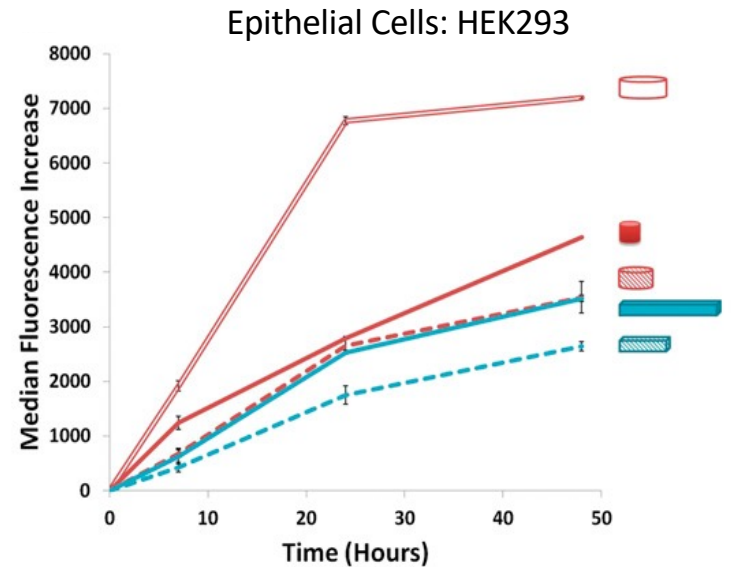
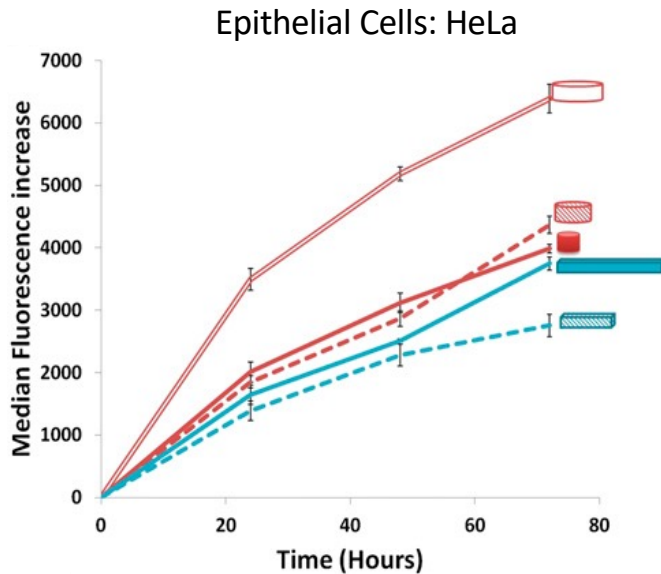


Fig. 3. SEM images of S-FIL imprinted (100% w/v, MW 3400) PEGDA nanoparticles: (A) 50 nm squares (scale bar=100 nm), (B) 100 nm squares (scale bar=200 nm), (C) 200 nm squares (scale bar=300 nm), (D) 200 nm triangles (scale bar=200 nm), (E) 400 nm triangles (scale bar=300 nm), and (F) 400 nm pentagonal particles (scale bar=200 nm).

Mammalian cells uptake high aspect ratio discs

80nm (d) x 70nm (h) 220nm (d) x 100nm (h) 325nm (d) x 100nm (h)

400nm x 100nm x 100nm 800nm x 100nm x 100nm

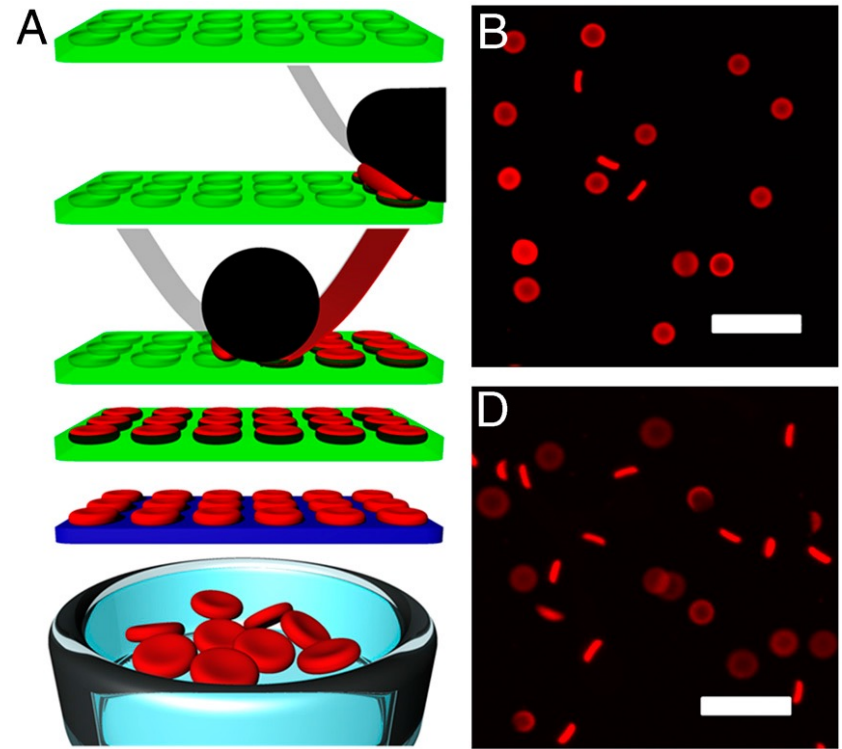


Charge of particle

- In body, particles encounter all kinds of charge
 - Cell membrane are slightly negatively charged
 - Serum proteins have all charges but predominantly anionic
- For delivery to mammalian cells in-vitro, positive charged particles are considered better
- In-vivo, positively charged particles are rapidly adsorbed by serum proteins and cleared
- Positively charged particles can also cause toxicity in the body
- Neutral and slightly negatively charged particles are able to circulate in the blood for long time
- However, the understanding is still not complete and certain applications may require use of positively charged particles

Elasticity of particle

- Elasticity has also been reported to have profound effect in circulation times in blood
- The best known example is RBCs, which are known to circulate in blood for months
- For low modulus, larger particles (6.4 μm) were able to circulate in blood longer compared with their smaller counterparts (780 nm)

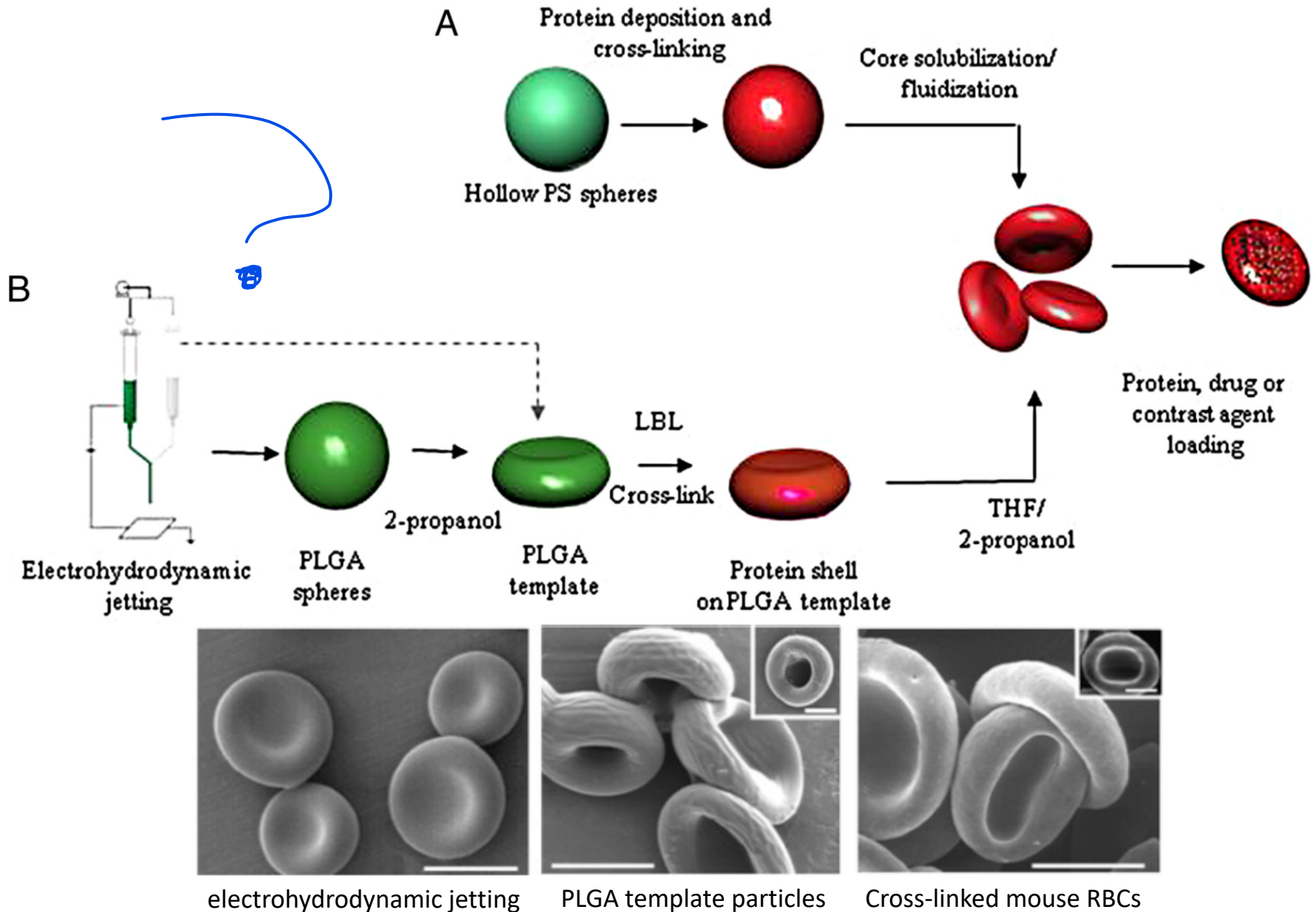


586–591 | PNAS | January 11, 2011 | vol. 108 | no. 2

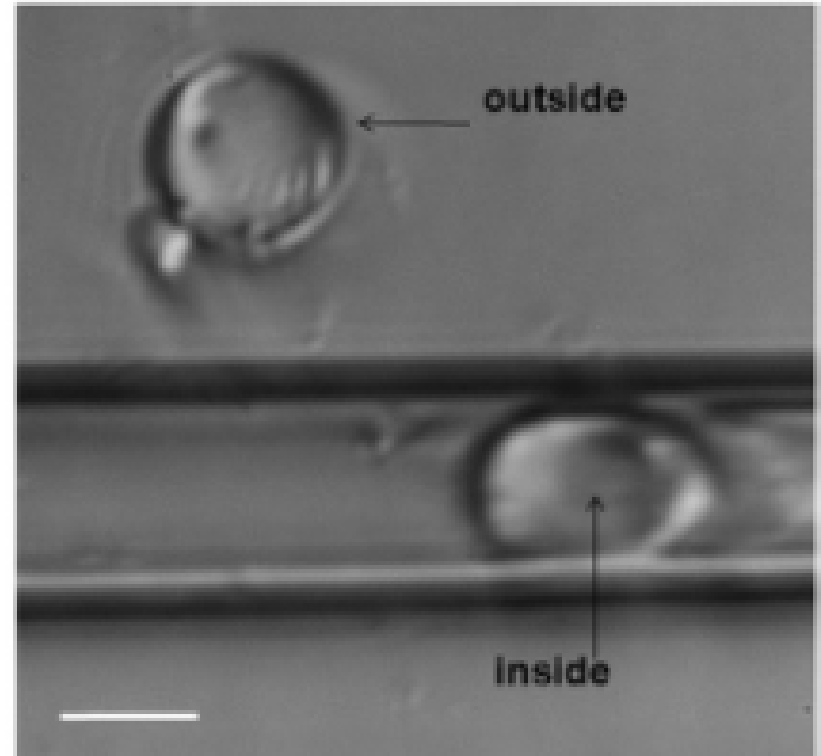
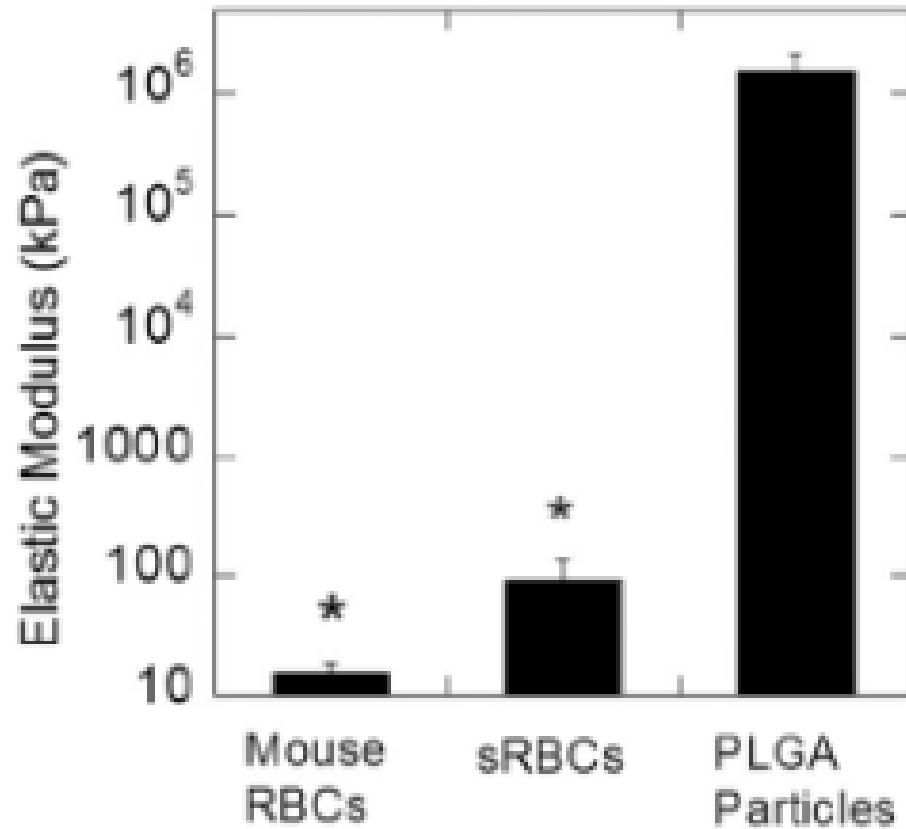
Table 1. Modulus values and compartmental analysis of RBCM particles from intravital microscopy experiments

% Cross-linker	Modulus of bulk material, kPa	Distribution half-life, h	Elimination half-life, h	AUC, fluorescence* h	Average R^2
10%	63.9 ± 15.7	0.038 ± 0.0012	2.88 ± 0.92	0.65 ± 0.14	0.8966
5%	39.6 ± 10.4	0.066 ± 0.036	5.12 ± 2.17	0.76 ± 0.57	0.9029
2%	16.9 ± 1.7	0.15 ± 0.025	7.12 ± 0.82	1.35 ± 0.26	0.9468
1%	7.8 ± 1.0	0.35 ± 0.13	93.29 ± 31.09	15.44 ± 15.63	0.9330

Elasticity of particle

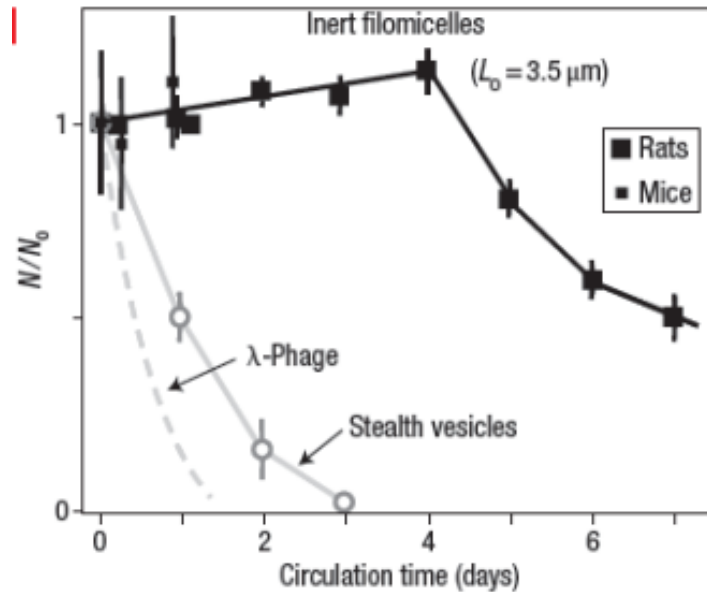


Elasticity of particle



Elasticity of particle

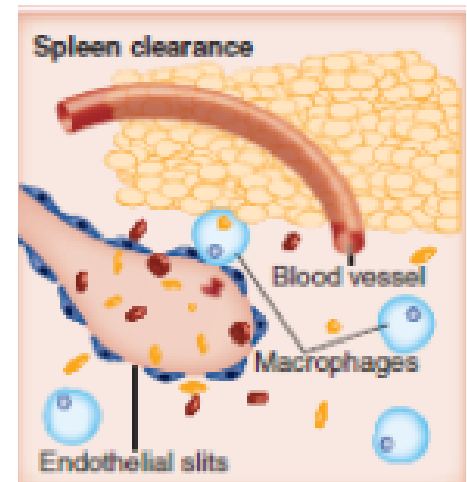
20-30
diameter,
5-8 μm
length



Crosslinked and rigid structures of similar dimensions clear rapidly

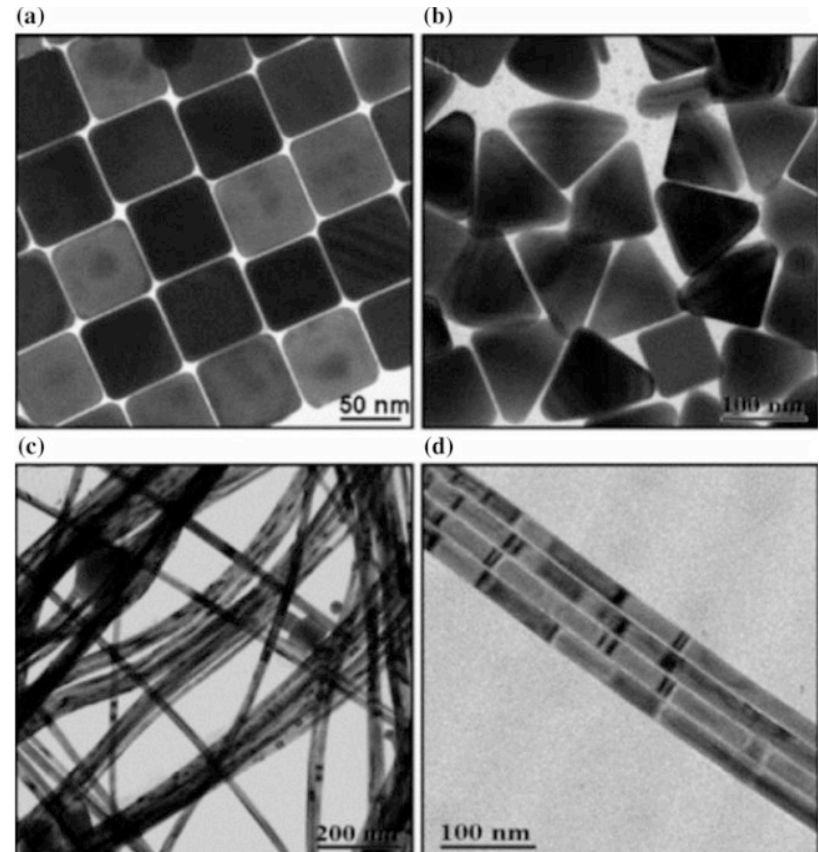
Reason?

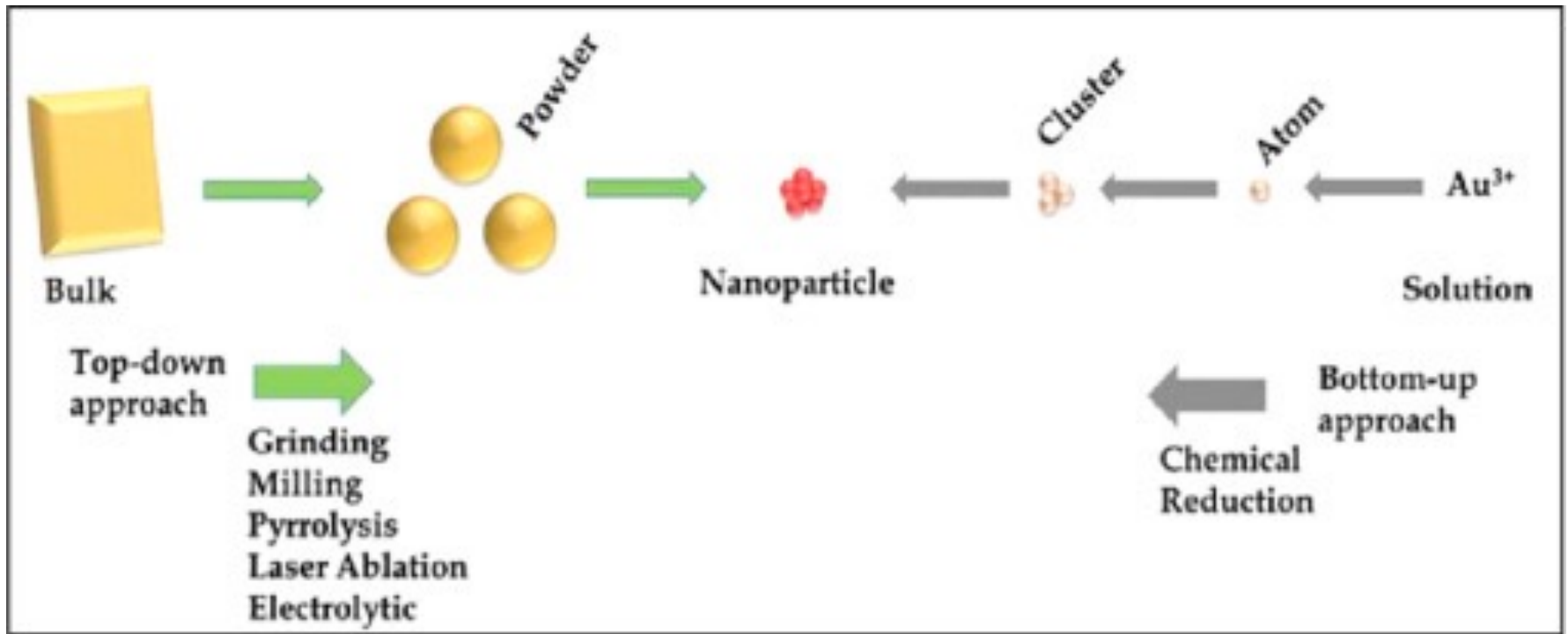
Particles are able to squeeze through the fenestrations in spleen and avoid uptake by immune cells



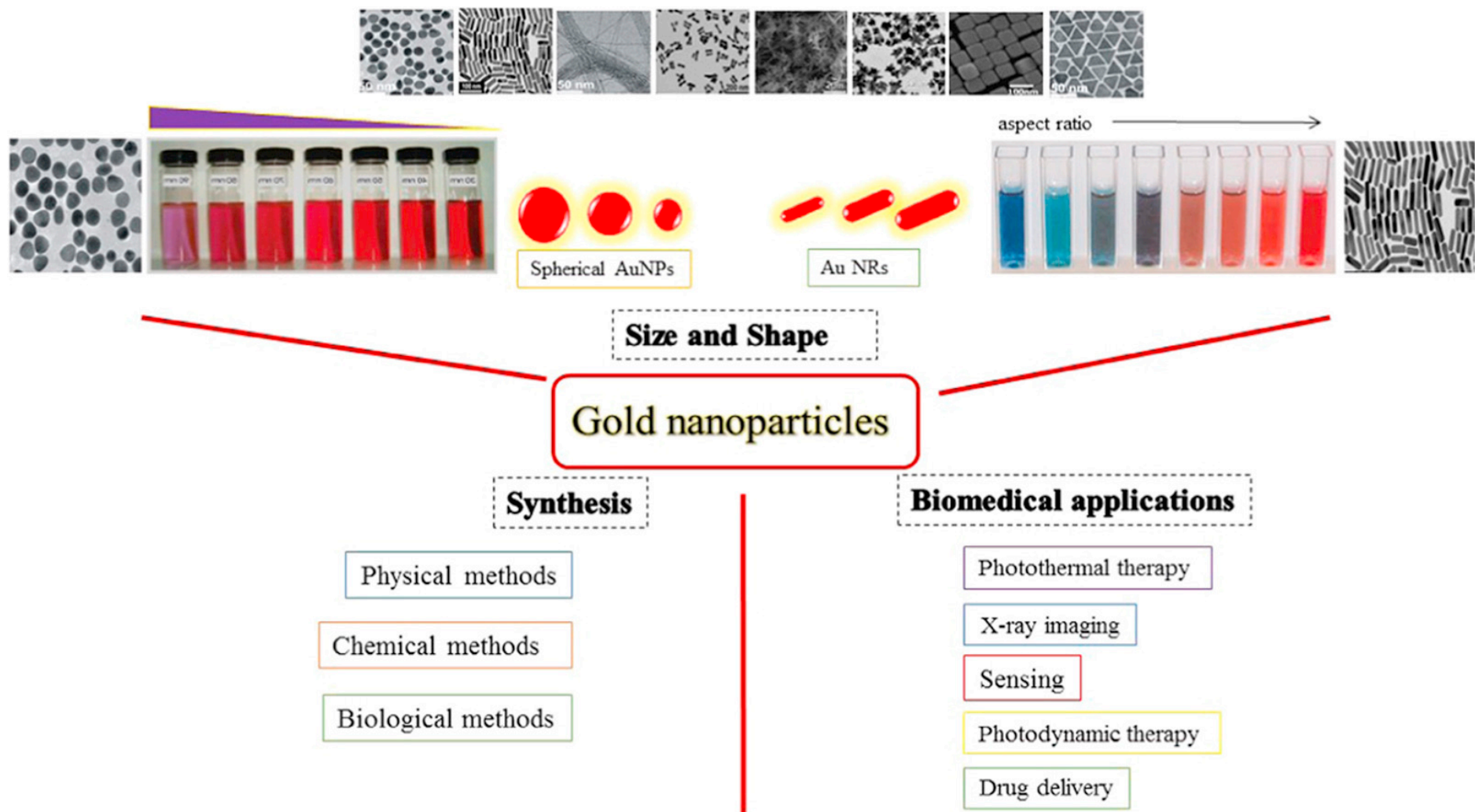
Metal particles

- Same concepts apply for metal particles
- Silver, gold, iron-oxide, quantum dots
- Widely used for imaging application
- Optical response is tunable based on size and shape of the particles: Heating, fluorescence
- Drug can also be coated/conjugated on the particle surface
- Non-degradable



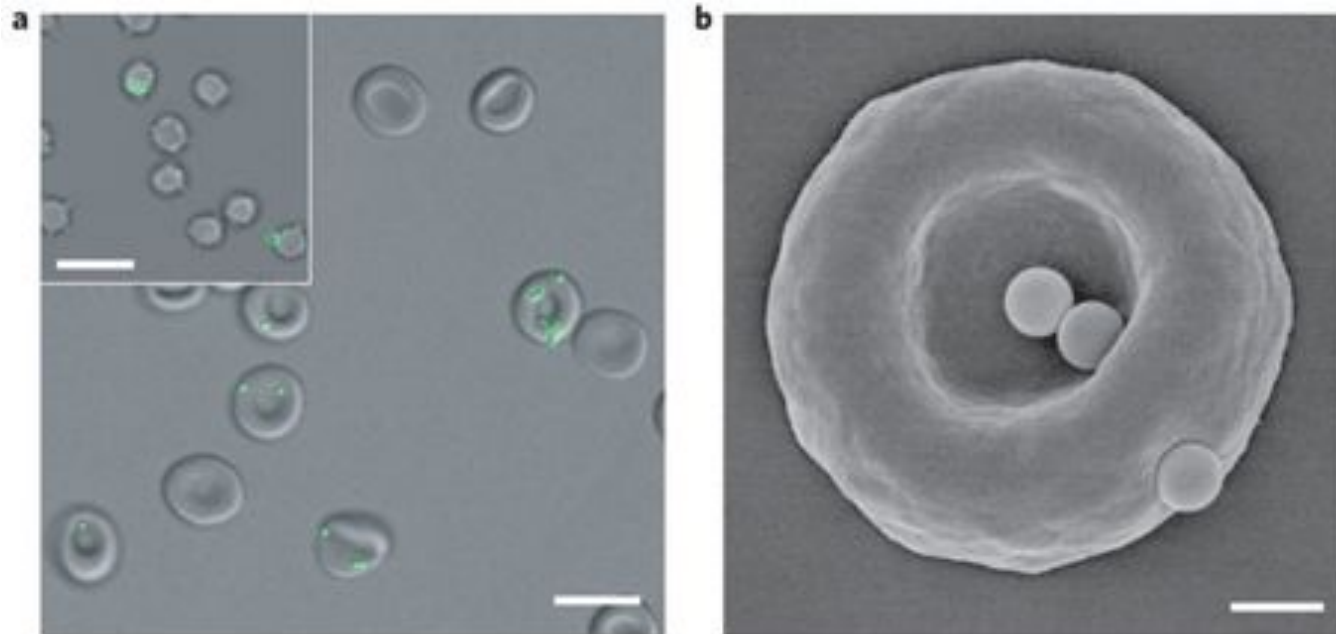


- Synthesis is extremely monodisperse
- Well established techniques to synthesize various shapes
- Top down approaches include grinding, milling etc
- Bottom up approaches typically use chemical reduction



Particle hitch-hiking

- Particles can be conjugated to a target cell that prevents its clearance
- “Imparts” biological homing activity to particles
- RBCs and immune cells are the attractive targets
 - Adsorption
 - Chemical conjugation
- Need to be careful not to affect the host cell function



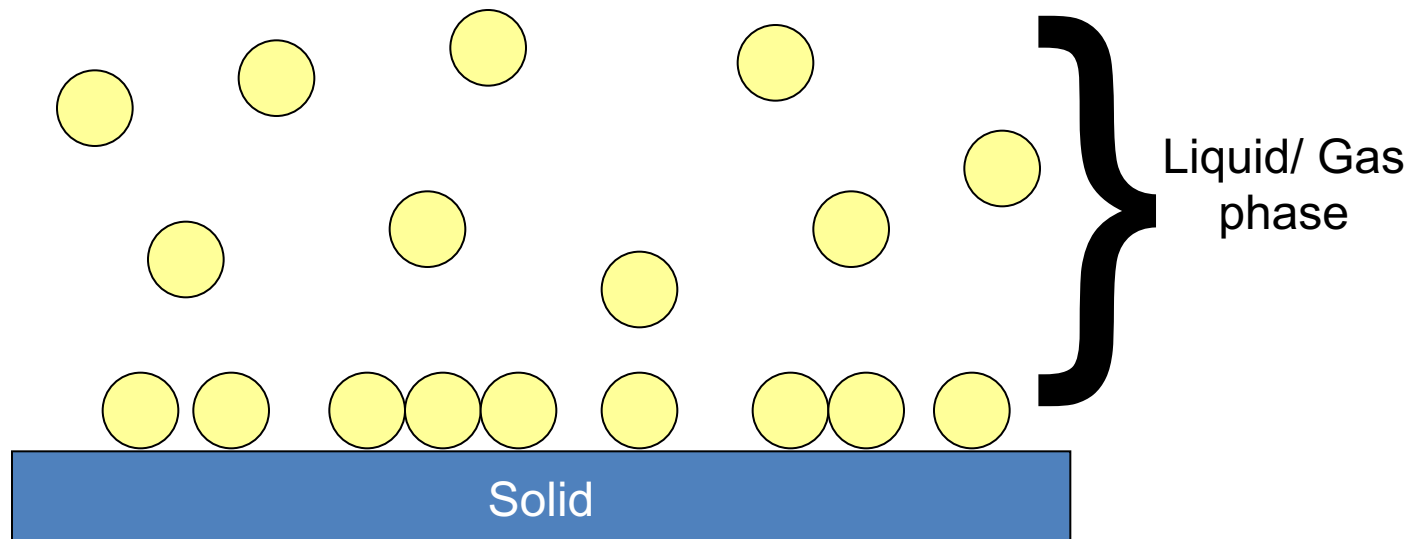
Protein Adsorption

What is a surface/ an interface?

The outermost region of a material that is chemically and/or energetically different from rest of the bulk by virtue of being located at a boundary.

What is adsorption?

Adsorption is formally defined as the partitioning of a chemical species between a bulk phase and an interface



adsorption vs. absorption

Adsorption vs. Absorption: Surface vs. Bulk phenomena

Both phenomena are relevant to biomaterials

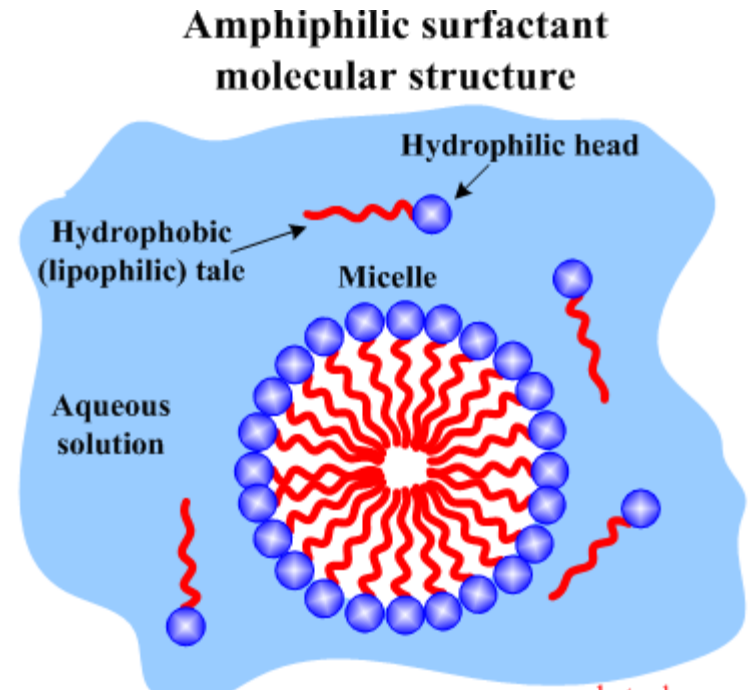
For example,

- When a dry crosslinked polymer network is put in water, it will absorb water to form a hydrogel
- Proteins adsorb to a biomaterial surface

What is a surfactant?

Surfactants are usually compounds that are amphiphilic: i.e., contain both hydrophobic groups (their *tails*) and hydrophilic groups (their *heads*).

Common examples:
Detergents and soaps



Why protein adsorb?

Proteins are *weak* surfactants (relative to synthetic detergents)

Because certain regions of a protein are richer in polar and charged amino acid residues (hydrophilic) whereas other regions are richer in non-polar residues (hydrophobic). There is a relative difference in hydrophobicity but not a large difference. So, they can easily change their structure to adapt to the surface encountered

Protein adsorption to biomaterial

Most bodily fluids (in vivo) and cell culture medium (in vitro) contain several soluble proteins

For example, blood contains > 400 different proteins at different concentrations

The sequence of events when a material is placed in a biological environment:

1. Interaction with water (wetting)
2. Interaction with proteins
3. Interaction with cells

CONSEQUENCE: *The adsorbed protein layer mediates the biological response to a biomaterial*

Protein may denature on adsorption

Protein structures are not highly stable.

Change in environment (heat and chemical agents) can cause a change in structure (denaturation).

When proteins adsorb (interaction with a solid) there is change in the surrounding chemical environment, thus potential for change in protein conformation (denaturation)

Some protein structures are more stable than others, so wide spectrum of denaturation depending on the protein and the surface with which it is interacting

How proteins denature on adsorption?

Denaturation will depend on biomaterial surface chemistry and water-wettability.

At a hydrophilic biomaterial surface (rich in charged and/or groups)-Hydrophilic amino acid rich regions of the protein will preferentially interact with the surface

So low denaturation as hydrophilic domains are on the outside

At a hydrophobic biomaterial surface (rich in non-polar groups)-Hydrophobic amino acid rich regions will preferentially interact with the surface

In water-soluble globular proteins hydrophobic amino acid residue-rich regions are in the protein core. These regions may prefer to interact with the hydrophobic biomaterial surface.

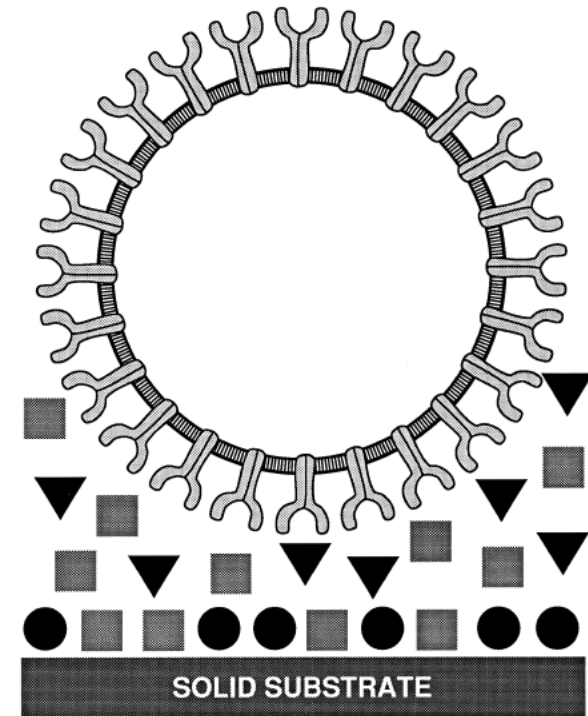
Why is protein adsorption important?

The adsorbed protein layer mediates the biological response to a biomaterial

Cell membrane has receptor proteins including “integrins” which bind to extracellular matrix (ECM) proteins in vivo

Cells attach to/ recognize biomaterials through binding of integrins to the adsorbed proteins.

- Immune response/ blood platelets
- Cell attachment/ growth for bio-integration of implants
- Cell attachment/ growth into 3D tissue scaffolds



Pre-adsorbing proteins to surface

Pre-adsorbing of cell adhesive proteins (such as fibronectin, fibrinogen, vitronectin) by incubating of the biomaterial in protein solution increases attachment of cells to the biomaterial surface.

Whereas, pre-adsorbing with non-cell adhesive proteins (such as albumin) decreases cell attachment to the biomaterial.

Often used to define the material signalling in-vivo so as to prevent the non-specific protein adsorption

Kinetics of protein adsorption

Very rapid initial rate of protein adsorption; rate limited by diffusion rates

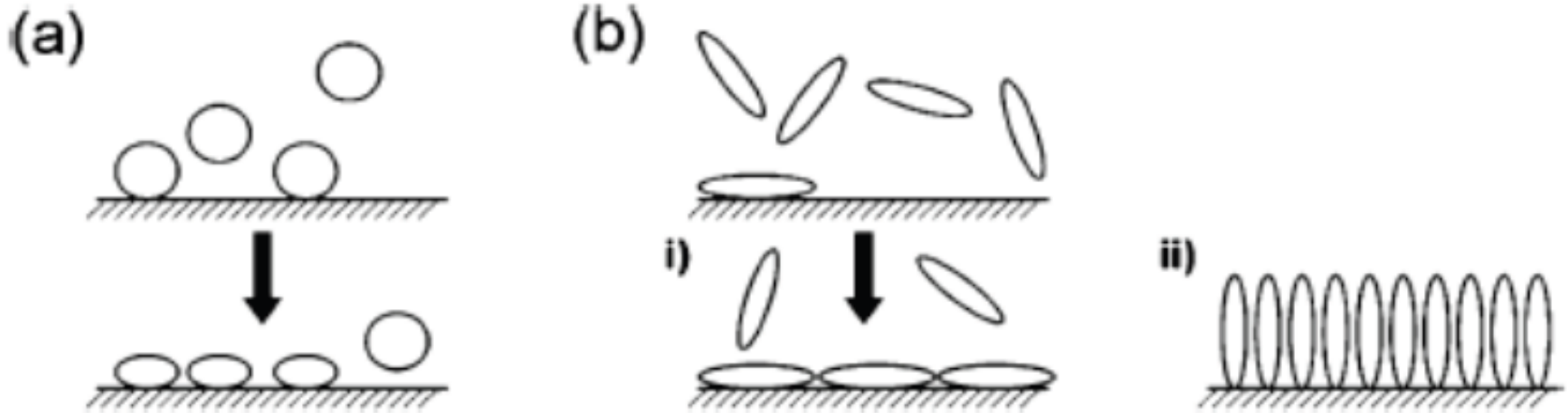
Then slower phase, more difficult for the arriving proteins to find and fit into an empty spot on the surface.

“Hard” and “Soft” proteins

“Hard proteins”: molecules with high internal stability for example; α -chymotrypsin, ribonuclease, lysozyme and β -lactoglobulin (β -Lg.)

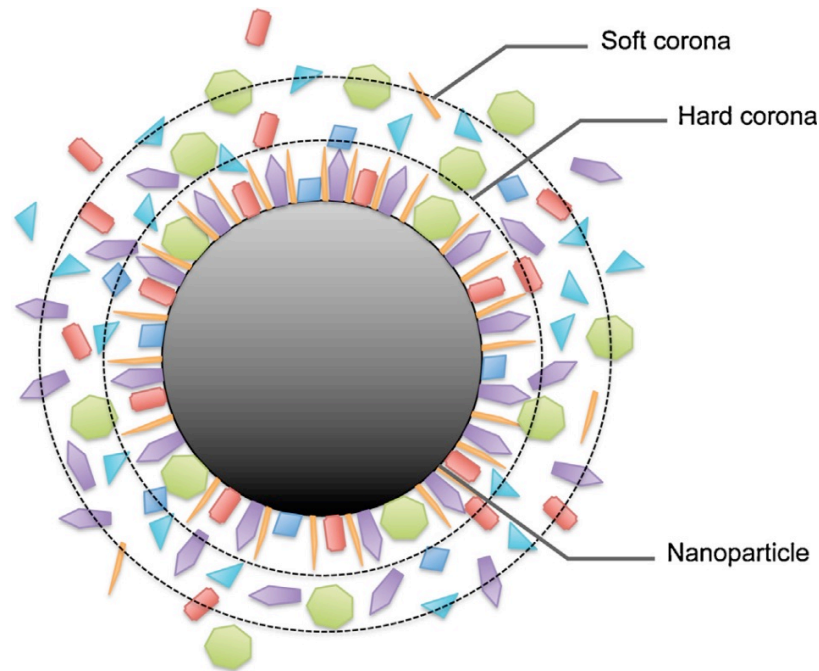
“Soft proteins”: have a low internal stability for example; α -lactoalbumin, β -casein, haemoglobin, catalase and phytase.

Change in protein orientation



Multilayer model

A 3D protein-rich “interphase” region is formed by the adsorbed proteins



Competitive adsorption

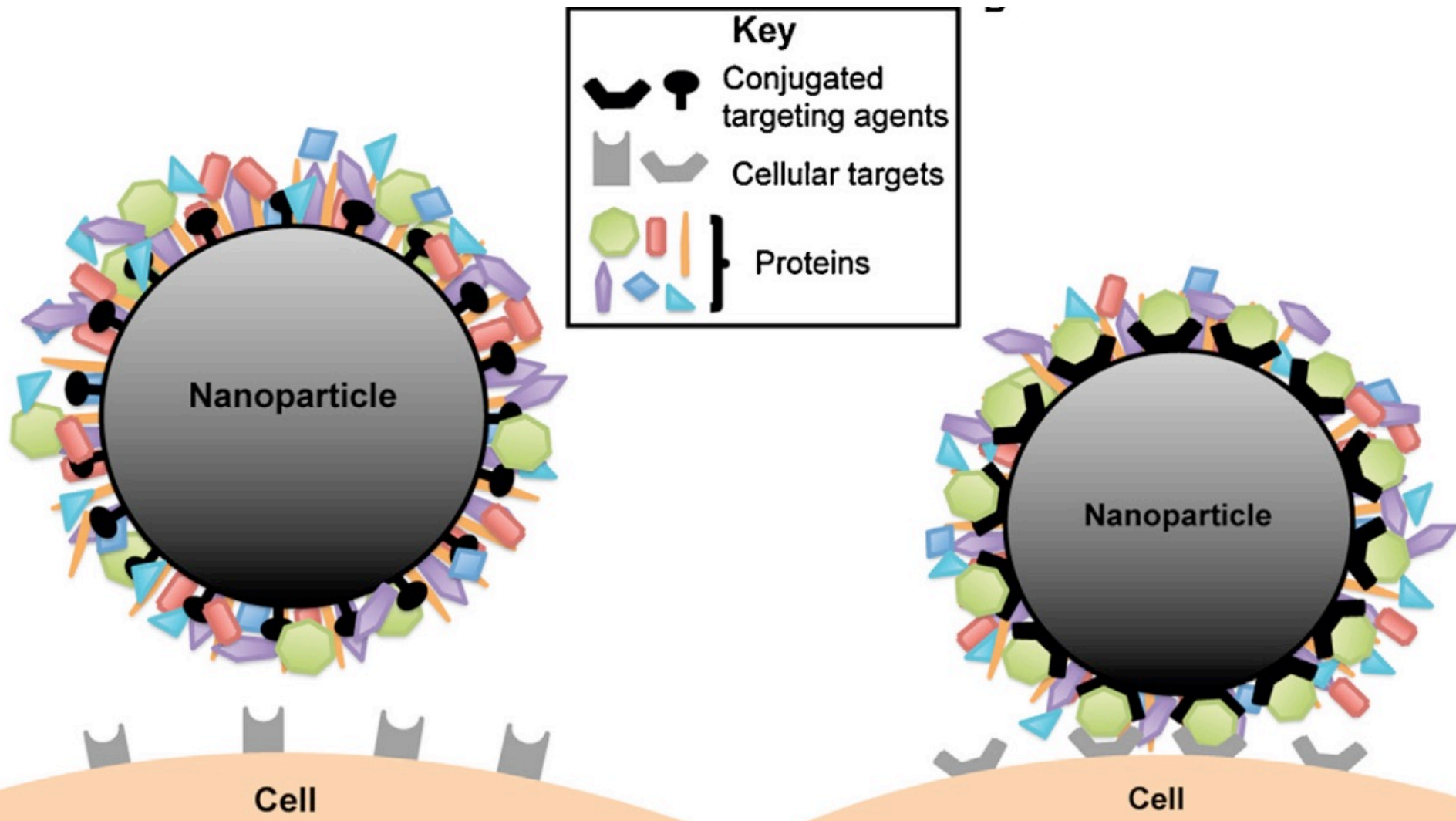
The equilibrium adsorbed protein layer composition determined by total and relative protein concentrations. Very complex phenomena and poorly understood.

Kinetics is also important. The composition of the adsorbed layer may change over time.

Adsorbed protein layer composition is dynamic

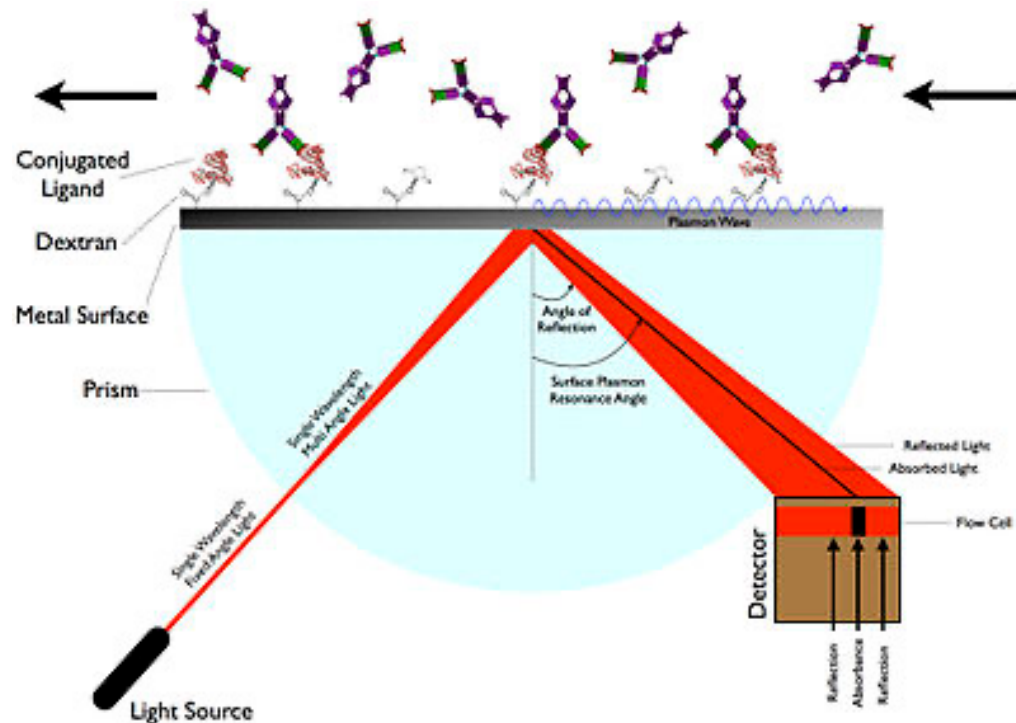
Initially-adsorbed proteins (kinetically favored) displaced by other proteins over time (minutes-to-hours-to-days) determined by thermodynamics

This is known as as Vroman effect



Protein adsorption quantification

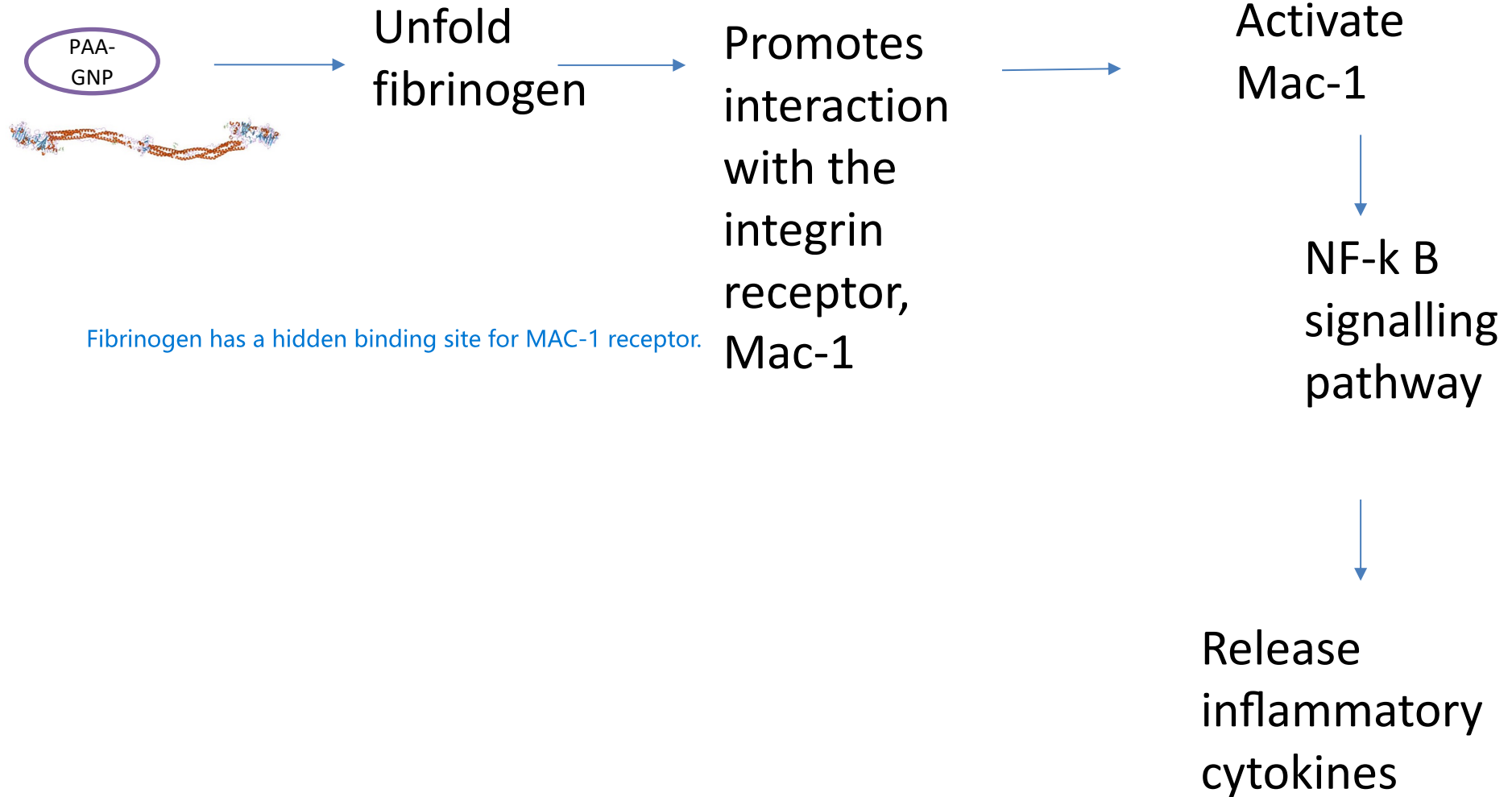
- Incubate, centrifuge followed by SDS PAGE gel
- Western blotting
- Surface Plasmon Resonance is widely used: Sensitive to dielectric of the surface which changes with adsorbed molecules

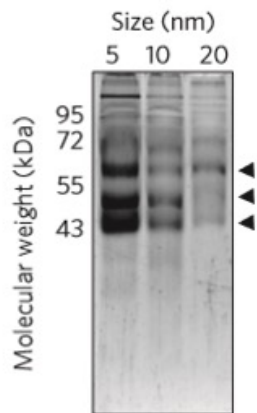
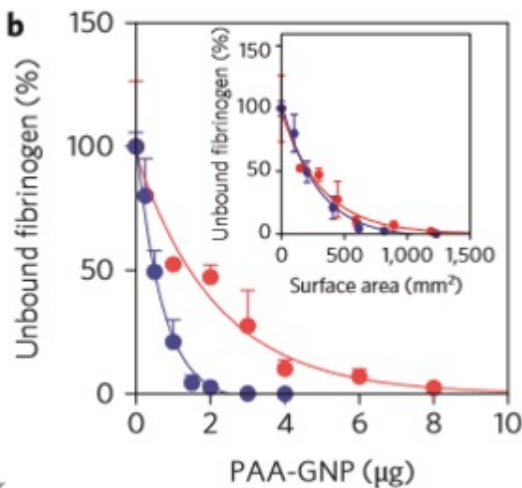
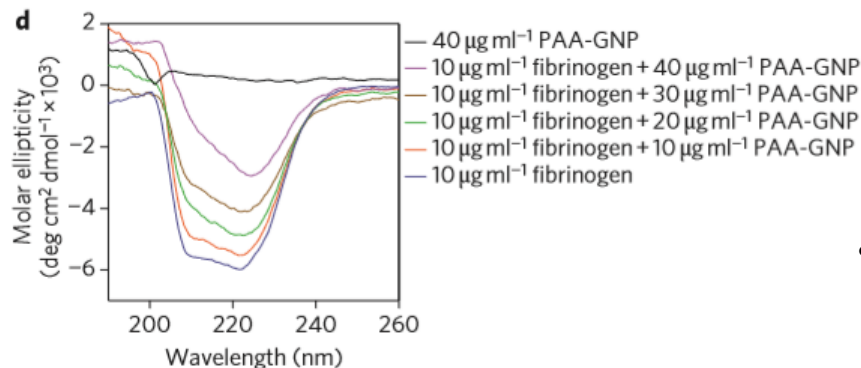


Nanoparticle-induced unfolding of fibrinogen promotes Mac-1 receptor activation and inflammation

Zhou J. Deng¹, Mingtao Liang^{2,3}, Michael Monteiro⁴, Istvan Toth^{2,3} and Rodney F. Minchin^{1*}

“Nanoparticles bound with proteins result in physiological and pathological changes”

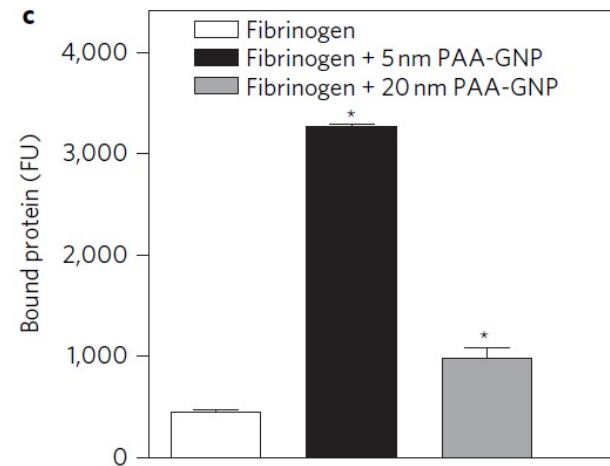
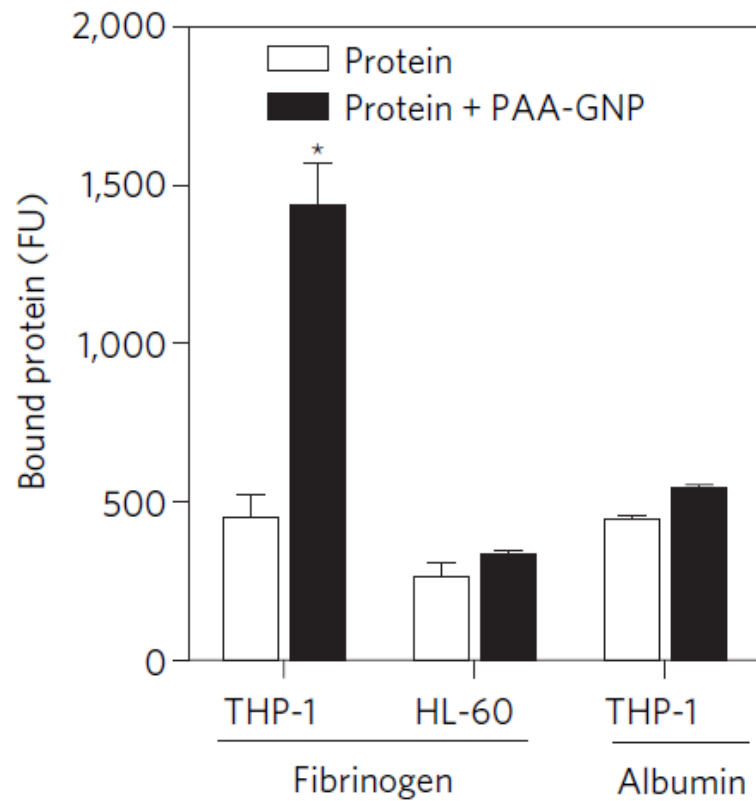


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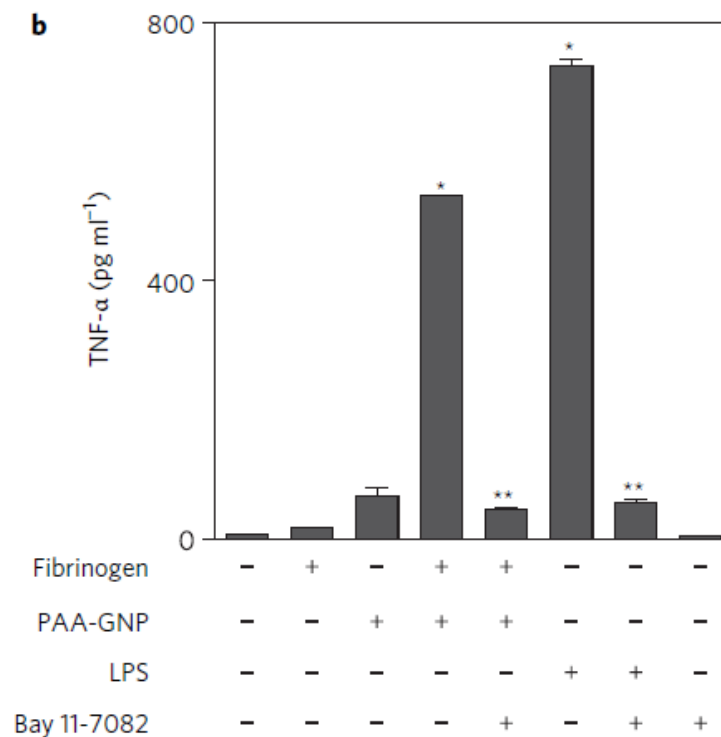
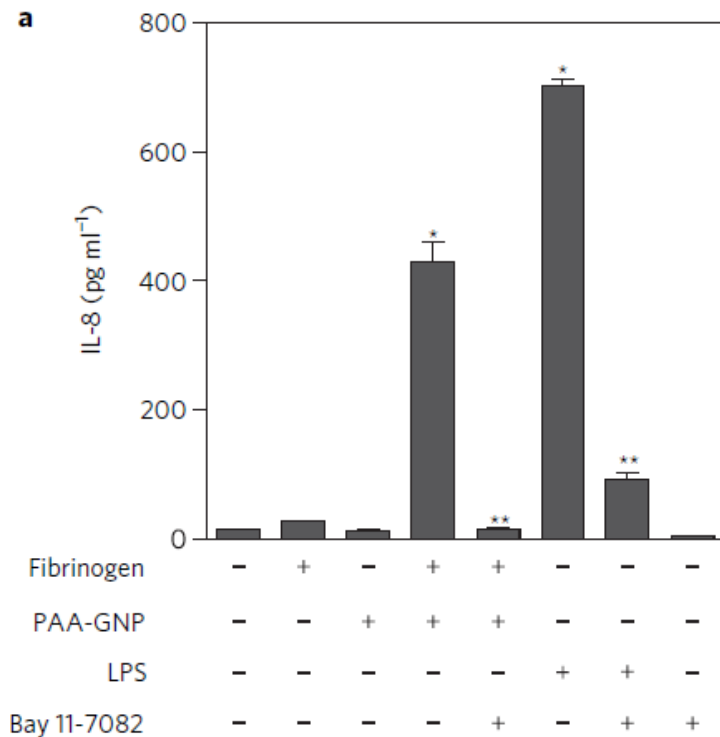
- SDS-PAGE of human plasma proteins bound to PAA-GNP with diameters of 5, 10 and 20 nm, three protein bands were observed at 65, 55 and 45 kDa.

The binding capacity of the nanoparticles for the purified human fibrinogen was determined by ultracentrifugation.

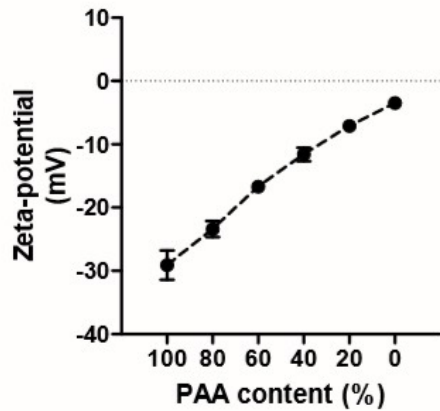
- 20 nm PAA-GNP bound significantly less fibrinogen than the 5 nm particles on a mass basis. But, when adjusted for total surface area, the binding was found to be similar
- The addition of PAA-GNP resulted in a loss of protein secondary structure, as indicated by the progressive increase in ellipticity



- Experiments done for 20nm GNP did not show any significant bound protein due to lack of exposure of γ C terminus

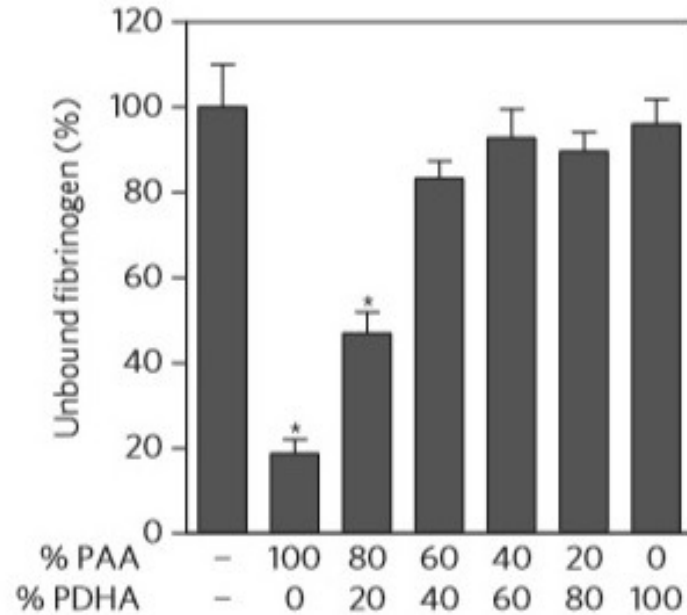


- As a result of NF-κB signalling, inflammatory cytokines are released by the cells.
- THP-1 cells were subjected to GNP-fibrinogen, non conjugated GNPs and fibrinogen. BAY 11-7082 which is an inhibitor of NF-κB pathway was used as a negative control and LPS treated cells were used as a positive control.



Zeta-potentials of PAA-PDHA-GNPs

Neutral polymer: Poly(2,3-hydroxyl-propylacrylamide) (PDHA)



Fibrinogen binding decreased significantly with 20% PDHA and was completely inhibited at higher proportions of PDHA.

Tissue Engineering

Tissue engineering is the use of a combination of cells, engineering and materials methods, and suitable biochemical and physicochemical factors to improve or replace biological tissues



Controversial image

- In August 1997, Joseph Vacanti's group published the ground-breaking paper
- Ear (made from cow cartilage) was grafted onto the mouse back
- The technology has improved since then including use of human cells



Tissue Engineering

- Generating whole tissue in-vitro and replacing damaged tissues in patients
- Generating part tissue in-vitro and replacing damaged portions of the tissues in patients
 - Putting in scaffolds to regenerate damaged part of the tissue in-vivo
- Adding support/filler material to support the function of damaged tissue

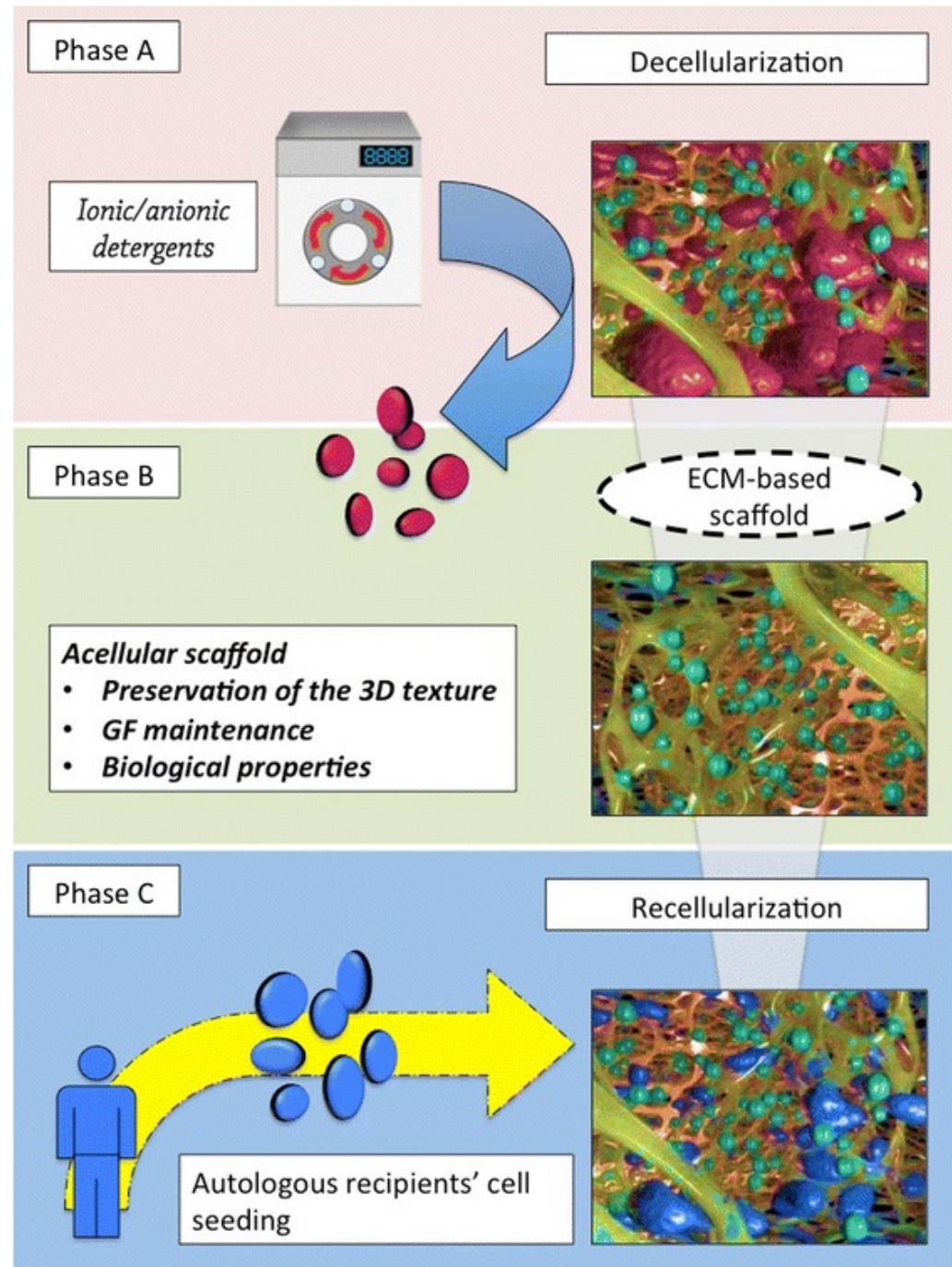
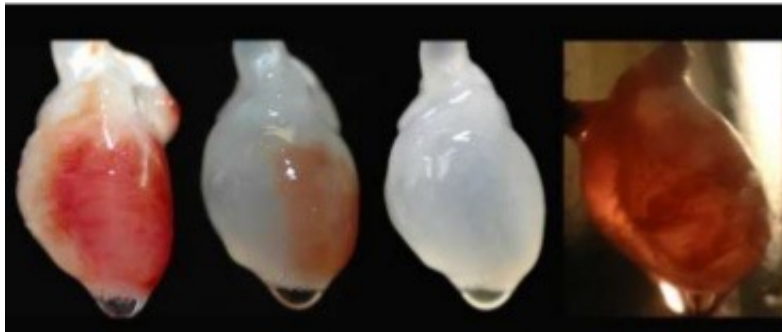
Whole tissue replacement

- Transplant from another human is the most clinically used practice
 - Lack of donors
 - Spread of disease
 - Immune rejection: half life of organs is around 10-15 years
- Transplant from genetically modified pigs has gained attention
 - However, immune rejection is still a major issue

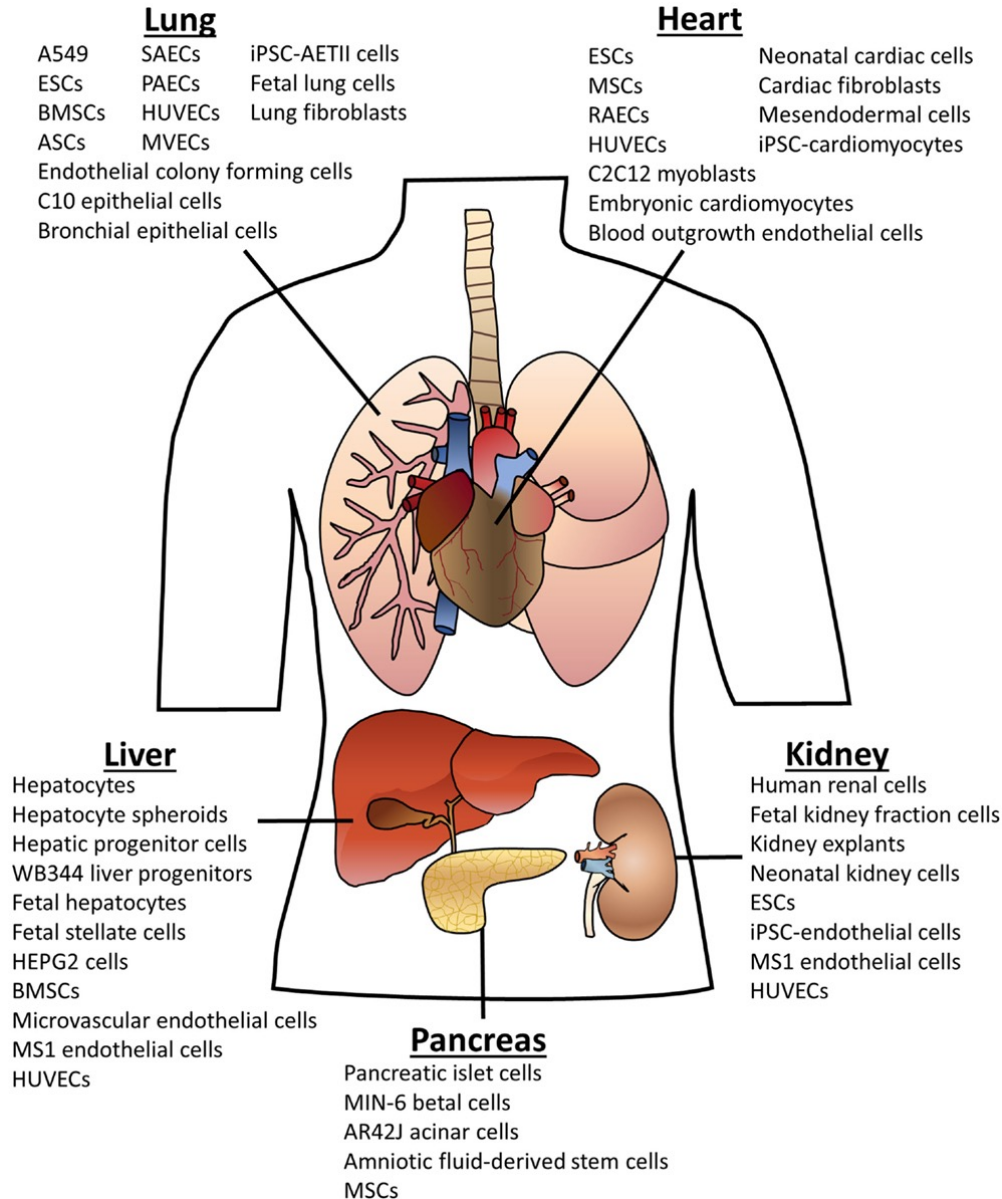
Whole tissue replacement

- Not many bioengineering approaches are available
- Major limitation is lack of blood vessels and hence the tissues are thin
- 3D printing of tissues has gained momentum but is still nascent
- Most widely used is decellularization of tissues from cadavers and animals

- Remove cells via detergents
- Seed autologous cells
- Implant in the body



Recellularization of scaffolds



Recellularization with primary cell is important

Difficult to ascertain the fate of these implanted cells

- Another approach is not recellularize at all
- Implant just the decellularized scaffold
- Rely on ability of primary cells and stem cells to repopulate the tissue
- One limitation is that small capillaries don't survive this process



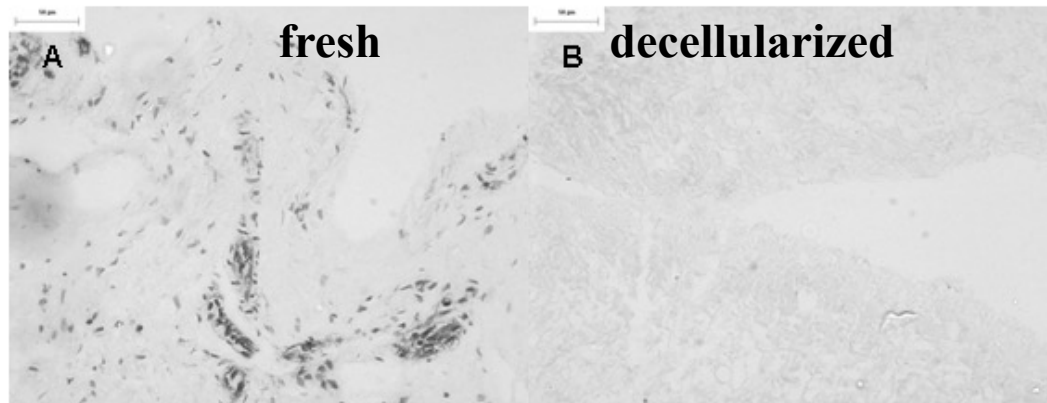
Clinical Applications:
dCELL® Vascular Patch &
Dermapure for Advanced Woundcare

Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

Decellularization: Considerations

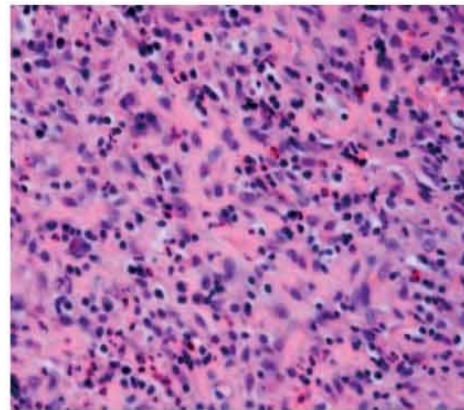
- Mammalian cells have surface antigens called alpha-gal
- If grafts are derived from non-human source then these antigens can produce strong immune response
- Important to check the absence of alpha-gal from decellularized implant

alpha-gal immunoperoxidase stain

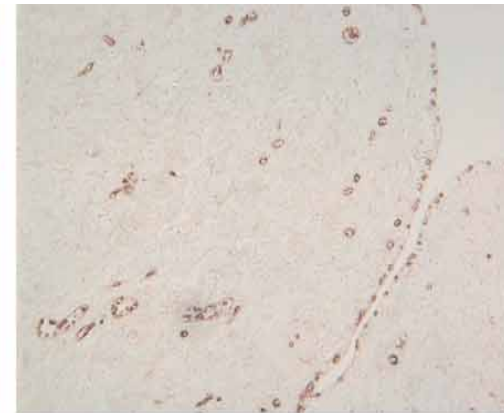


Complication following TE graft implantation: hyperacute rejection

Spark et al. *Br J Surg* 2008; 95: 582-585



b Haematoxylin and eosin



c Immunostain for α -Gal

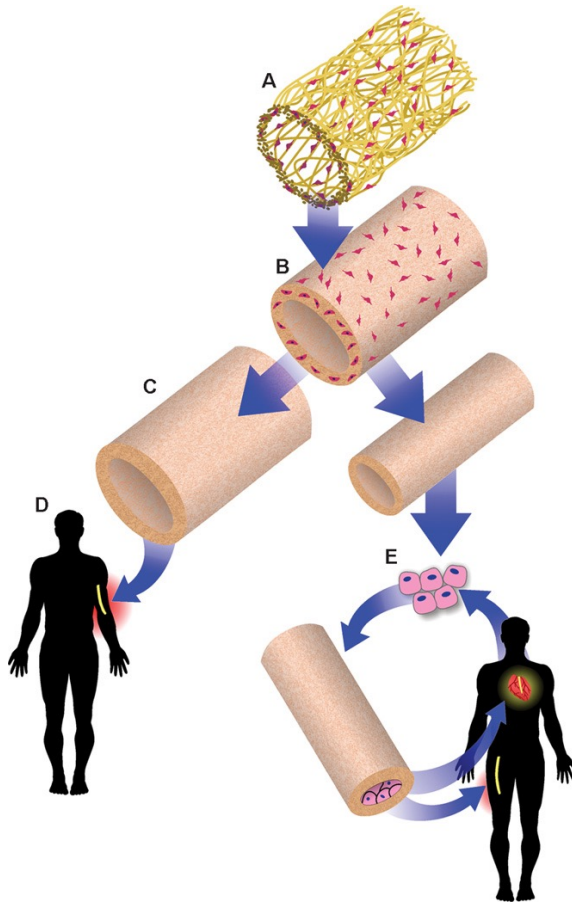
Vascular graft to bypass blockage

Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

Tissue engineered vascular graft (TEVG)



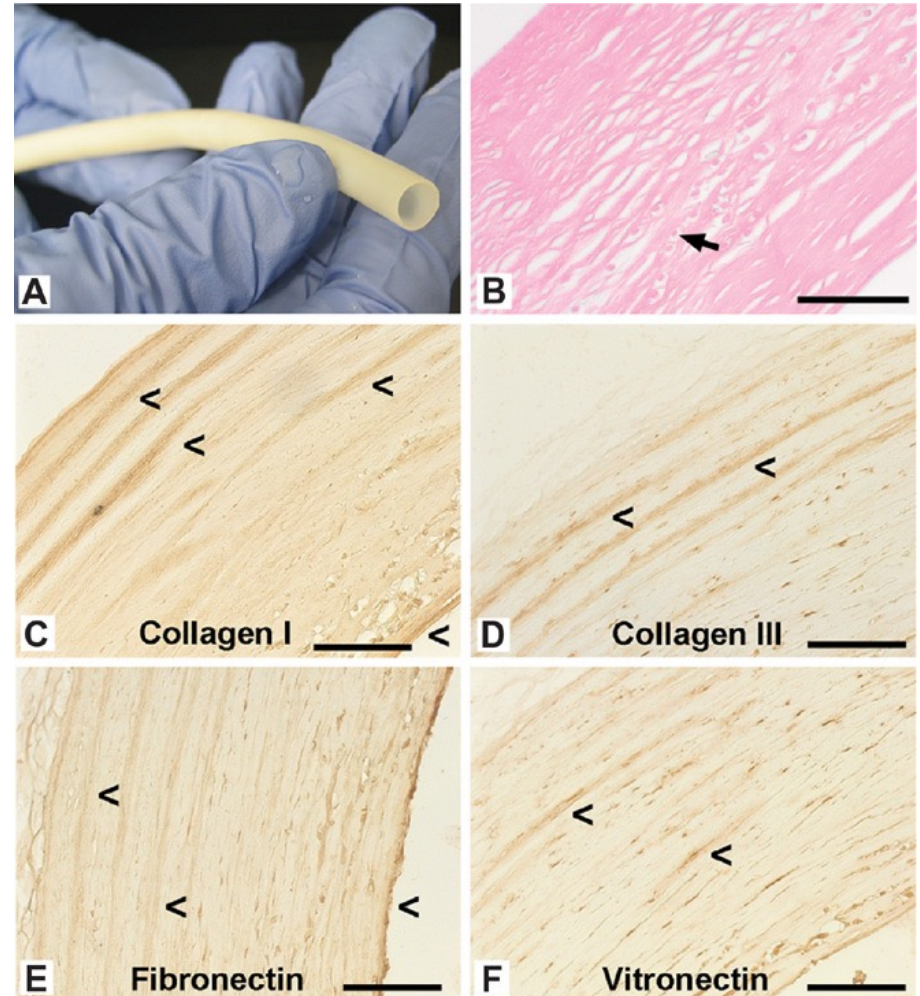
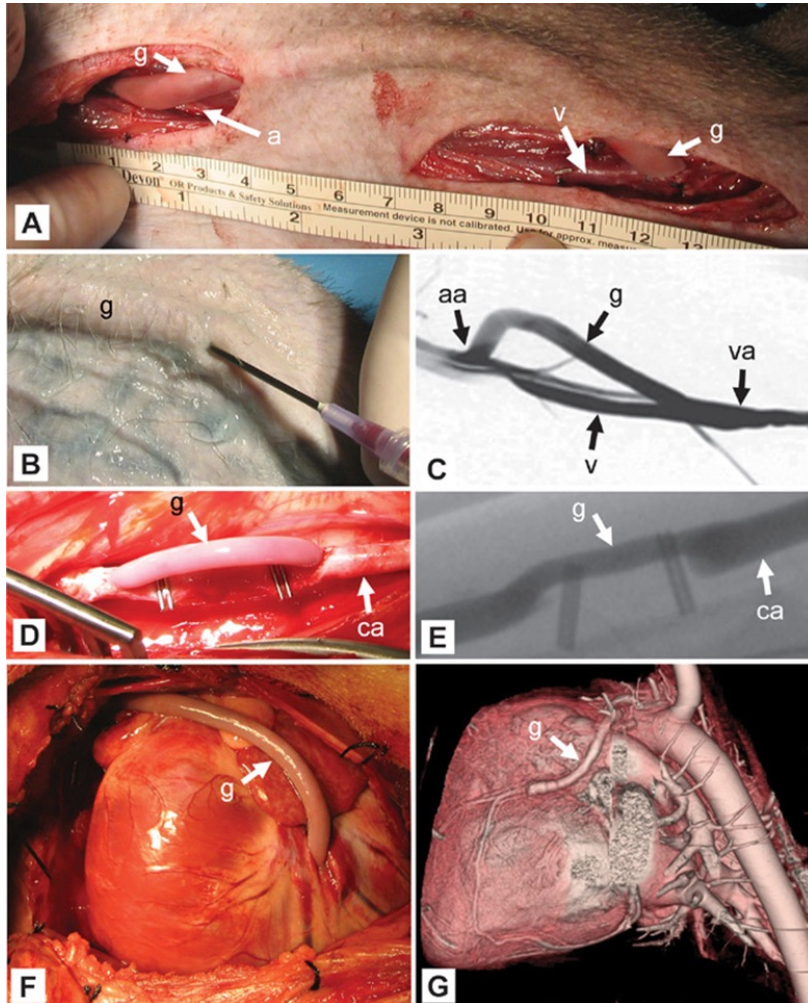
HUMACYTE



- Seed human cells on platform (A)
- Cells produce ECM (B)
- Cells decellularized and implant ready to in human (C)
- Shown to recellularize better in humans

Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

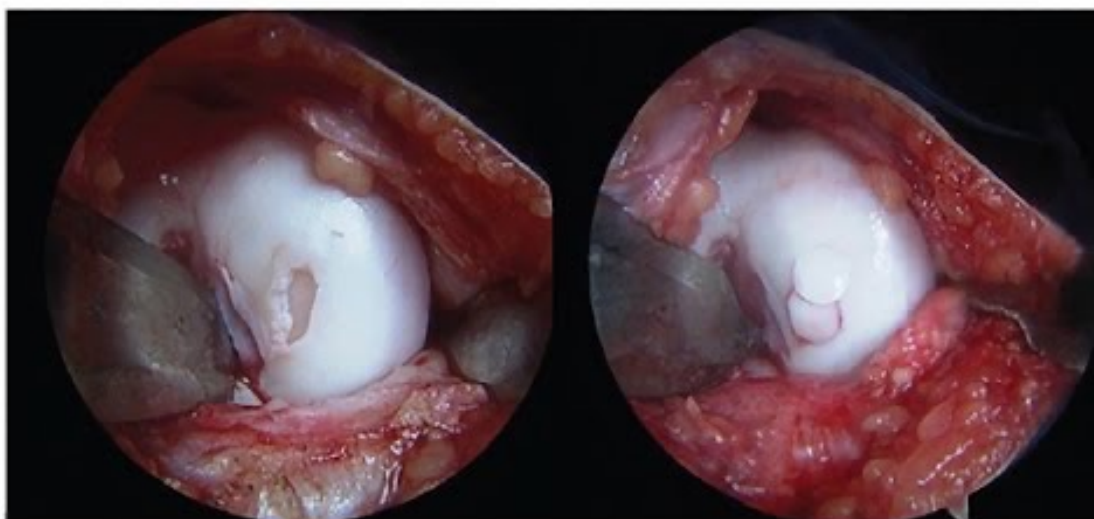
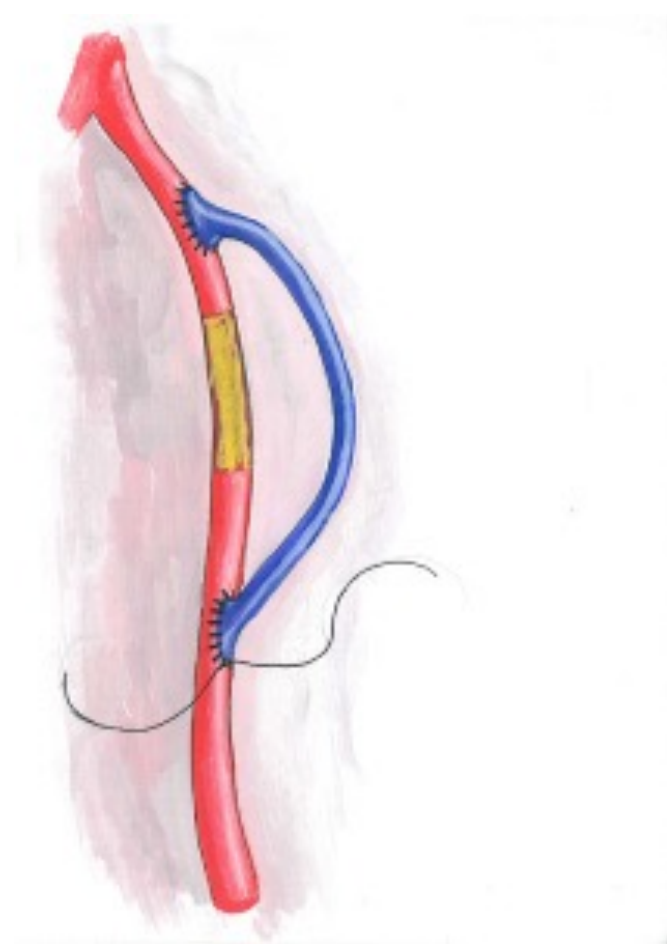
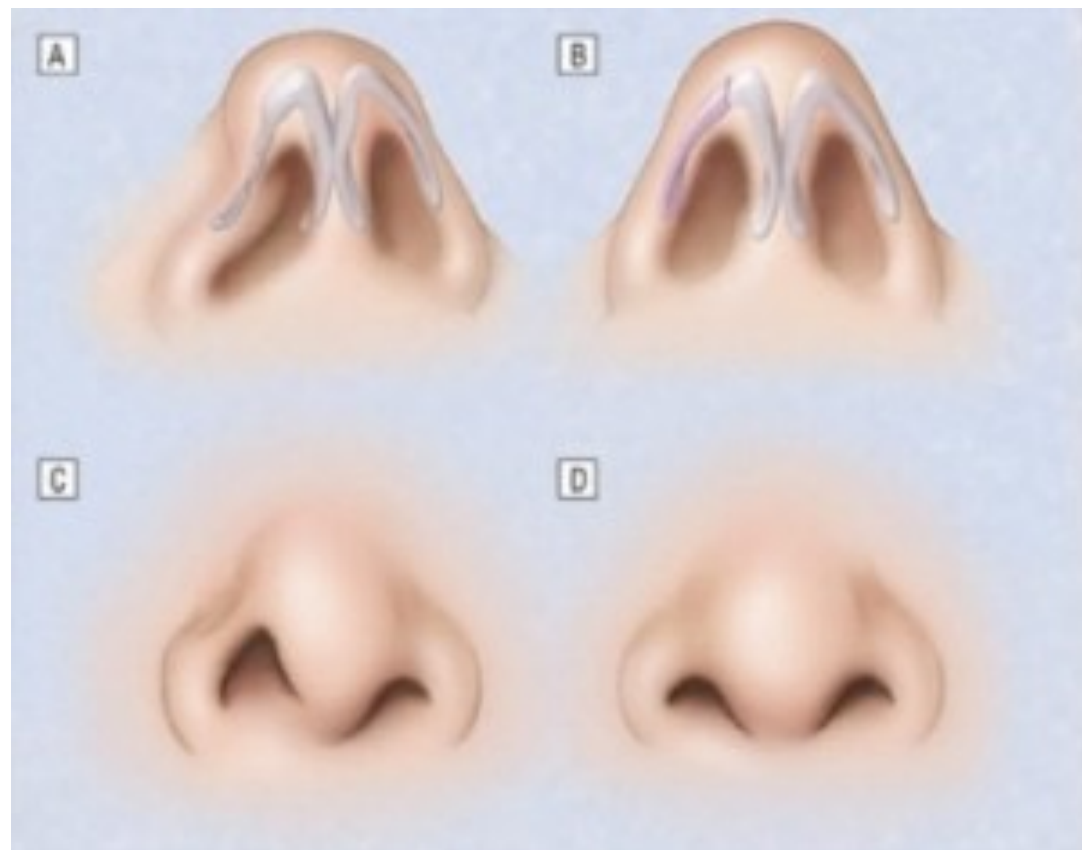
Humacyte used for bypassing blocked vessels



Slide acknowledgements: Prof. Shervanthi Homer-Vanniasinkam

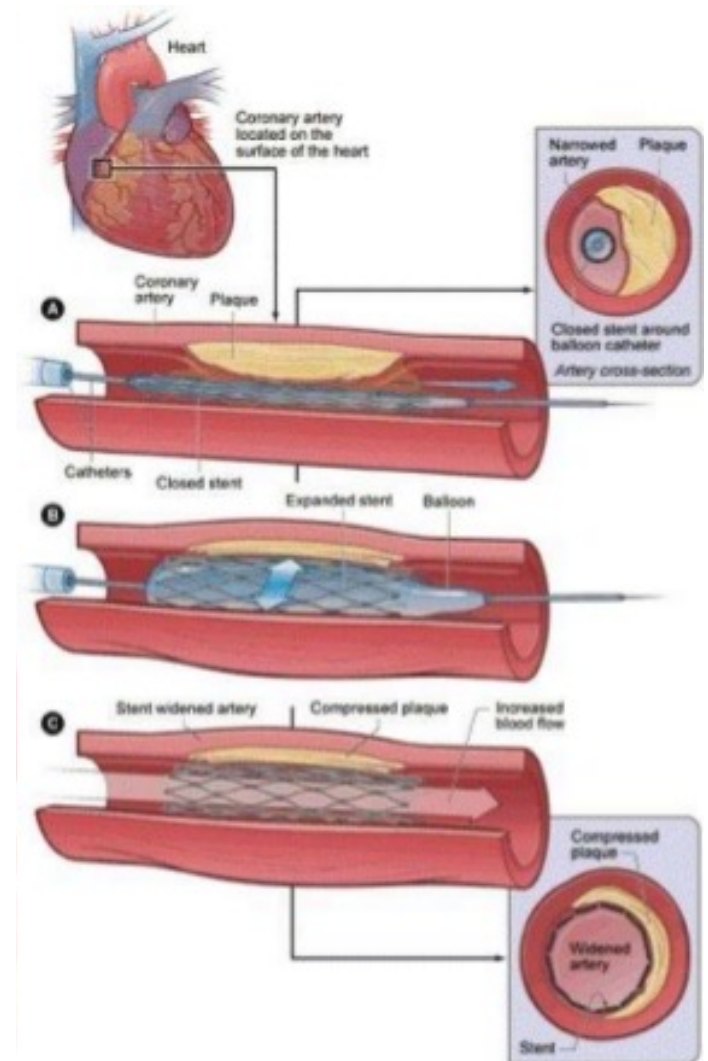
Tissue Engineering – regenerating part of a tissue

- Much more successful
- Two approaches
 - Material + cells in-vitro followed by implantation
 - Material implantation for scaffolding
- Ultimate goal is to get the tissue back to its native state
- Applications:
 - Cartilage defects
 - Vascular grafts
 - Heart patch



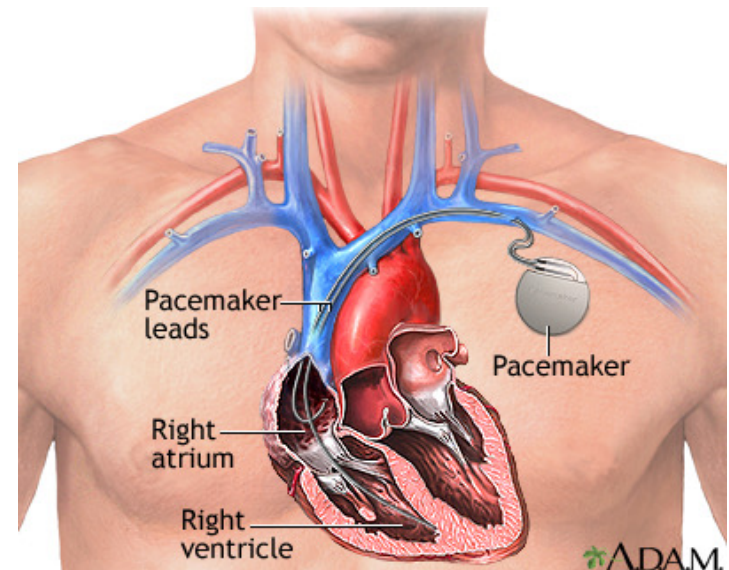
Tissue Engineering – Support function of damaged tissue

- Mainly to provide support/filler material for proper function of damaged tissue
- Stents: Remove occlusions from vessels



Tissue Engineering – Support function of damaged tissue

- Dental and Orthopaedic metal implants
 - Knee
 - Long bones
- Pacemaker
- Contact lens: Widely used



Drug Delivery in Tissue Engineering

- Depending on the application most of the these implants would need to release chemicals and drugs for effective functioning
 - VEGF and PDGF for blood vessel formation
 - Antibiotic for prevention of infections
 - Immunosuppressants for acceptance of graft
 - Adsorbed proteins and growth factors
 - Other small molecules

Review

Polymeric Growth Factor Delivery Strategies for Tissue Engineering

Ruth R. Chen¹ and David J. Mooney¹⁻⁴

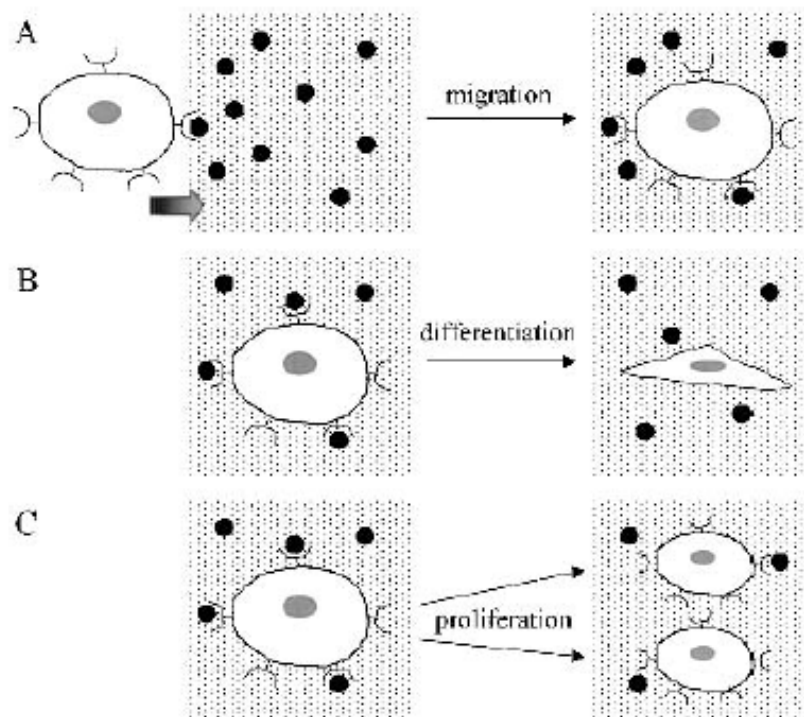


Fig. 1. Growth factor binding to a cell surface receptor can elicit different types of responses to regenerate tissue in a defect site. A, Chemotactic responses to growth factor binding result in cell migration into a defect site. B, Cell differentiation can occur to provide the needed cell population. C, Proliferation of cells within the site fill the tissue defect.

Table I. A List of Growth Factors Commonly Used in Tissue Engineering

Growth factor	Abbreviation	Molecular weight (kDa)	Known activities
Epidermal growth factor	EGF	6.2	Proliferation of epithelial, mesenchymal, and fibroblast cells
Platelet-derived growth factor	PDGF-AA	28.5	Proliferation and chemoattractant agent for smooth muscle cells; extracellular matrix synthesis and deposition
	PDGF-AB	25.5	
	PDGF-BB	24.3	
Transforming growth factor- α	TGF- α	5.5	Migration and proliferation of keratinocytes; extracellular matrix synthesis and deposition
Transforming growth factor- β	TGF- β	25.0	Proliferation and differentiation of bone forming cells; chemoattractant for fibroblasts
Bone morphogenetic protein	BMP-2	26.0	Differentiation and migration of bone forming cells
	BMP-7	31.5	
Basic fibroblast growth factor	bFGF/FGF-2	17.2	Proliferation of fibroblasts and initiation of angiogenesis
Vascular endothelial growth factor	VEGF ₁₆₅	38.2	Migration, proliferation, and survival of endothelial cells

rH, recombinant human.

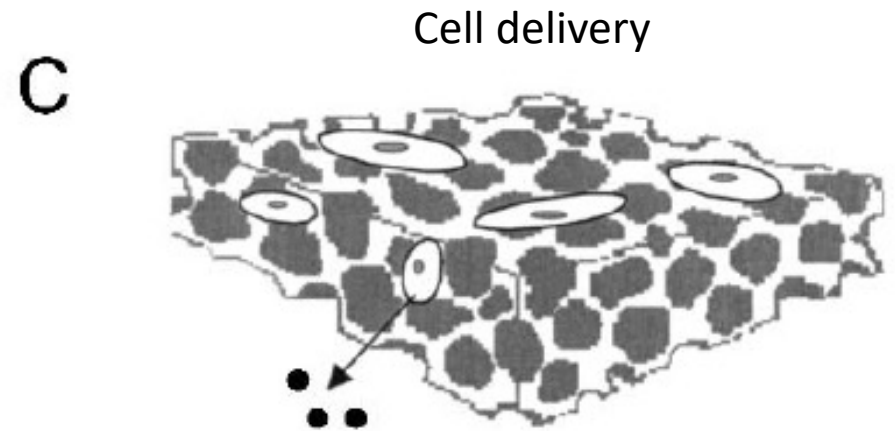
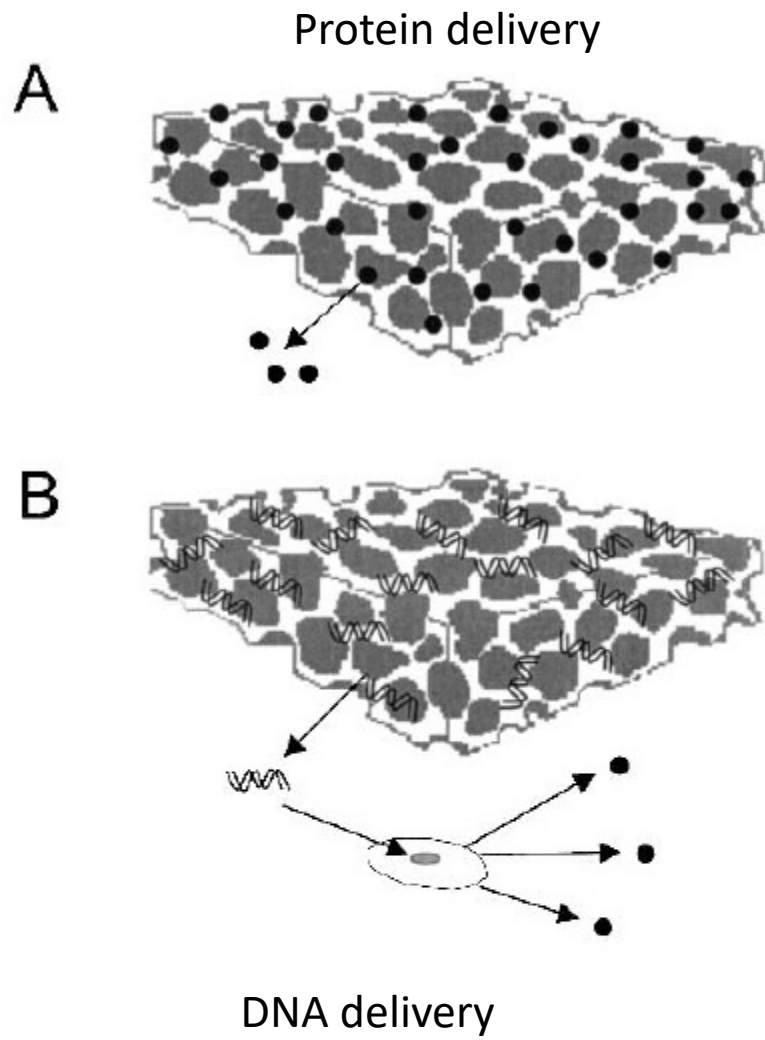
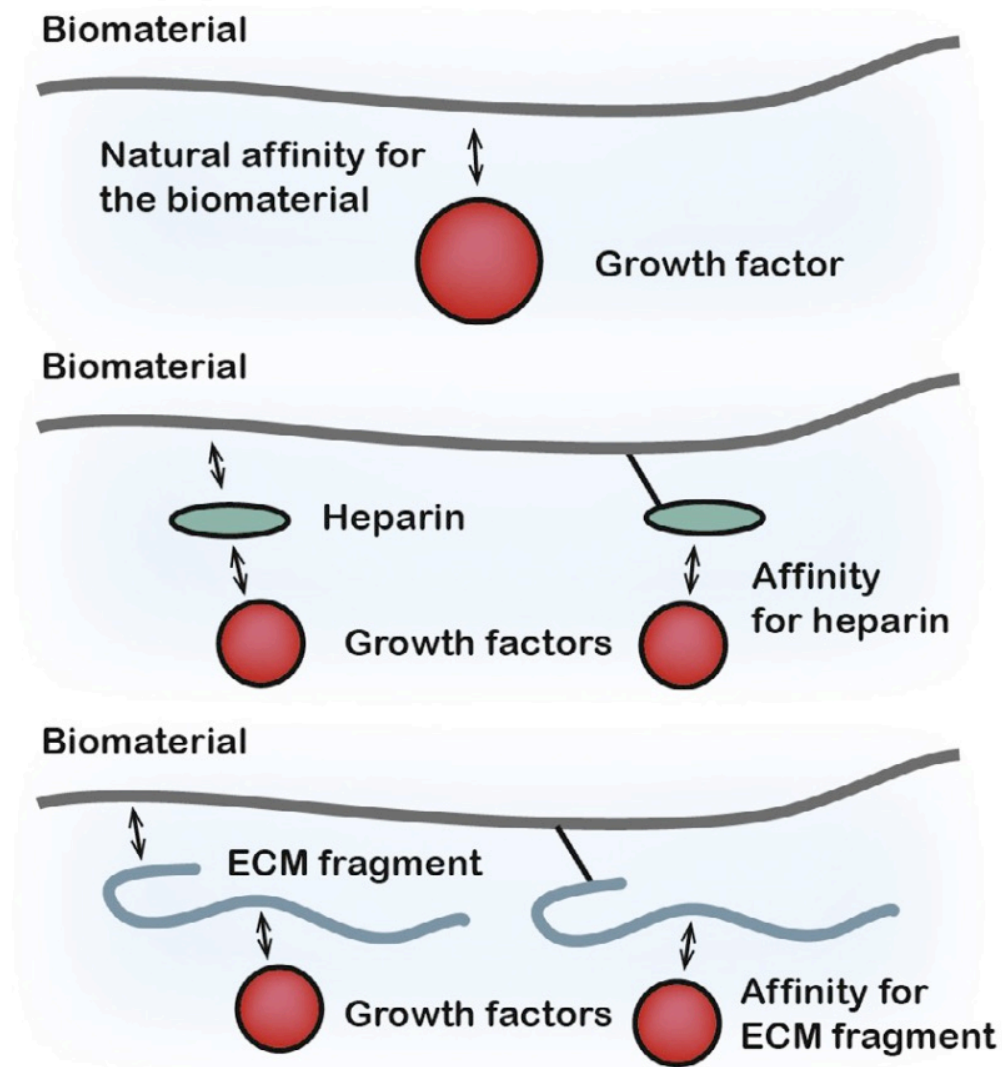
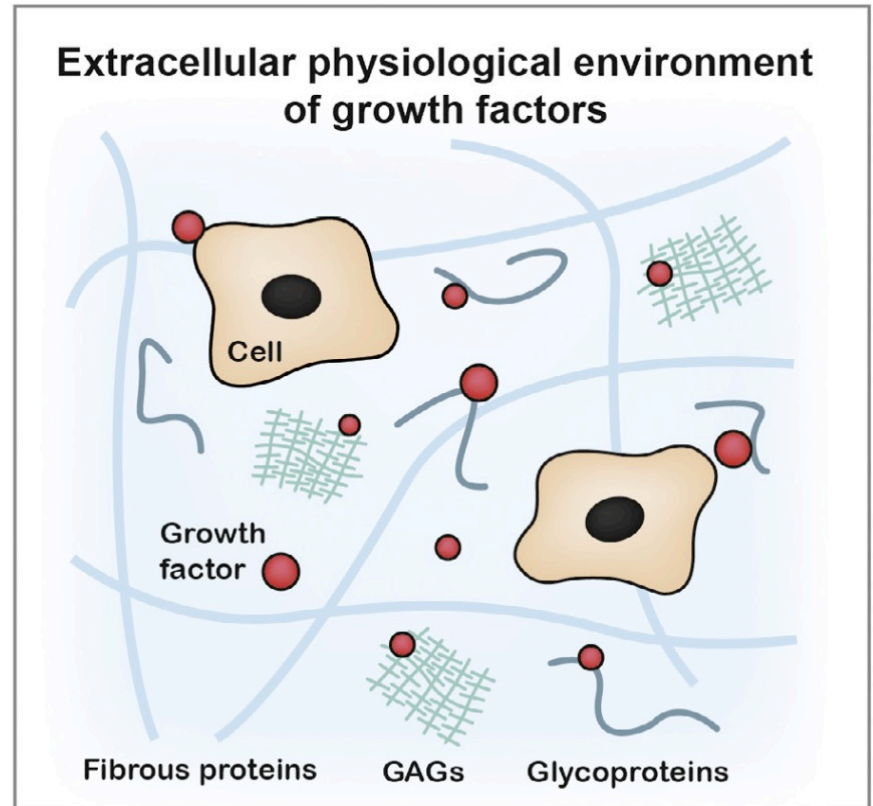
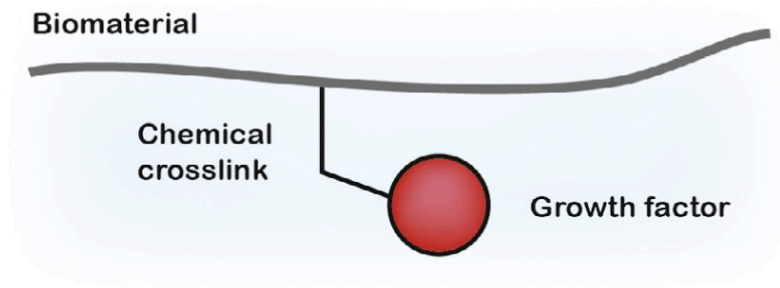


Fig. 3. Three primary polymeric growth factor delivery strategies. A, Growth factors are embedded within the polymer matrix and released. B, Genes encoding a growth factor are embedded within the polymer matrix and released, followed by cellular uptake and expression of the gene to produce growth factor. C, Growth factor is released from cells seeded on the polymer matrix that secrete the factor.

Non-covalent affinity for biomaterial matrices

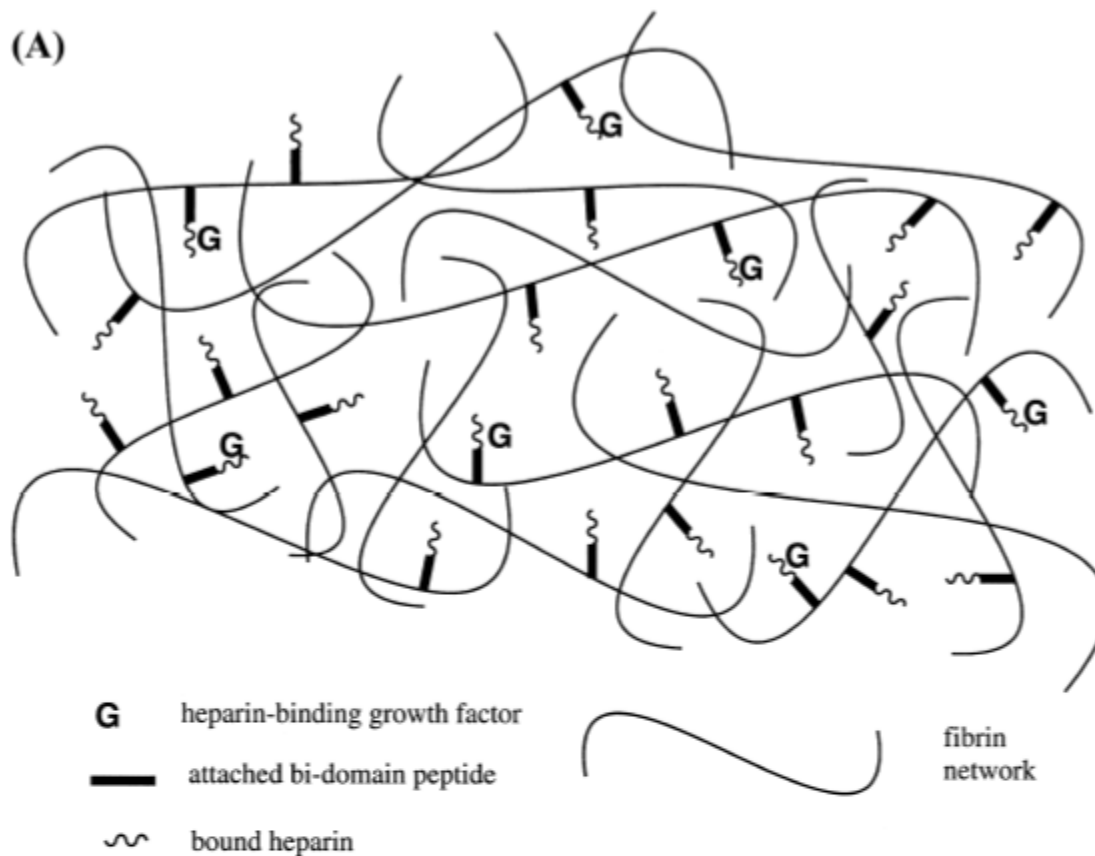


Covalent attachment to biomaterial matrices



Development of fibrin derivatives for controlled release of heparin-binding growth factors

Shelly E. Sakiyama-Elbert^{a,b}, Jeffrey A. Hubbell^{a,*}



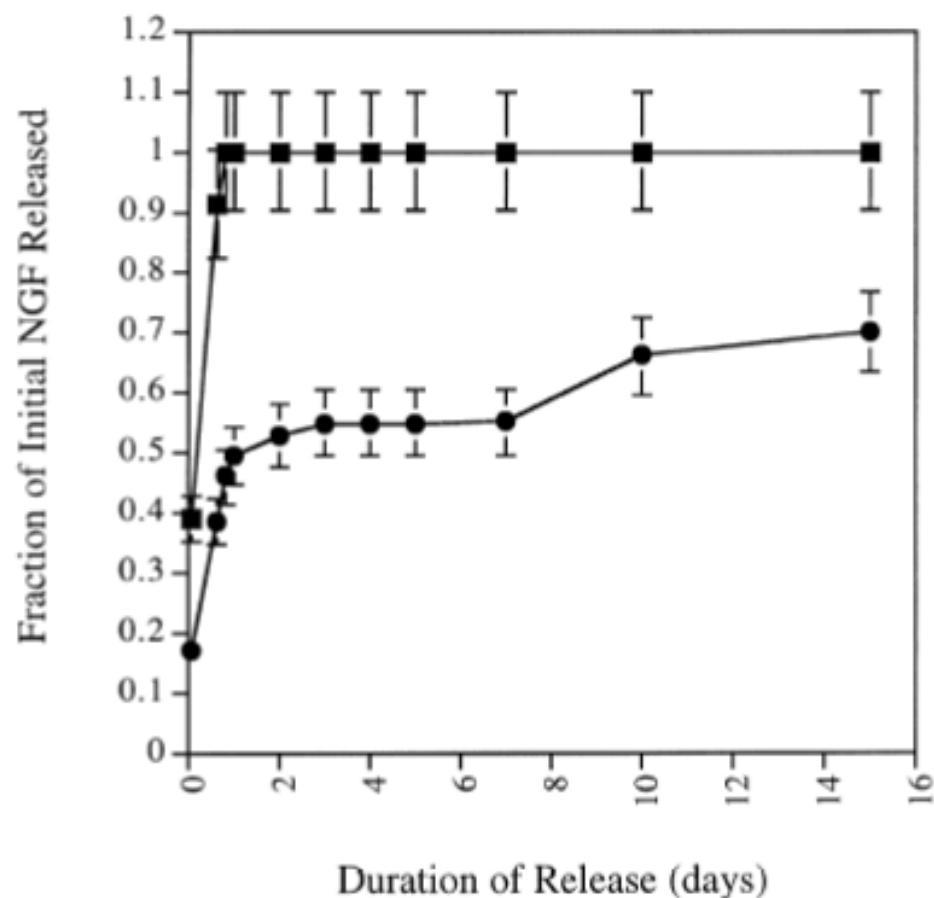


Fig. 2. Release profile of β -NGF diffusion from fibrin matrices in the presence or absence of the heparin-containing delivery system. (■) Unmodified fibrin with 100 ng/ml of β -NGF in the culture medium. (●) Heparin-containing matrices with 100 ng/ml of matrix-interacting β -NGF. Mean values and standard deviation are shown. These results demonstrate that despite β -NGF's weak heparin-binding affinity, the release of β -NGF from fibrin matrices is prolonged by a heparin-containing drug delivery system.

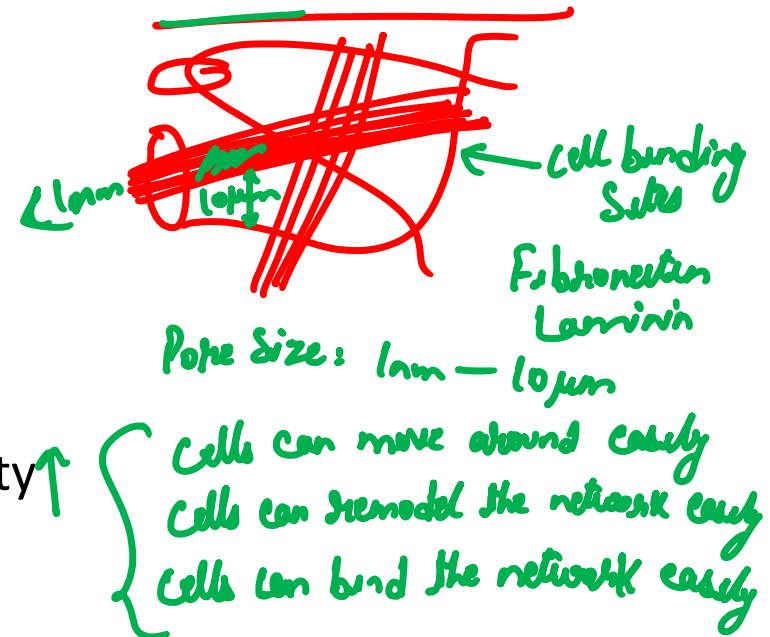
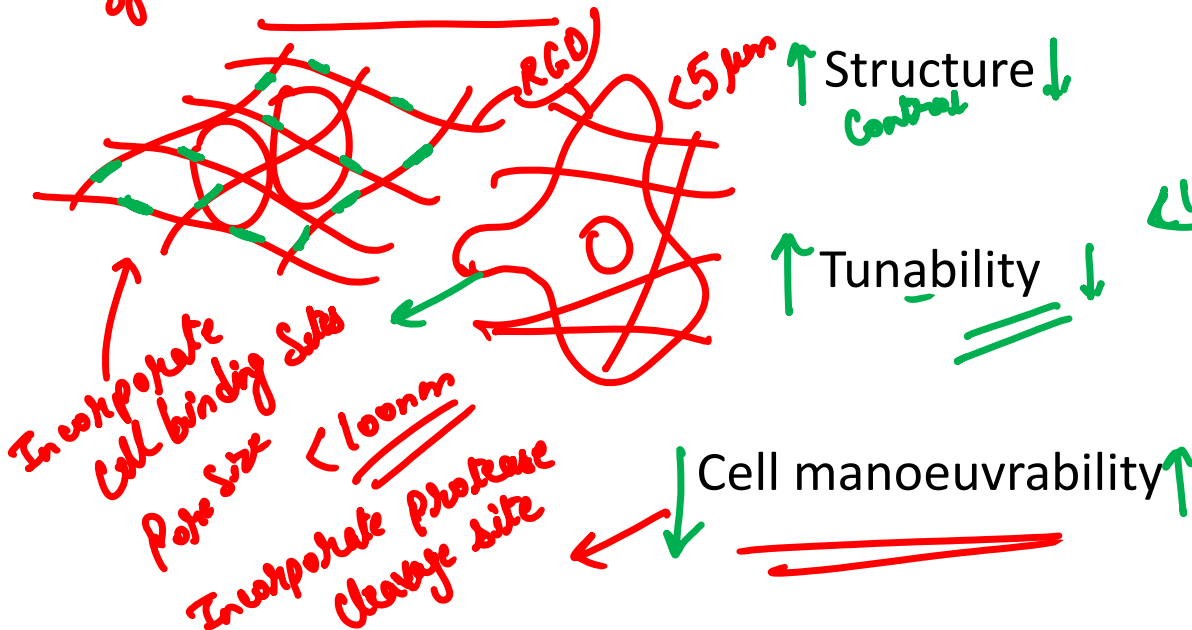
Natural hydrogels vs synthetic hydrogels for tissue engineering

PEG hydrogels

Collagen hydrogels

Synthetic material

Natural material

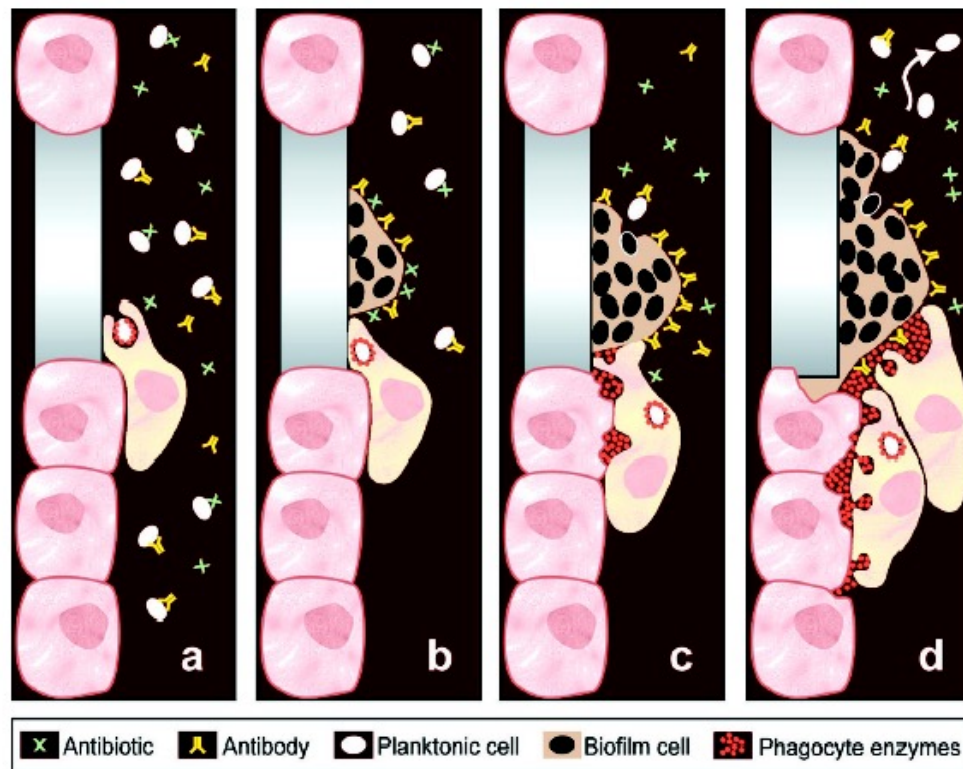


Implant Associated Infections

Definitions

- **Biofouling**
 - The phenomenon of protein, cellular and bacterial (microorganism) attachment to material surfaces that leads to loss of device function, infection and implant failure
- **Biofilms**
 - Microorganisms (bacteria) in colonies on surfaces form layers (2-100 organism thick) which are composed of organism, cellular material, extracellular polysaccharides, environmental adsorbates and debris. This surface composite is called biofilm or slime

Fig. 1. Diagram of a medical biofilm. (A) Planktonic bacteria can be cleared by antibodies and phagocytes, and are susceptible to antibiotics. (B) Adherent bacterial cells form biofilms preferentially on inert surfaces, and these sessile communities are resistant to antibodies, phagocytes, and antibiotics. (C) Phagocytes are attracted to the biofilms. Phagocytosis is frustrated but phagocytic enzymes are released. (D) Phagocytic enzymes damage tissue around the biofilm, and planktonic bacteria are released from the biofilm. Release may cause dissemination and acute infection in neighboring tissue.



Controlled release of antibiotics from plasma-deposited membranes

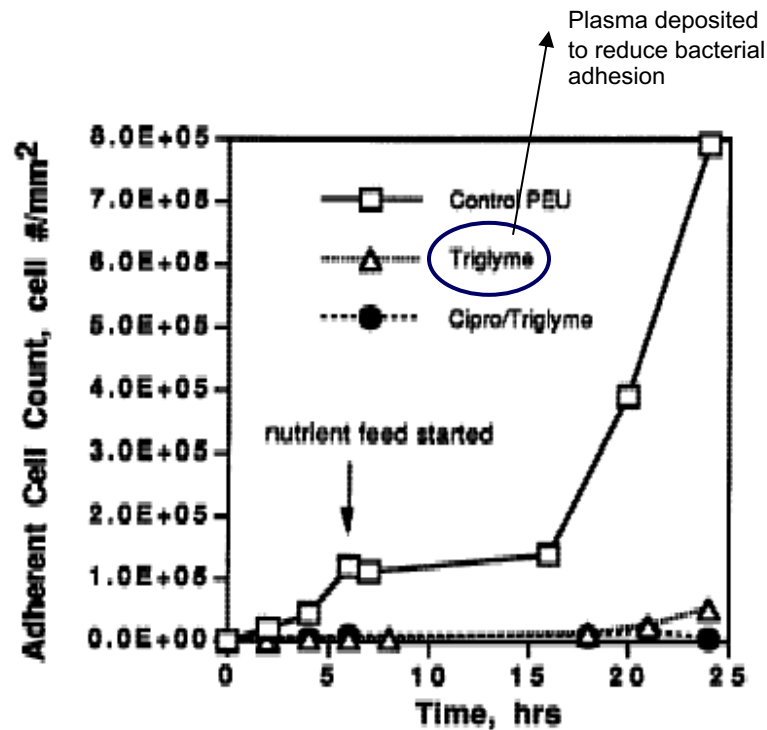


Figure 5. Adherent cell numbers per unit area with elapsed time for three different materials: PEU control, PEU with triglyme only, and PEU with pBMA barrier membrane releasing ciprofloxacinTM. Total experiment is shown.